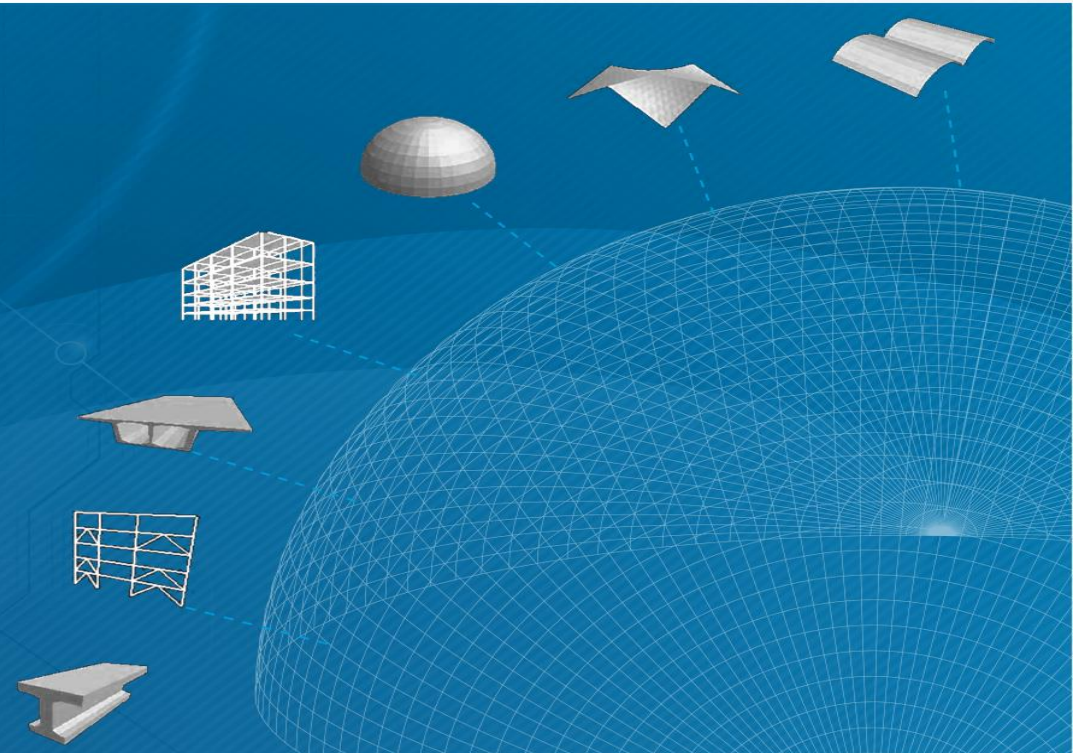
CE – 412

Design of Concrete Structures



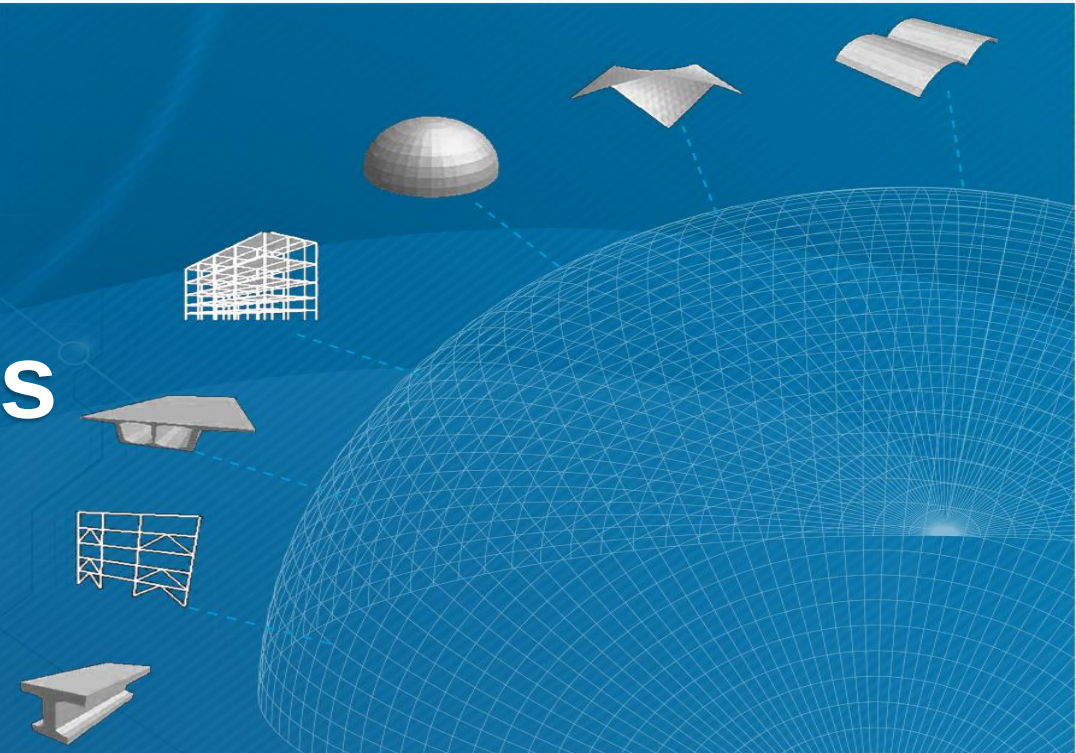
September – 2021

Fawad A. Najam
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CE – 412 Design of Concrete Structures

Lecture 1:

Introduction to Structures and Structural Design Process



September – 2021

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What is Structural Engineering

A branch of civil engineering dealing primarily with the design and construction of structures (such as bridges, buildings, dams).

Merriam-Webster Dictionary

Structural Engineering is the art of

modelling **materials** *we don't wholly understand*
into **shapes** *we cannot precisely analyze*
so as to withstand **forces** *we cannot properly assess*

in such a way that the public at large has no reason to suspect the extent of our ignorance.

- Dr. A R Dykes, British Institution of Structural Engineers, 1976

The Goal of

Structural Engineering



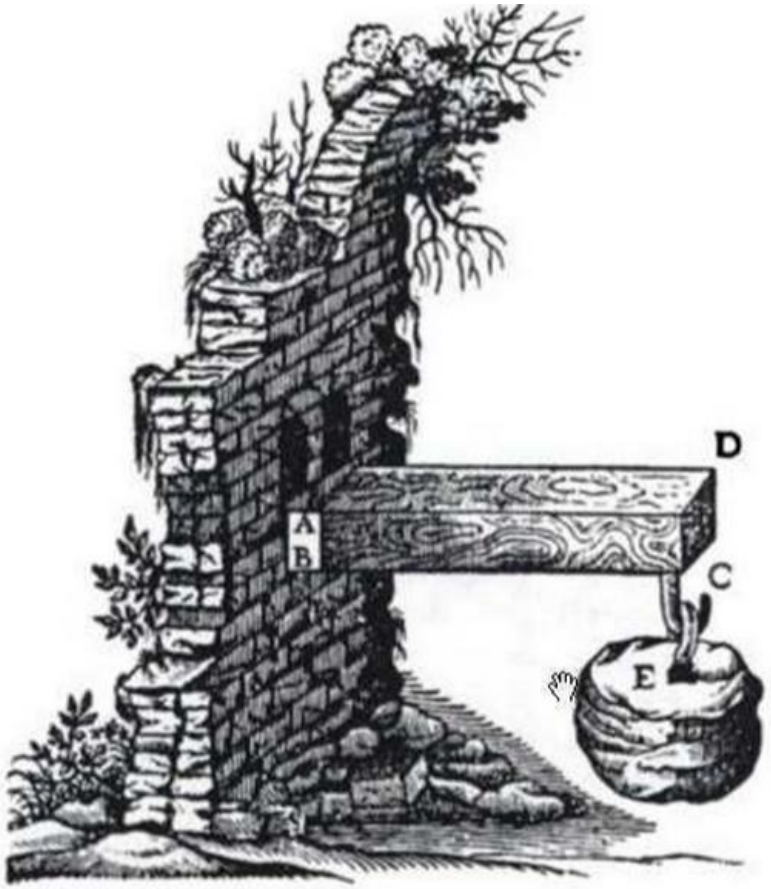
The Role of Structural Engineers

- To help in the creation of the safe built environment.
- Nothing can function, if structural engineers do not do their job well.
- Every other professional “Depends” on the role of structural engineers.

All systems absolutely need structures

- Transportation systems
- Telecommunication systems
- Electricity and power systems
- Water supply, waste water systems
- Irrigation and agricultural systems
- Mining, oil, gas, offshore platforms
- Industrial plants, piping and process
- Housing, commercial Businesses
- Public Buildings

The beginning of Structural Engineering

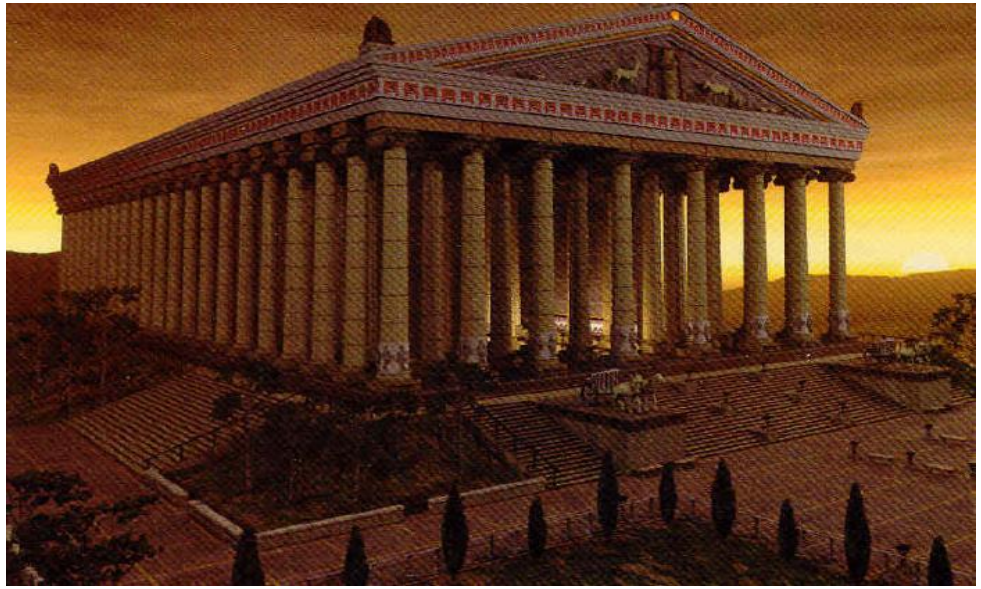


recognition of the loads,

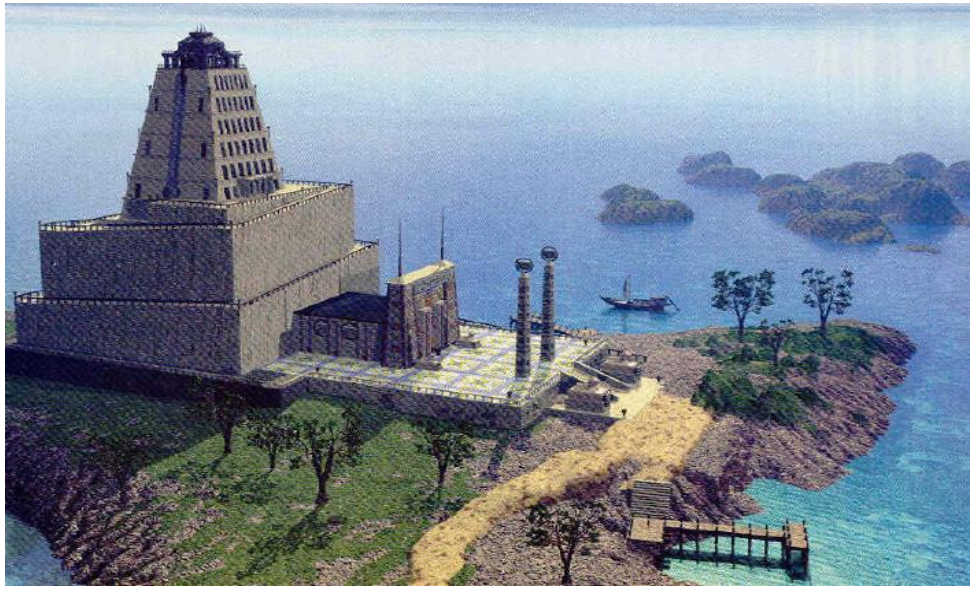
and fact that loads have an effect on members and materials

and that there is a resistance within materials to resist the loads effects

and that there is some relationship between them



The Temple of Artemis at Ephesus



The Lighthouse of Alexandria



The Mausoleum at Halicarnassus



The Hanging Gardens of Babylon



The Colossus of Rhodes

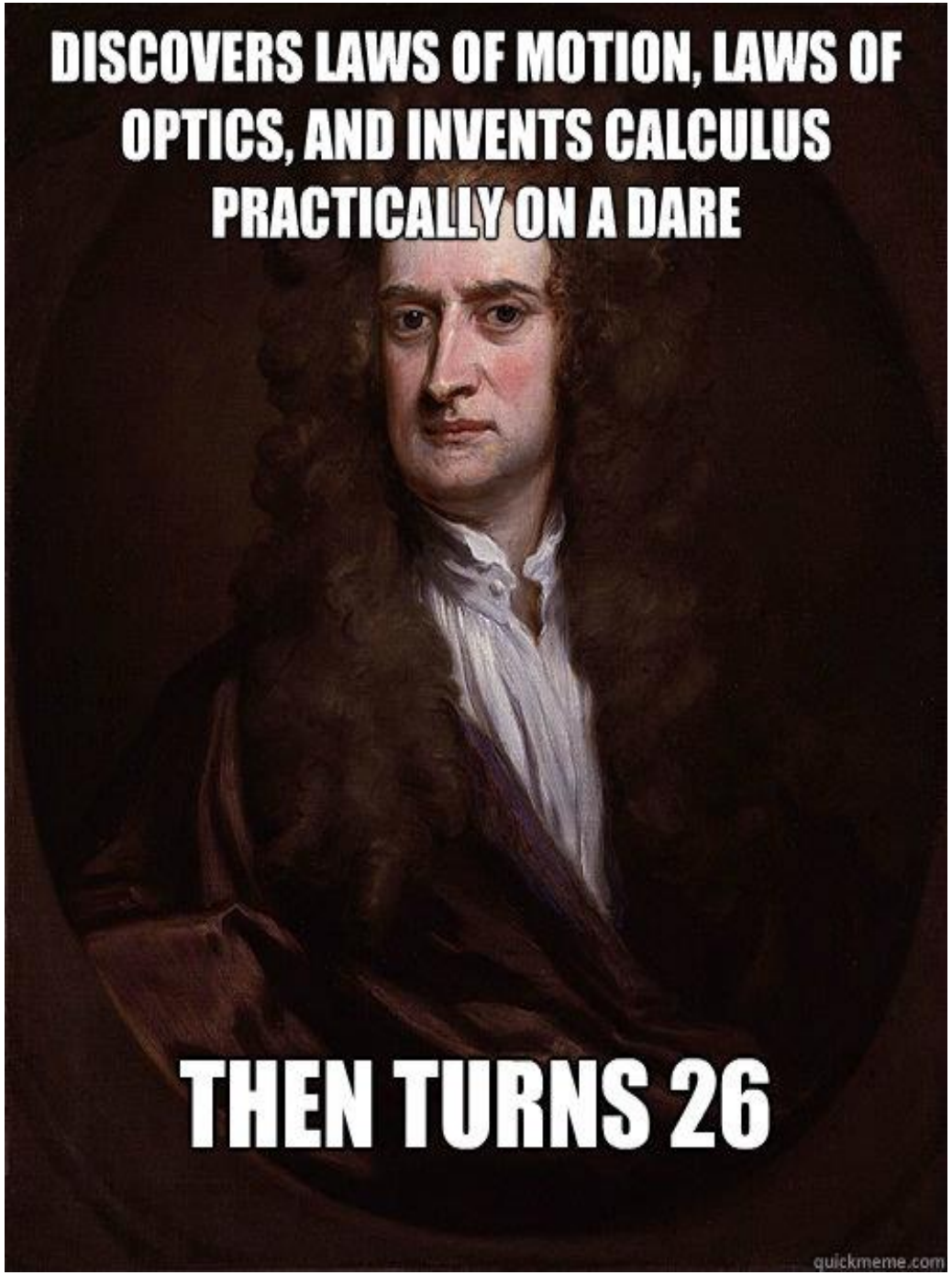


The Statue of Zeus at Olympia





The Great Pyramids of Egypt

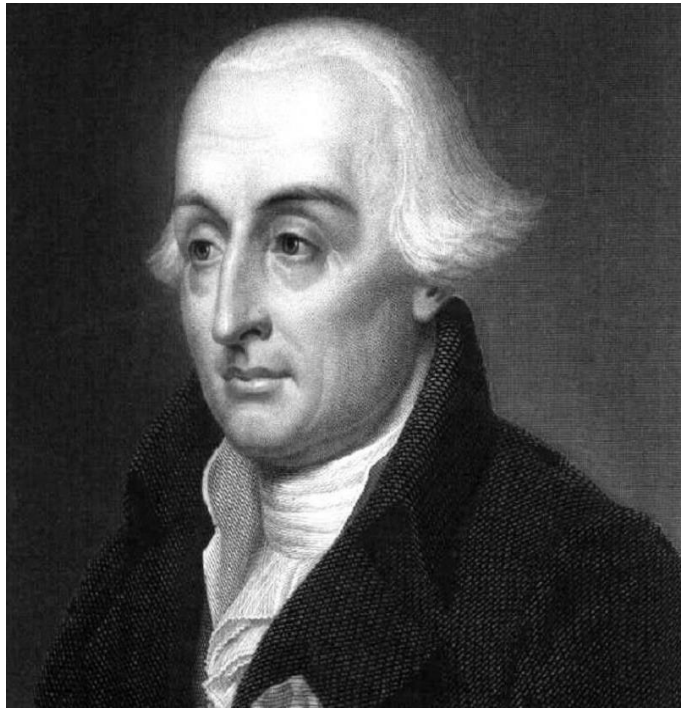

DISCOVERS LAWS OF MOTION, LAWS OF OPTICS, AND INVENTS CALCULUS PRACTICALLY ON A DARE
THEN TURNS 26
quickmeme.com

(1643 – 1727)

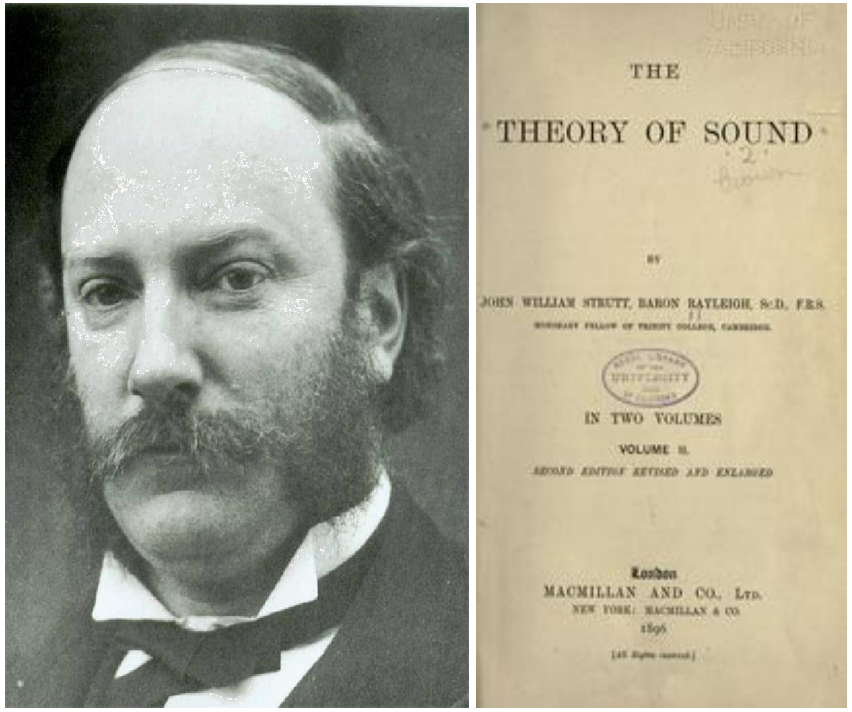
From where it all started …



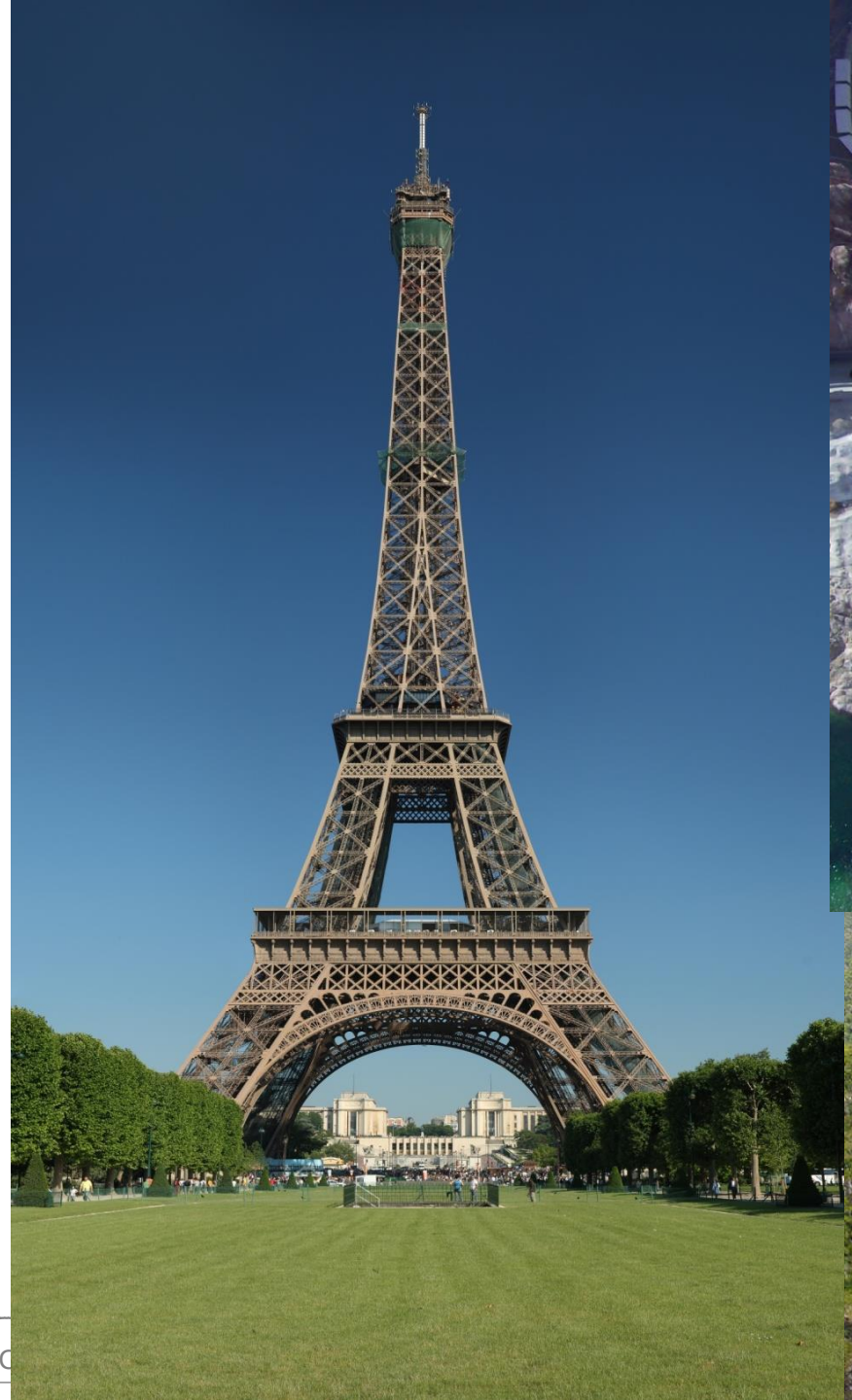
D’Alembert (1717–1783)



Joseph-Louis Lagrange (1736 - 1830)



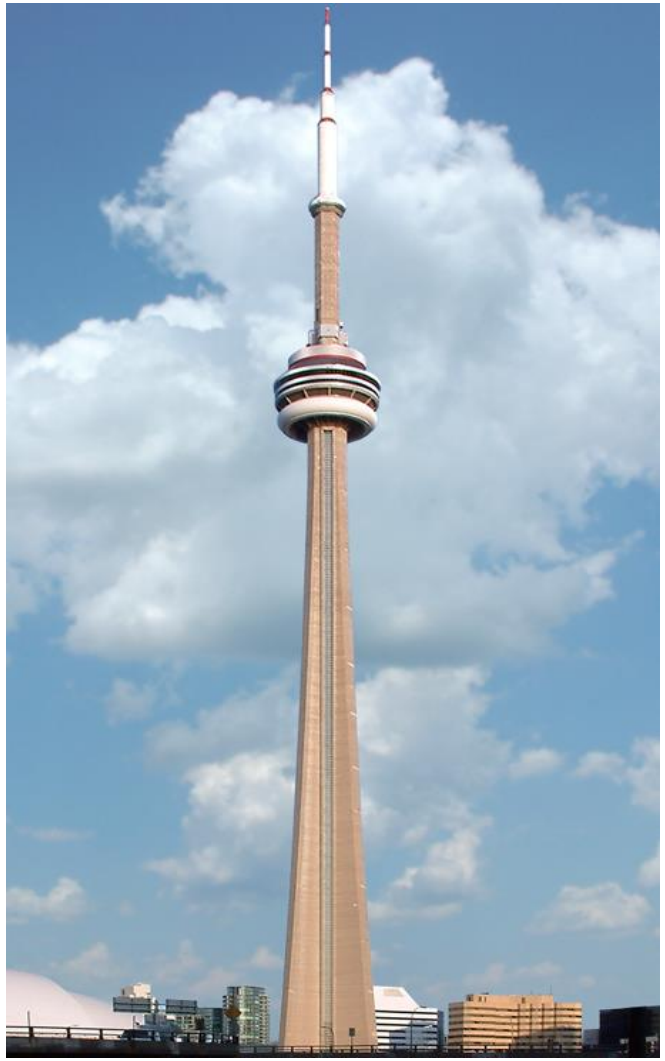
Lord Rayleigh (John William Strutt, 1842 - 1919)












HowardDigital.com














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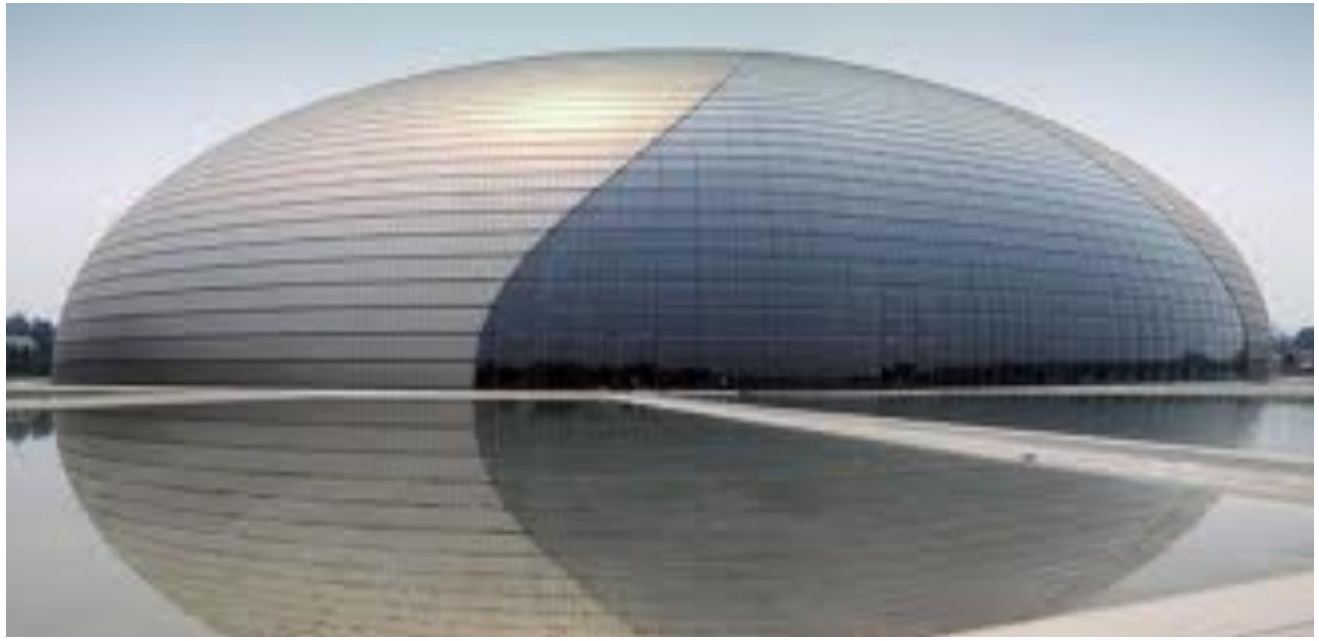






Prof. Dr. GÜnther Filz (student of the master program), Visualization





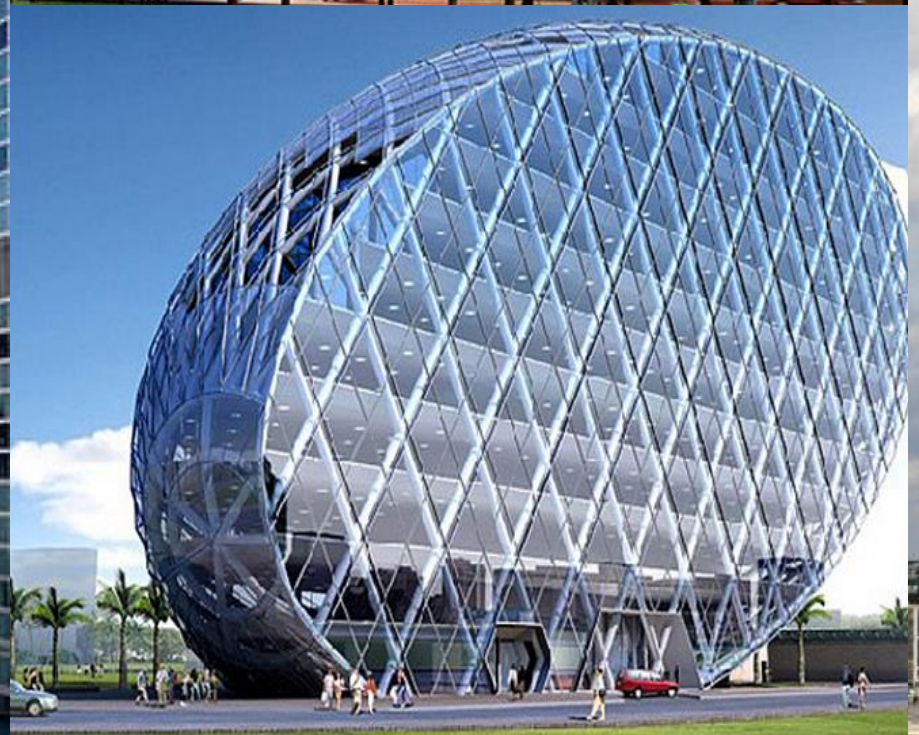








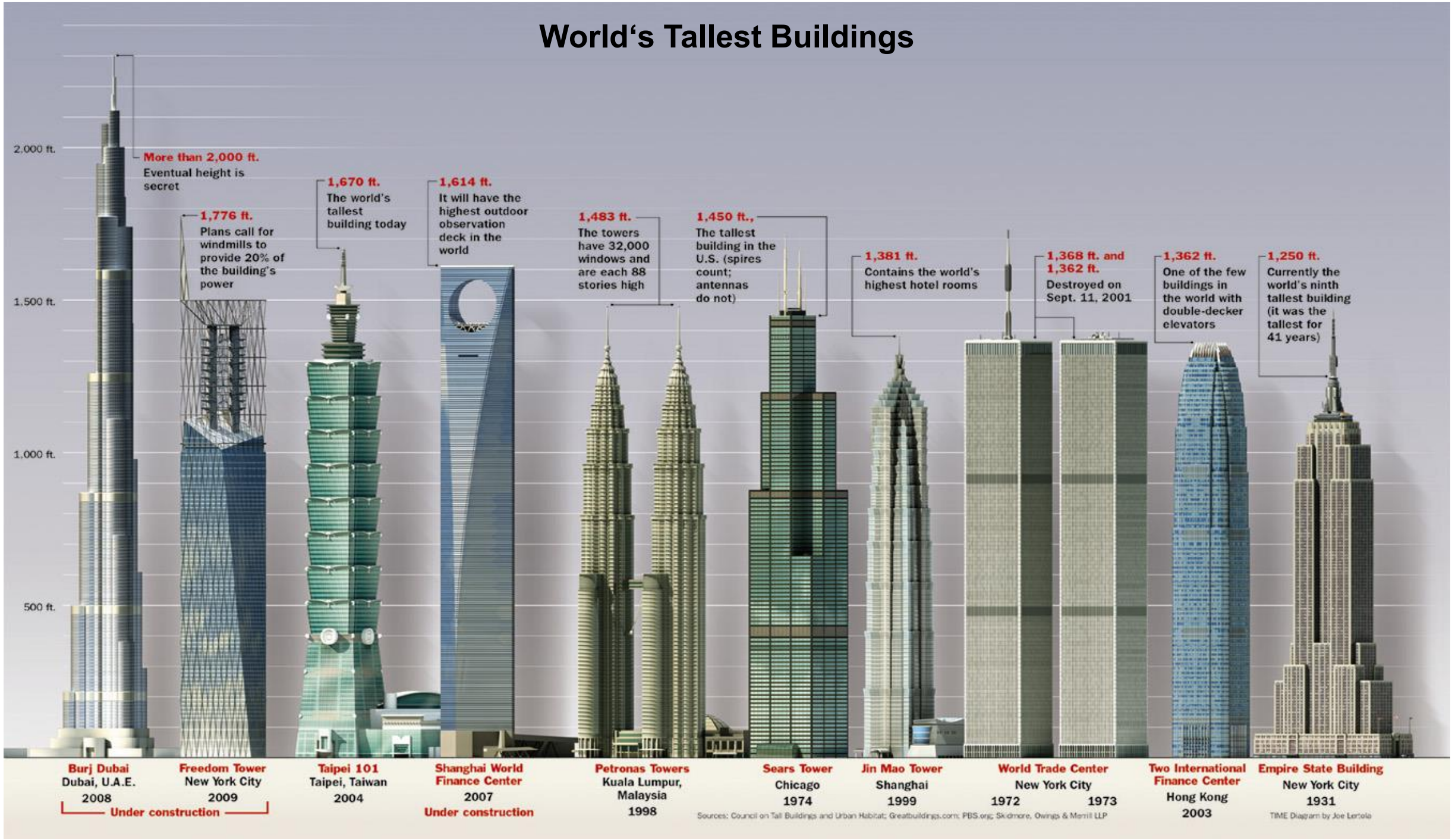









World's Tallest Buildings



Top 10 Tall Buildings of World

Rank	Building	City	Country	Height (m)	Floors	Built
1	Burj Khalifa	Dubai	UAE	828 m	163	2010
2	Makkah Royal Clock Tower Hotel	Mecca	Saudi Arabia	601 m	120	2012
3	Taipei 101	Taipei	Taiwan	509 m	101	2004
4	Shanghai World Financial Center	Shanghai	China	492 m	101	2008
5	International Commerce Centre	Hong Kong	Hong Kong	484 m	118	2010
6	Petronas Tower 1	Kuala Lumpur	Malaysia	452 m	88	1998
7	Petronas Tower 2	Kuala Lumpur	Malaysia	452 m	88	1998
8	Zifeng Tower	Nanjing	China	450 m	89	2010
9	Willis Tower	Chicago	USA	442 m	108	1973
10	Kingkey 100	Shenzhen	China	442 m	100	2011

Evolution of High-rise Architecture



Monadnock Building,
Chicago



Woolworth Building,
New York City



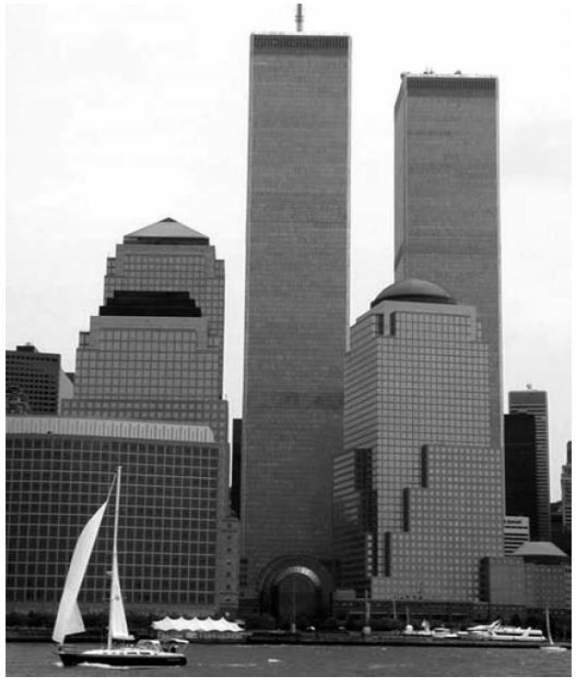
Chrysler Building,
New York City



Sears Tower,
Chicago



Empire State Building, New York



World Trade Center Towers, New York



Petronas Towers, Kuala Lumpur





Burj Khalifa, Dubai

Evolution of High-rise Architecture

- The development of high-rise architecture can be divided into the following eras.

The first stage [early 1940s]

- Before the advent of air conditioning and fluorescent light fixtures, the building form was controlled by the need for **natural daylight and ventilation**.
- A building width of 55–60 ft. (16.7-18.3 m) with office spaces on either side of a double-loaded corridor was common.



Evolution of High-rise Architecture

The second phase

- Emerged as a result of interactions between the desire to create an **increasing amount of rentable area** in a given space and the advent of air conditioning and fluorescent lighting.
- Stressed the aesthetic value of simplicity in facade treatment by using simple cubic shapes, such as rectangles, squares, circles, and ovals.



Evolution of High-rise Architecture

The third phase

- A result of **interaction between marketing experts and a mild boredom of the architectural community**
- The simplest prismatic shape has only four corners, and therefore could offer, at best, four corner offices.
- To capture the market, the trend in high-rise floor planning is to create **as many corner offices as possible** which is achieved by undulating the exterior by providing nicks, notches, and other contortions at the perimeter of the building.



Evolution of High-rise Architecture

The fourth phase

- It is known as **postmodern architecture** and is currently sweeping through the profession, bringing in **daringly articulated buildings**.
- These buildings not only have setbacks, angles, notches, and curves, but the resulting articulations are so severe as to preclude the use of any one type of structural system.



Evolution of High-rise Architecture

The fifth phase

- It can be looked upon as a modification of the current building shapes in the **energy conservation context**.
- Energy conservation efforts have brought about an understanding of spaces as a whole.





Structural Engineering at its Best – Tubular Systems

Emergence of Tubular Systems

- The combination of long-span moment frames and bracing systems (used in phase 2) when placed within the confines of the service core, were **no longer economical** for modern tall buildings.
- A natural structural response to the economic requirements was to move the bracing from the **interior to the exterior** of the building, thereby creating maximum separation between the windward and leeward walls.



Structural Engineering at its Best – Tubular Systems

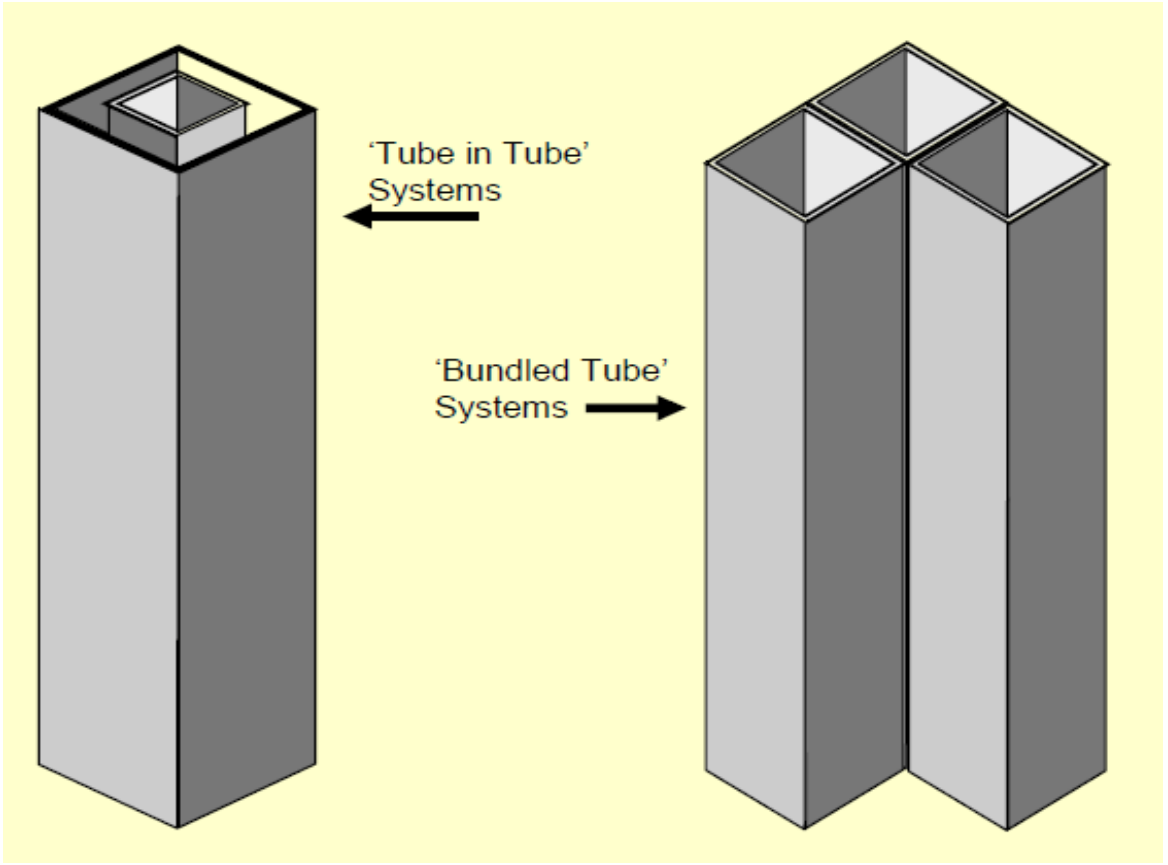
- To make the windward and leeward walls work as integral parts of a three-dimensional system, it was necessary to introduce a shear-resisting element between the two faces of the building.

- This was achieved either by
  - **providing diagonal bracing between the exterior columns**
  - **a system of closely spaced columns and deep spandrels along the building periphery**

- **Thus a new concept termed a “tubular system” was introduced into the structural vocabulary.**



Development of Tube Approach





Dr. Fazlur Rahman Khan

Examples of Tube, Framed (Braced) Tube, Tube in Tube and Bundled Tube Systems







Current Stage of High-rise Architecture



Bank of China, Hong Kong



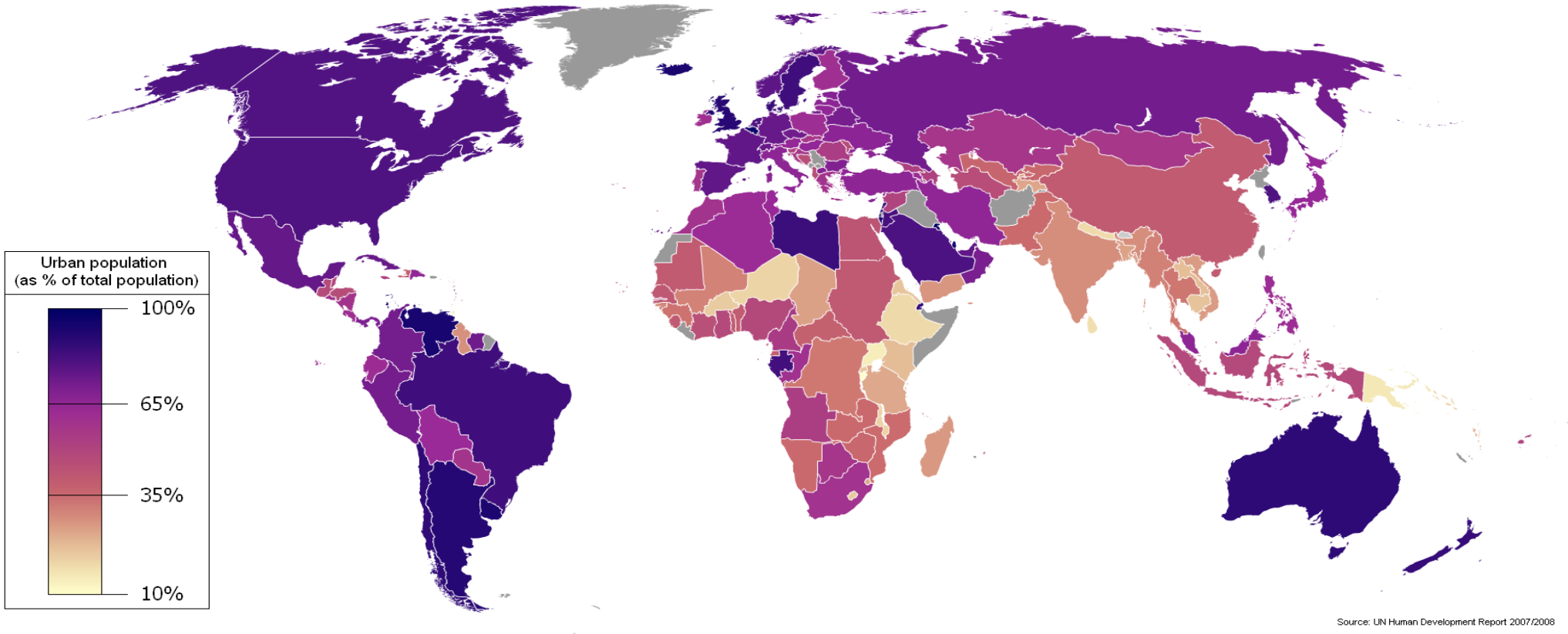
Jin Mao Tower, Shanghai



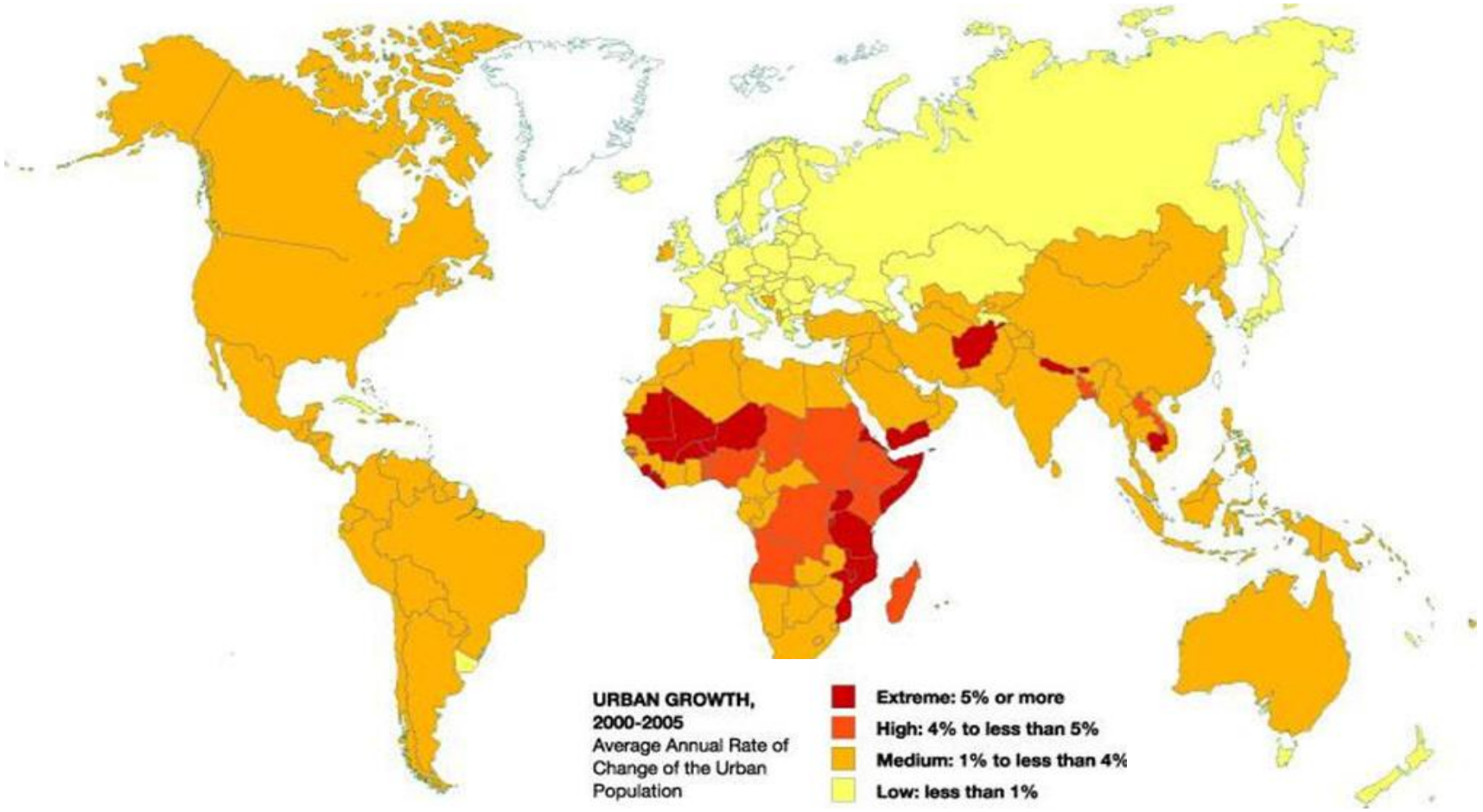
John Hancock Building, Chicago

The Other Side of the Picture

Percentage of Urbanized World

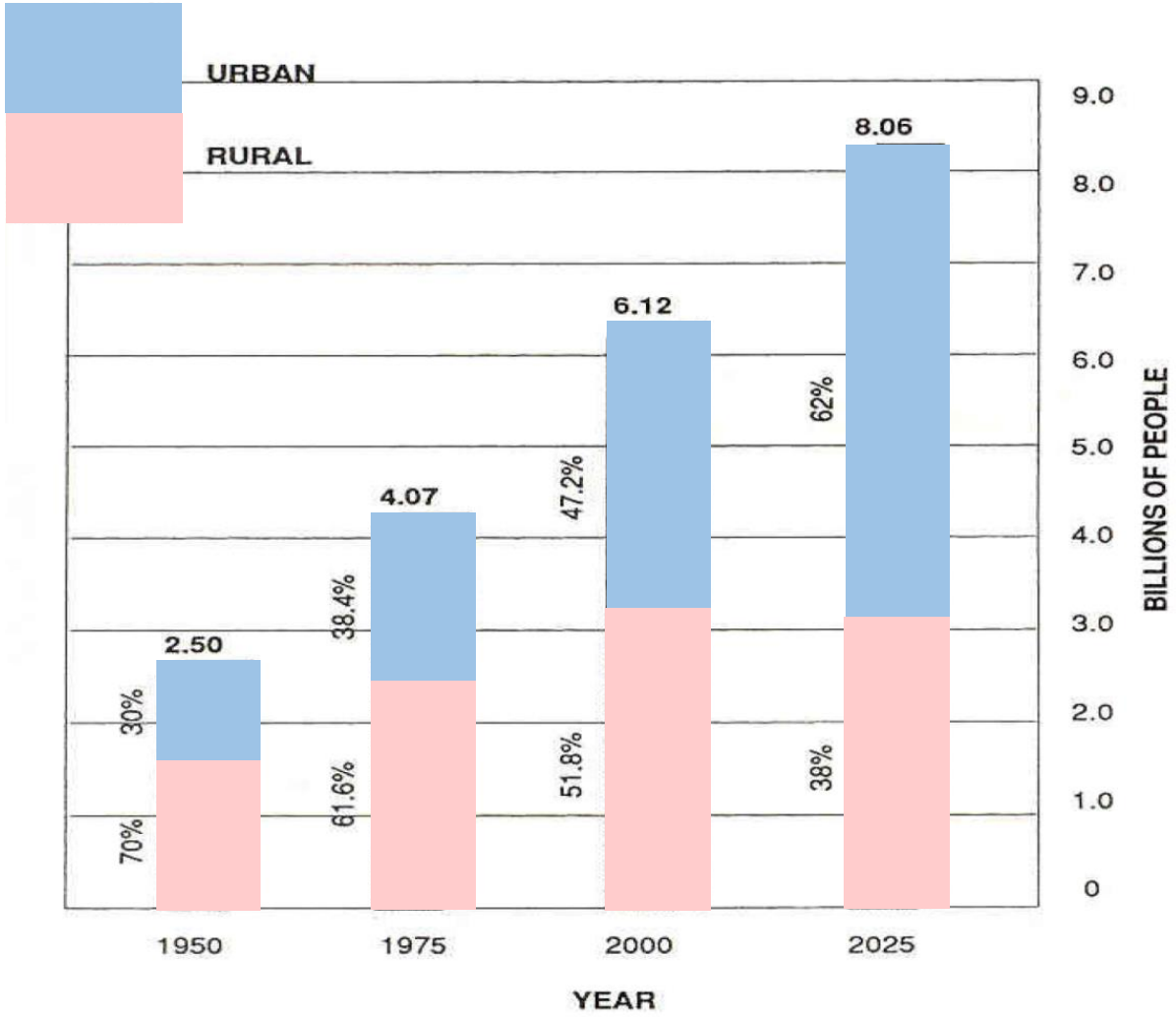


Rate of Urban Population Growth



www.globalchange.umich.edu

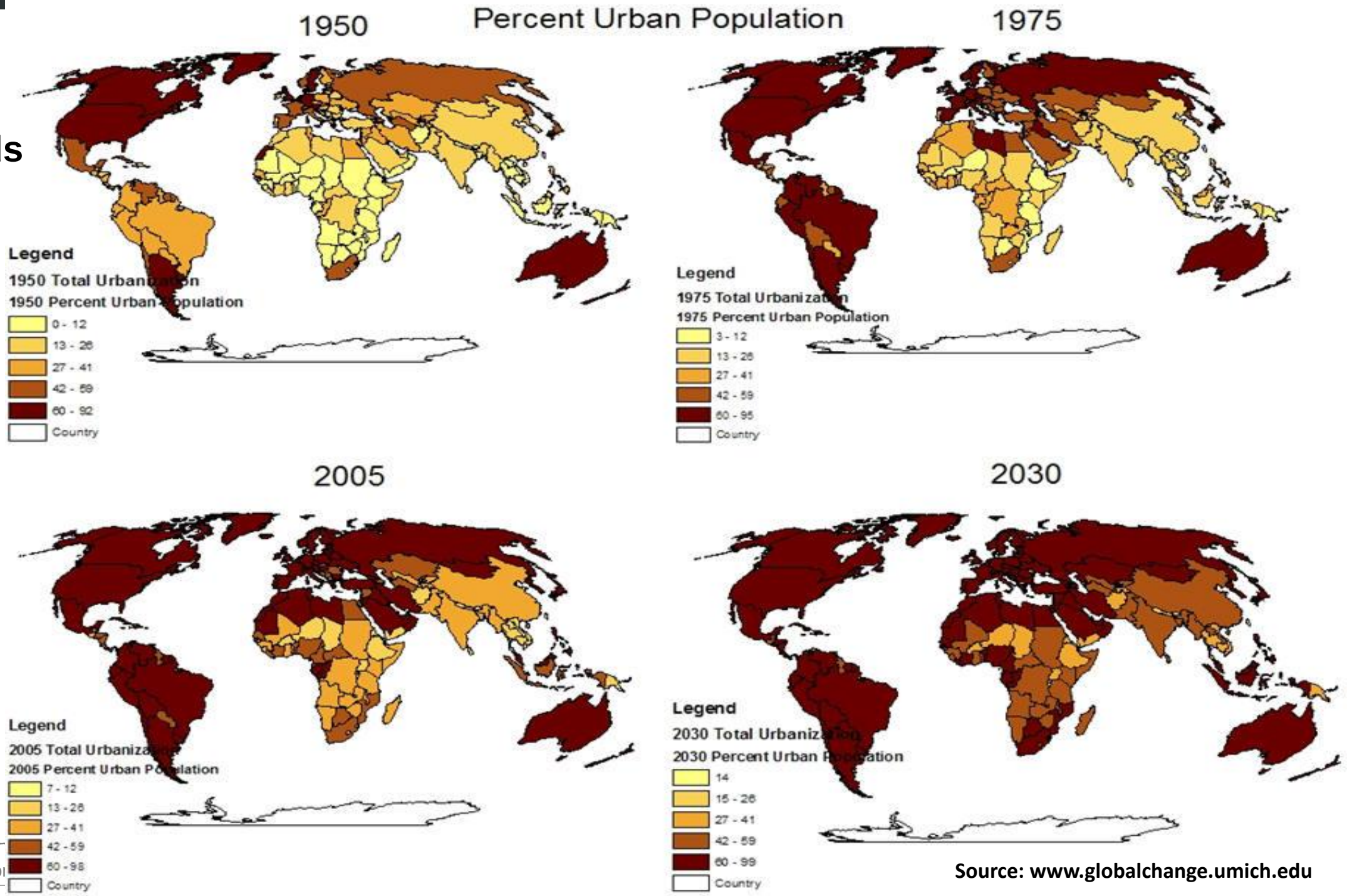
World's Population Urban-to-Rural Ratio



(www.un.org)

Urbanization

Future Trends



Source: www.globalchange.umich.edu



Urbanization → Growing need for built-up space



Kobe Earthquake (1995)

Magnitude = 7.2

Death Toll > 6,000





Kashmir Earthquake (Oct 8, 2005)

Magnitude = 7.7

Death Toll > 70,000

Balakot, Kashmir Earthquake (2005)





Yogyakarta Earthquake (2006)

Magnitude = 7.0

Death Toll: > 5,600

Wenchuan Earthquake (2008), China

Magnitude = 8.0









Haiti Earthquake (2010)

Magnitude = 7.0

Death Toll: 100,000 ~ 200,000

The ~~Goal~~ Challenge of

Structural Engineering

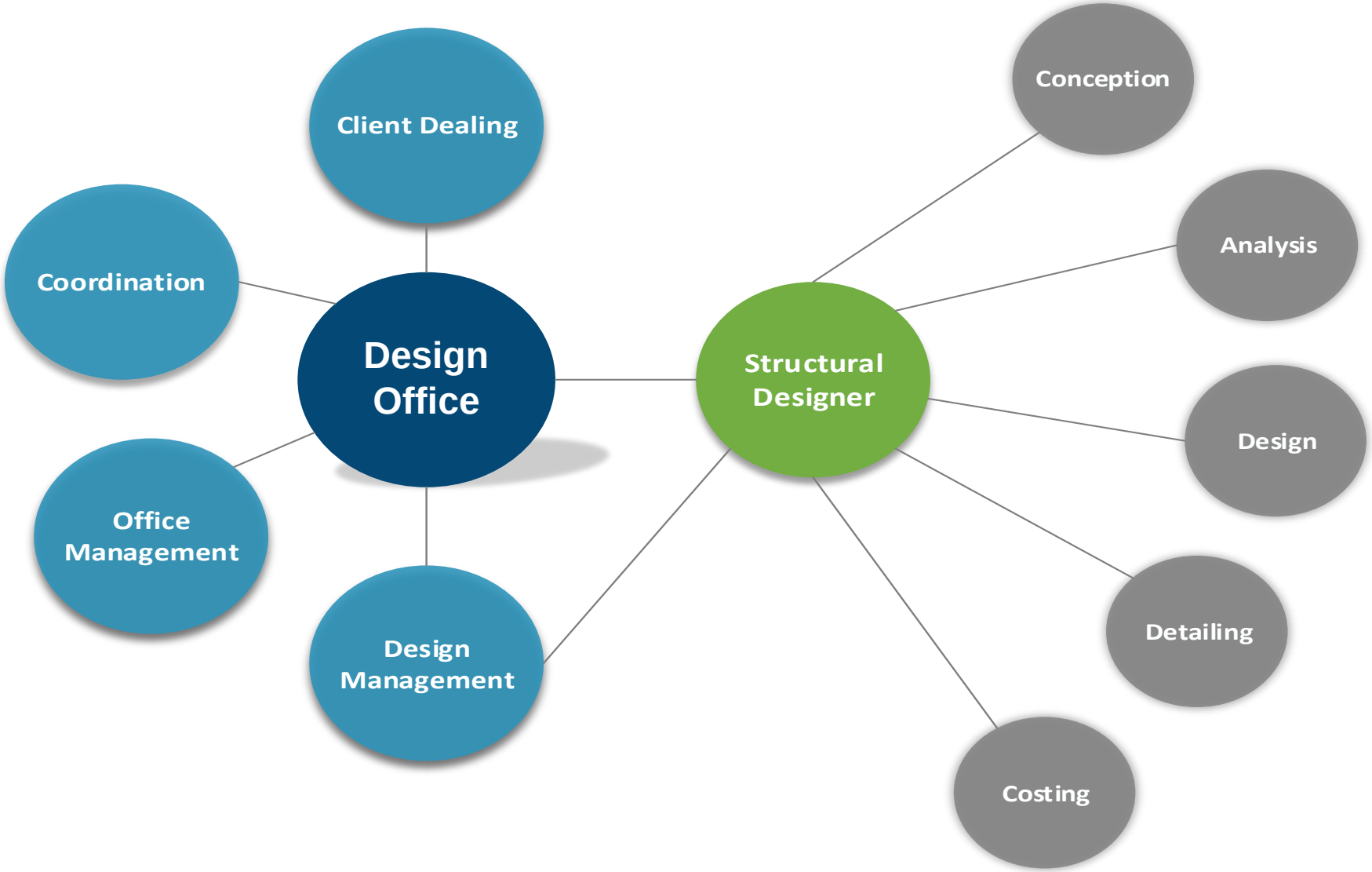


Introduction to Structural Design

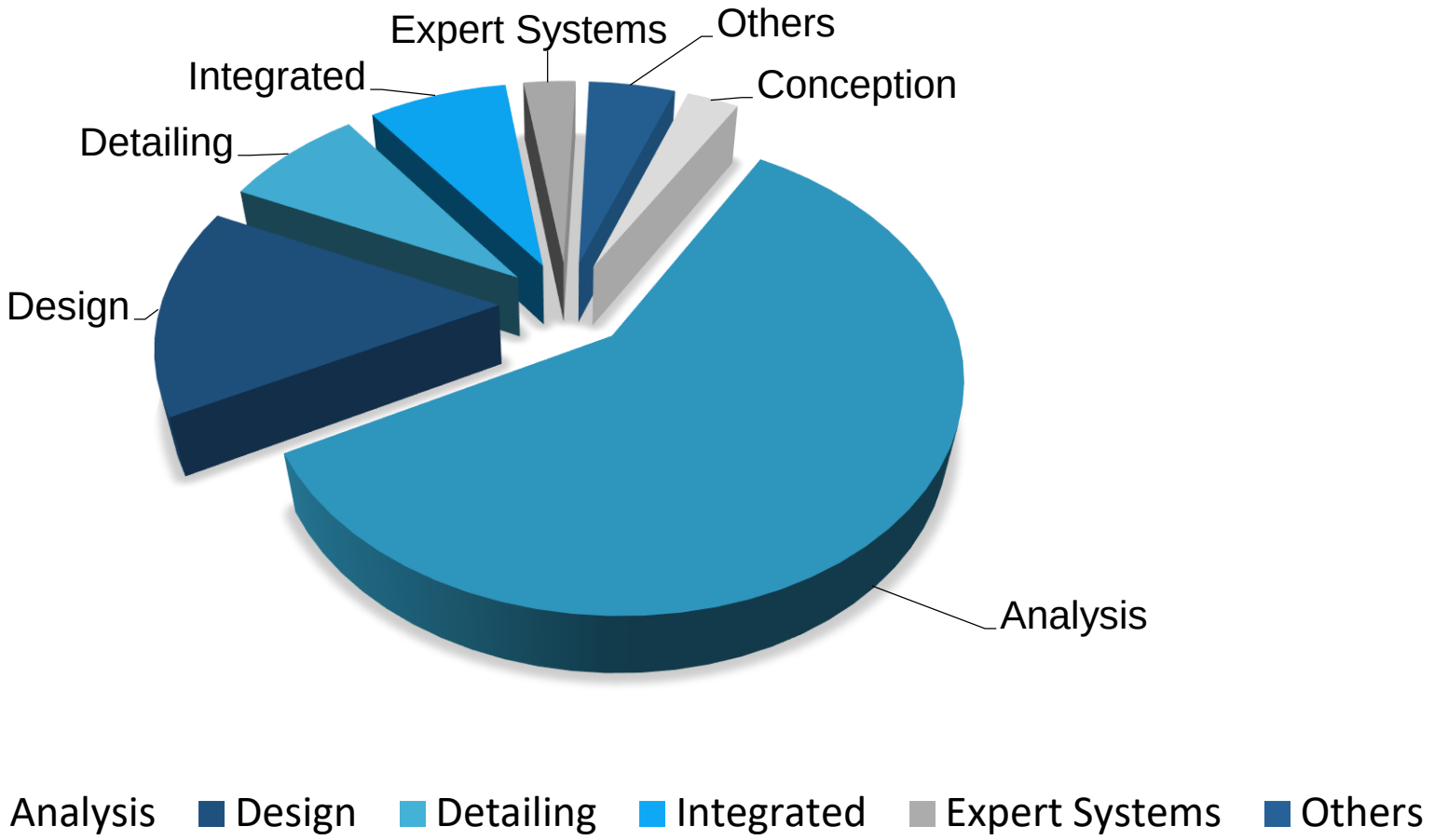
Structural Engineering Spectrum

Activity	Conception – Analysis – Design – Detailing ...
Structure	Buildings – Bridges – Trusses – Shells – Towers ...
Code	American – British – European – Japanese ...
Material	Concrete – PSC – Steel – Timber ...
Model	2D Frame/Truss – 3D Frame/Truss – Full 3D FEM ....
Analysis	Linear Static – NL Static – Linear Dynamic – Large Disp. ....
Solution	Equation Solution – Finite Elements – Programming ...

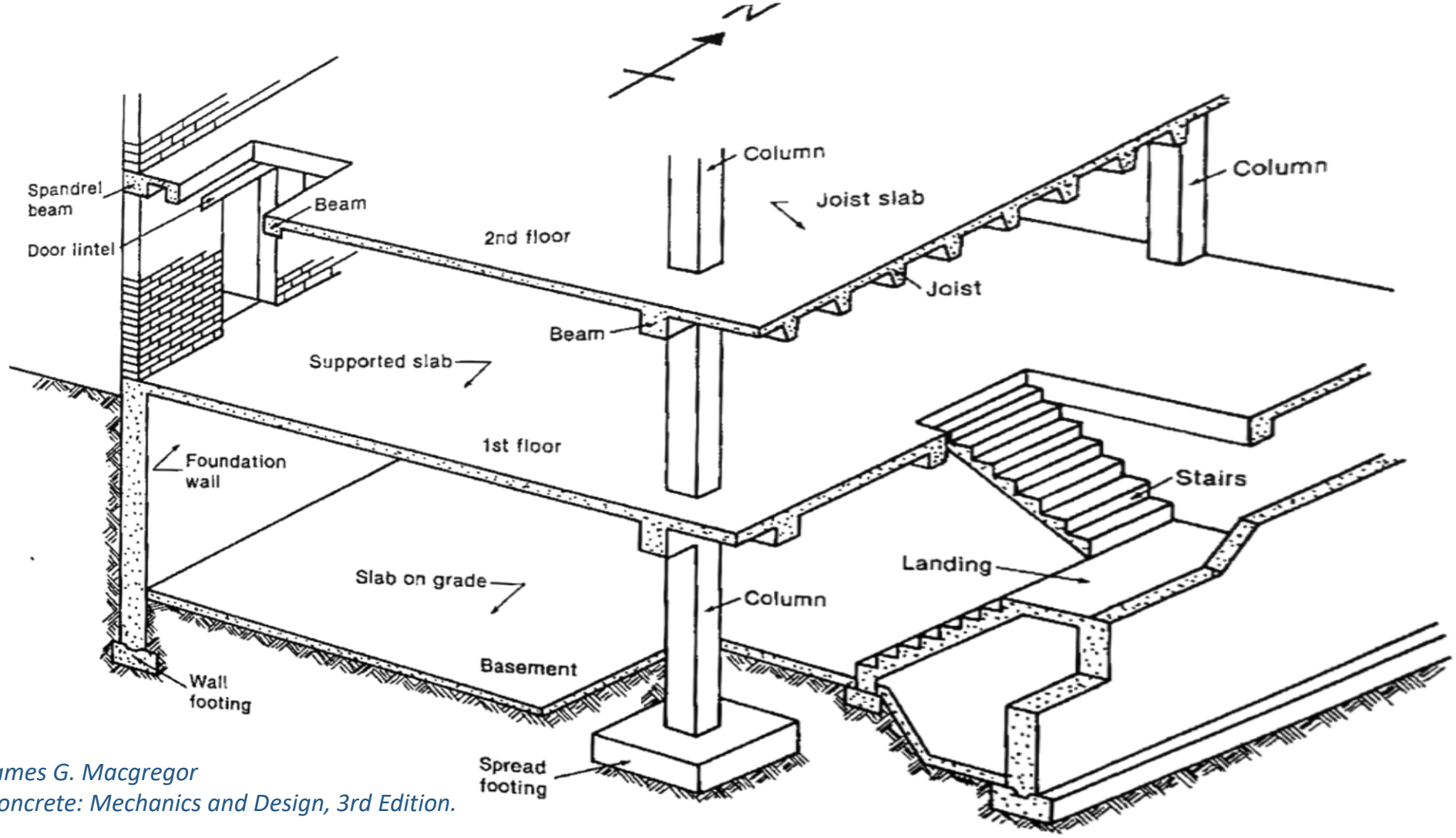
Structural Design Office: Activities



CAD in Structural Engineering

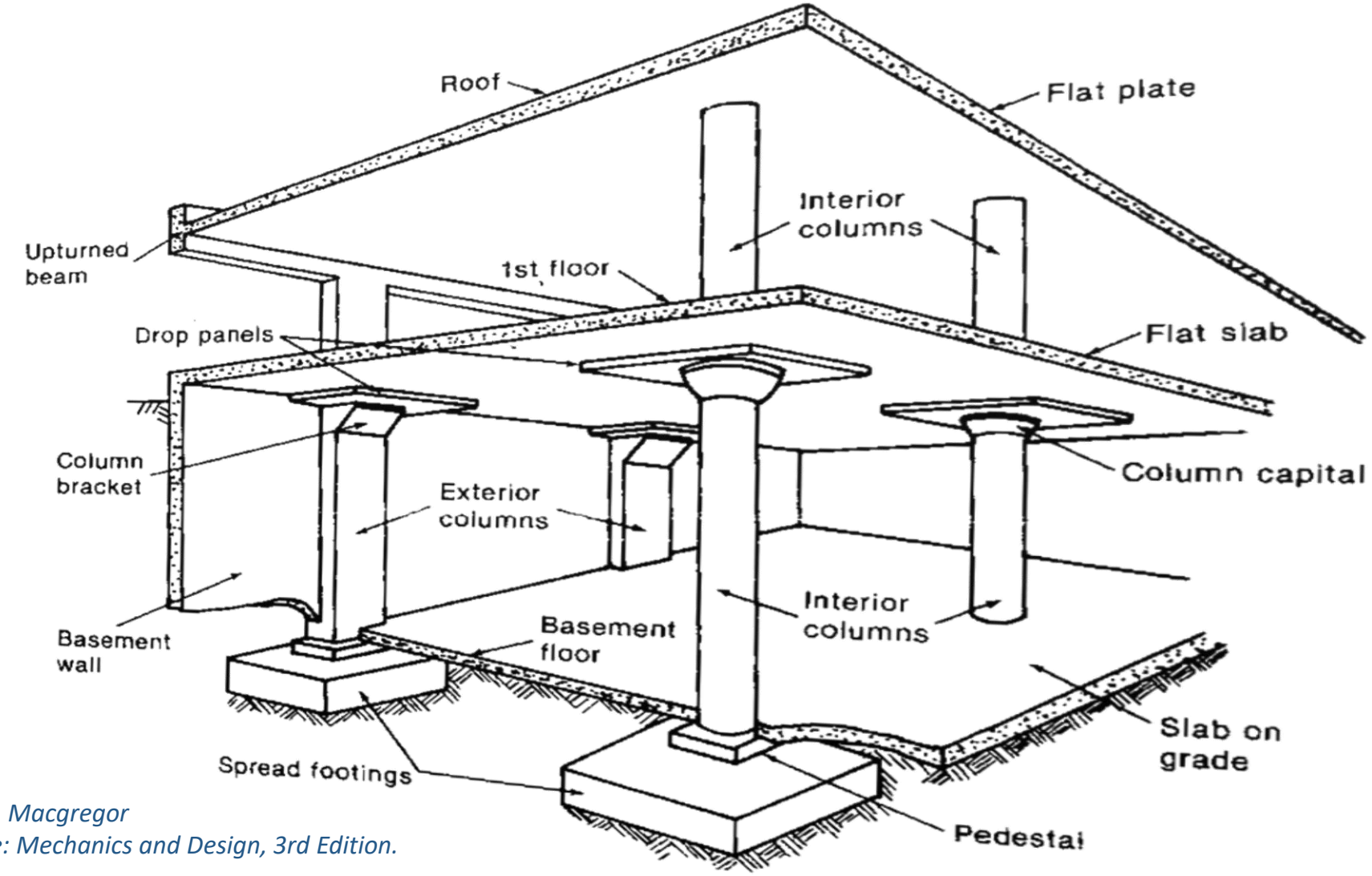


Reinforced Concrete Building Elements



Reference: James G. Macgregor
Reinforced Concrete: Mechanics and Design, 3rd Edition.

Reinforced Concrete Building Elements

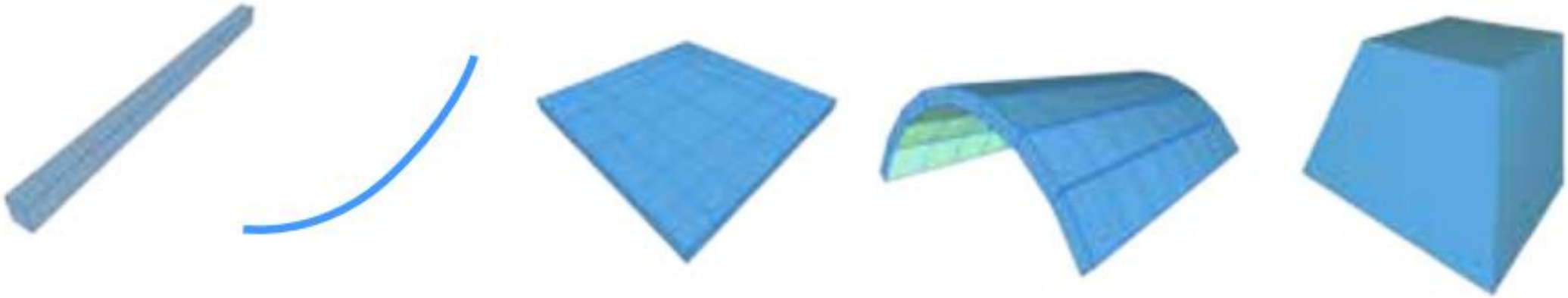


Reference: James G. Macgregor
Reinforced Concrete: Mechanics and Design, 3rd Edition.

The Hierarchy of Structures and its Components

- Physical Structures
    - When applied to the physical and the built environment, the term Structure means an assemblage of physical components and elements, each of which could further be a structure itself, signifying the complexity of the system

Basic Member Types and Forms



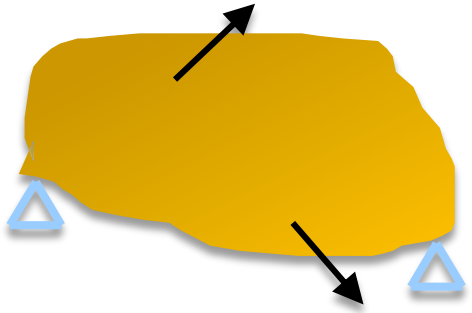
Characterization of Structures

- Based on the member types, the structures can be broadly categorized as
    - Cable structures
    - Skeletal structural
    - Spatial structures
    - Solid structures and
    - A combination of these

The Structural System

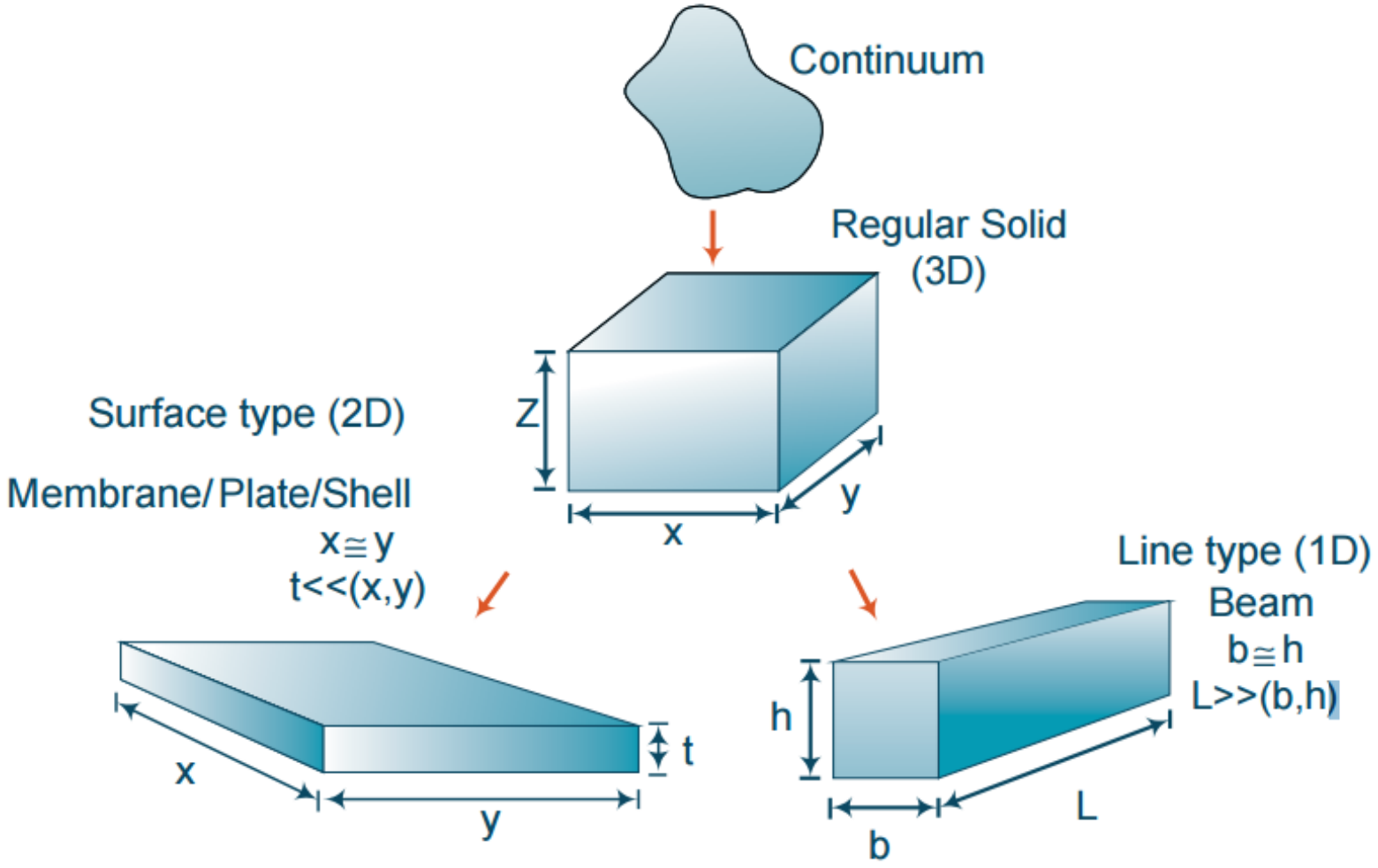
EXCITATION
Loads
Vibrations
Settlements
Thermal Changes

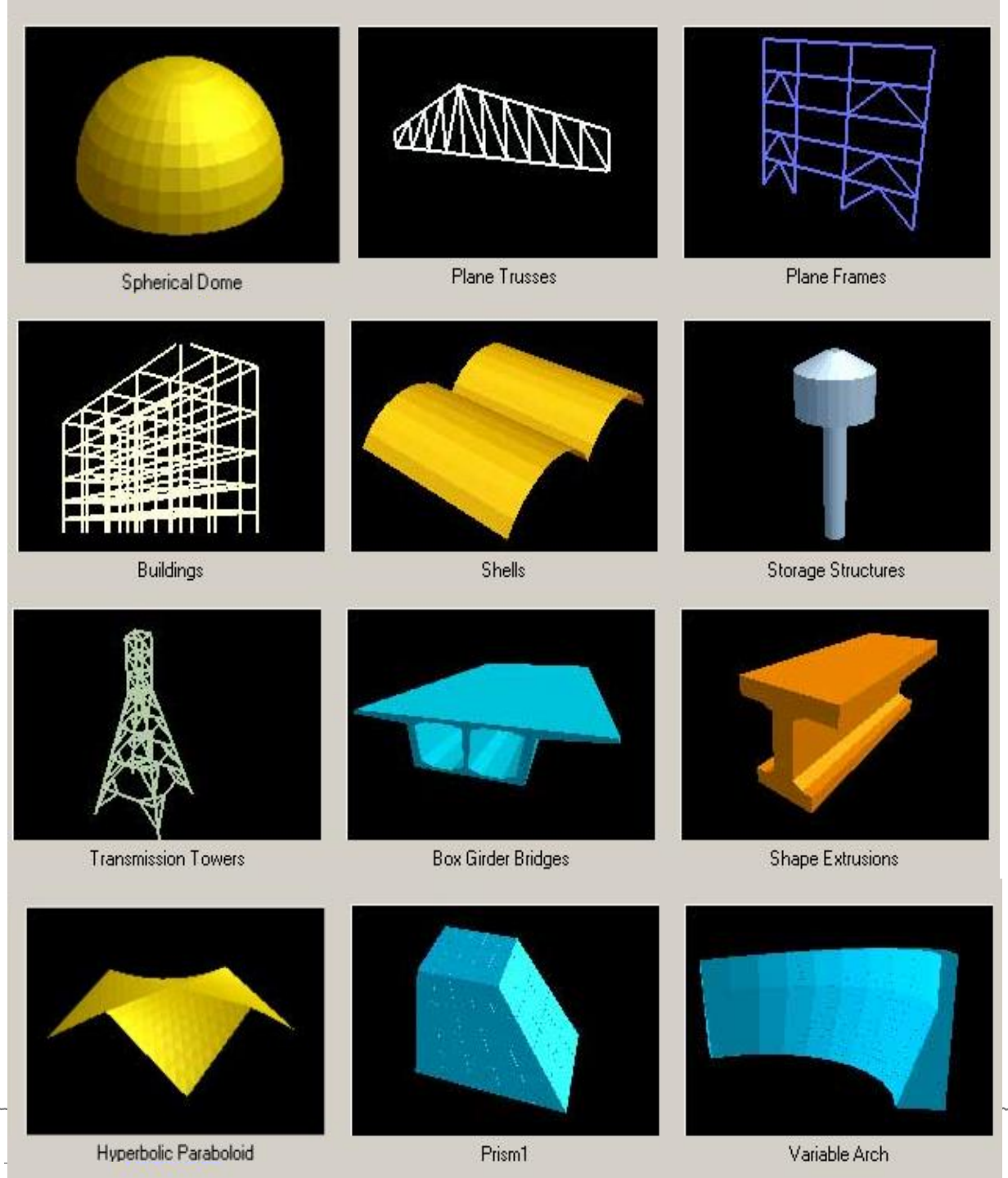
STRUCTURE



RESPONSES
Displacements
Strains
Stress
Stress Resultants

The Structural Elements





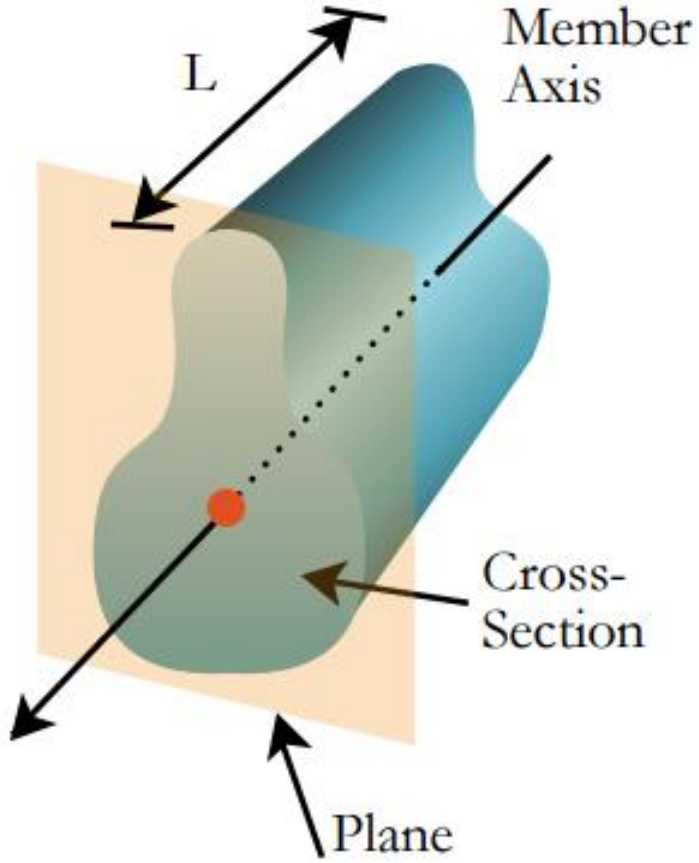
Various Types of Structures:

Special Design Procedures and

Concepts for each type and material

Member Cross-sections

- Cross-sections are generally associated with line type or beam type members, where the length is much longer than the other dimensions.
- The design of a line type member often means the design of its cross-section, which means the selection of appropriate dimensions, proportions and materials used at the cross-section level.



The Structural Materials

- Concrete
- Reinforcing Steel
- Pre-stressing Steel
- Hot Rolled Structural Steel
- Cold Formed Structural Steel
- Aluminum other metals
- Structural Timber

The Need for Analysis

- We need to determine the Response of the Structure to Excitations

Analysis

so that:

- We can ensure that the structure can sustain the excitation with an acceptable level of response

Design

Structural Analysis vs Design

Structural Analysis

- Fairly General, Unified (FEM, BEM ...)
- Output: Element/ Member Actions, Displacements ...

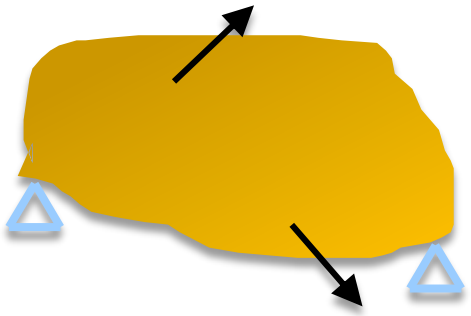
Structural Design

- Structural Material (RC, PSC, HRS, CFS, timber ...)
- Design Code (ACI, BS Codes, EuroCode, JIS ...)
- Design Approach (working stress, ultimate strength, limit state ...)
- Structural Members (beams, columns, slabs, footings ...)
- Local Construction Techniques and Practices
- Output: Element/ Member Cross-section, Reinforcement ...

Analysis of Structure

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + p_{vx} = 0$$

Real Structure is governed by “Partial Differential Equations” of various order.



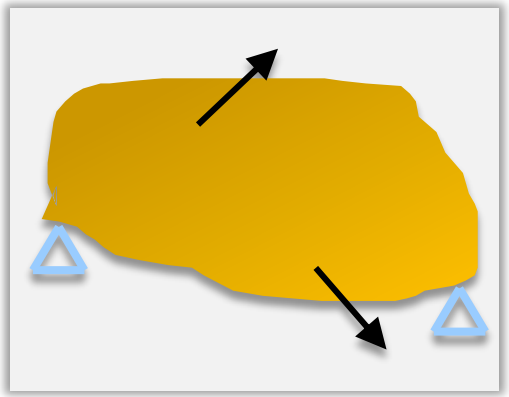
Direct Solution is only Possible for:

- Simple Geometry
- Simple Boundary
- Simple Loading

Need for Structural Model

EXCITATION
Loads
Vibrations
Settlements
Thermal Changes

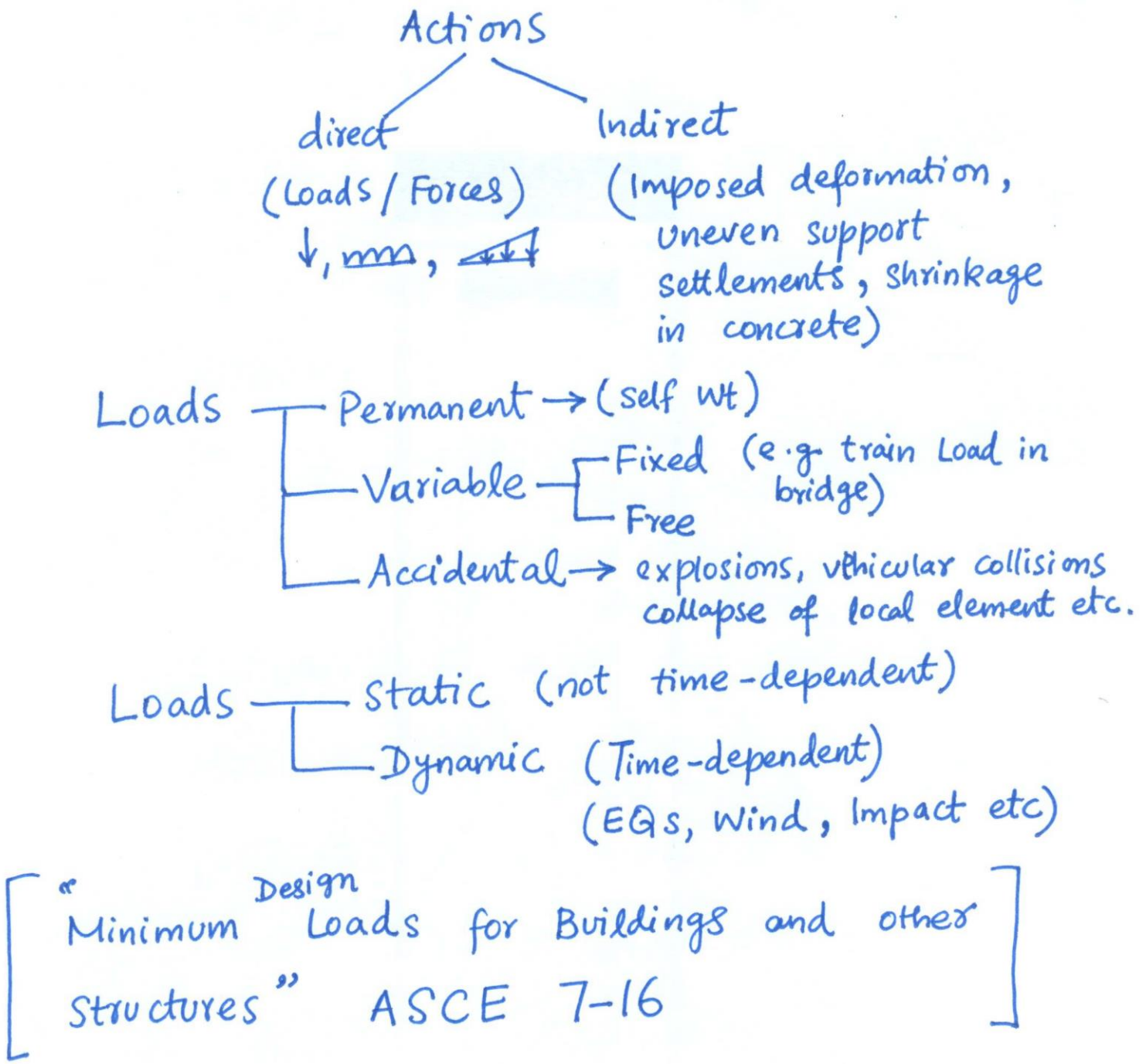
STRUCTURE



STRUCTURAL MODEL

RESPONSES
Displacements
Strains
Stress
Stress Resultants

Classification of Loads



Proportioning for Safety

- Prime Concern: “Balance External Actions with Internal Stress Resultants with adequate margin for safety”

$$S \geq FOS \times A$$

- And Check for
  - Deflections, Deformations, Vibrations, Crack Width
  - Fire Protections, Permeability, Chemical Attacks
  - Ductility and Other special considerations

Proportioning for Safety

Basic Design Equation

Resistance $\geq$ Load Effects

S > A

$$\frac{S}{FOS} = A$$

Working Strength Design

- ***A = Actions due to loads***
- ***S = Strength of member***
- ***FOS = Factor of safety***

$$\frac{S}{FOS_s} = A \times FOS_L$$

- **Ultimate Strength Design**
- **Limit State Design**
- **Partial FOS Design**

$$S = A \times FOS$$

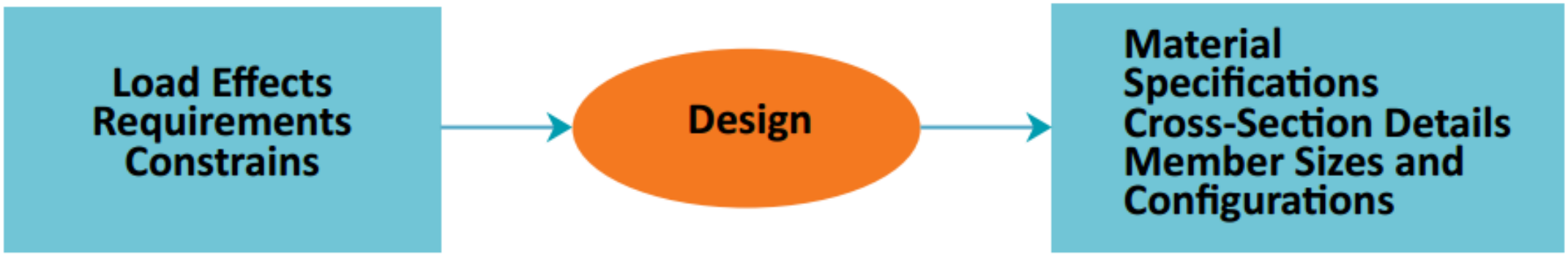
Load Factor Design

Structural Design Philosophy and Process



Design Philosophy and Process

"Structural Design is the process of proportioning the structure to safely resist the applied forces in the most cost effective and friendly manner"



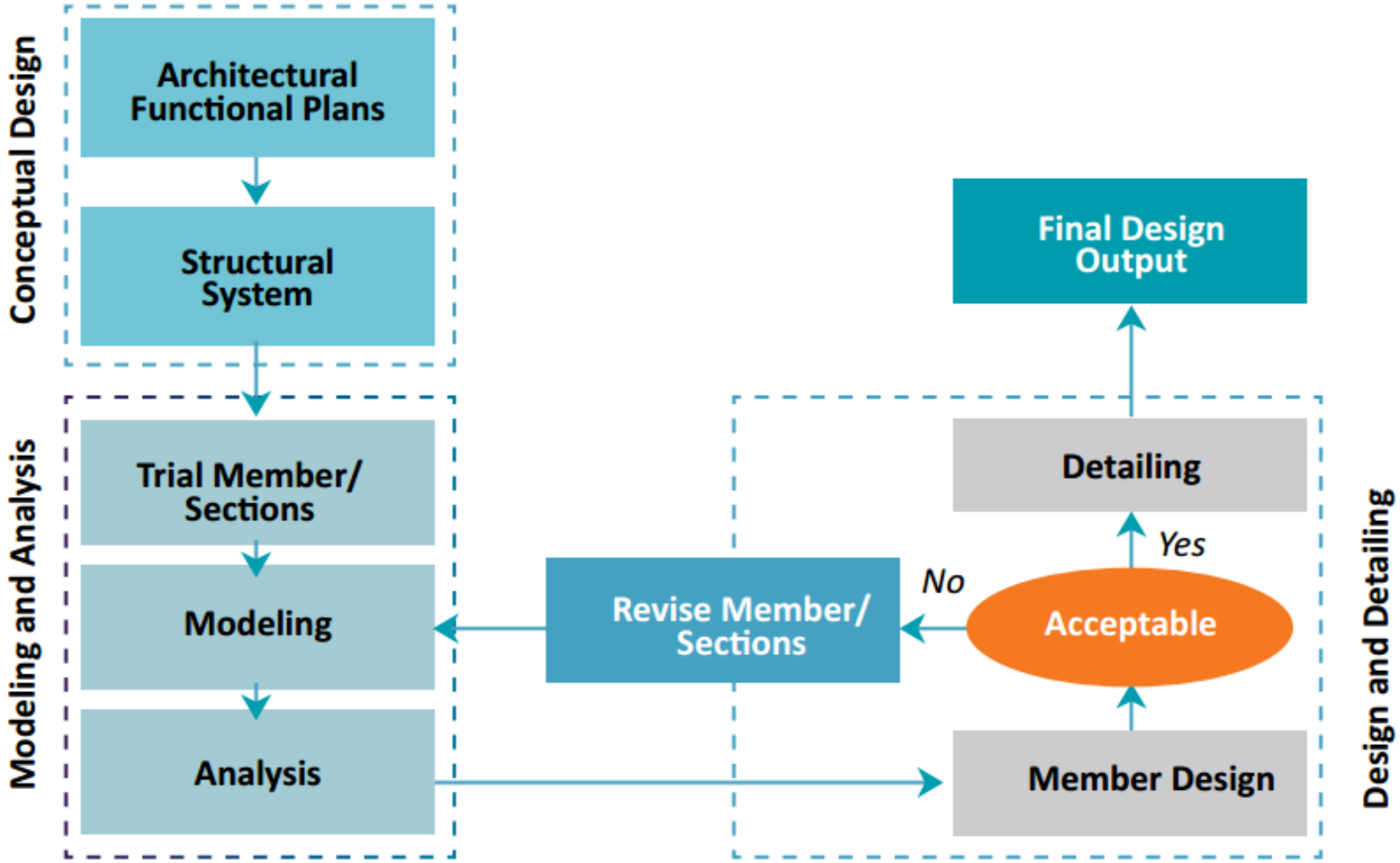
Overall Design Process

- Conception
- Modeling
- Analysis
- Design
- Detailing
- Drafting
- Costing

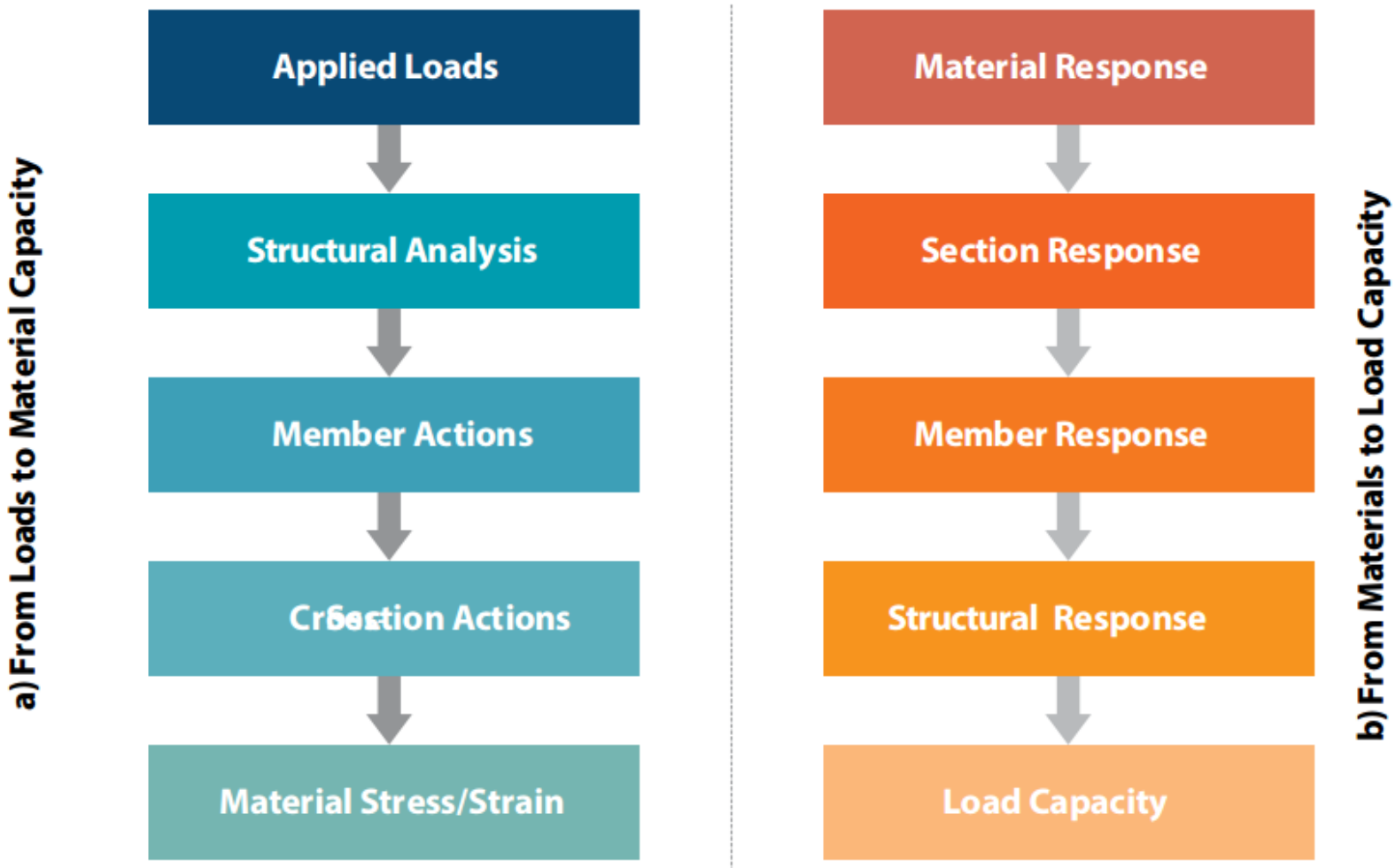


"Integrated Design Process"

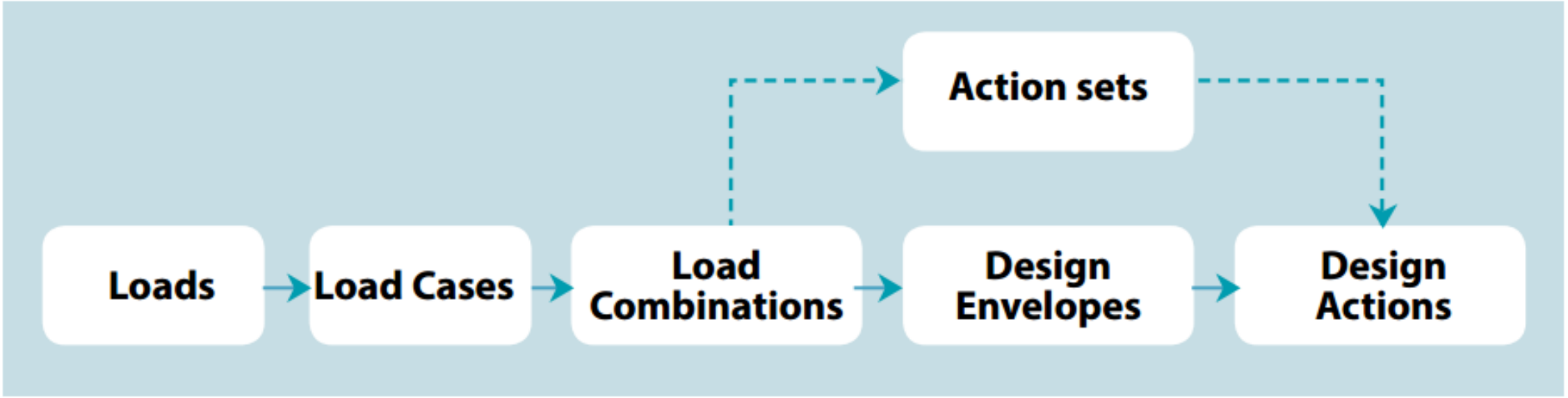
Structural Design Process



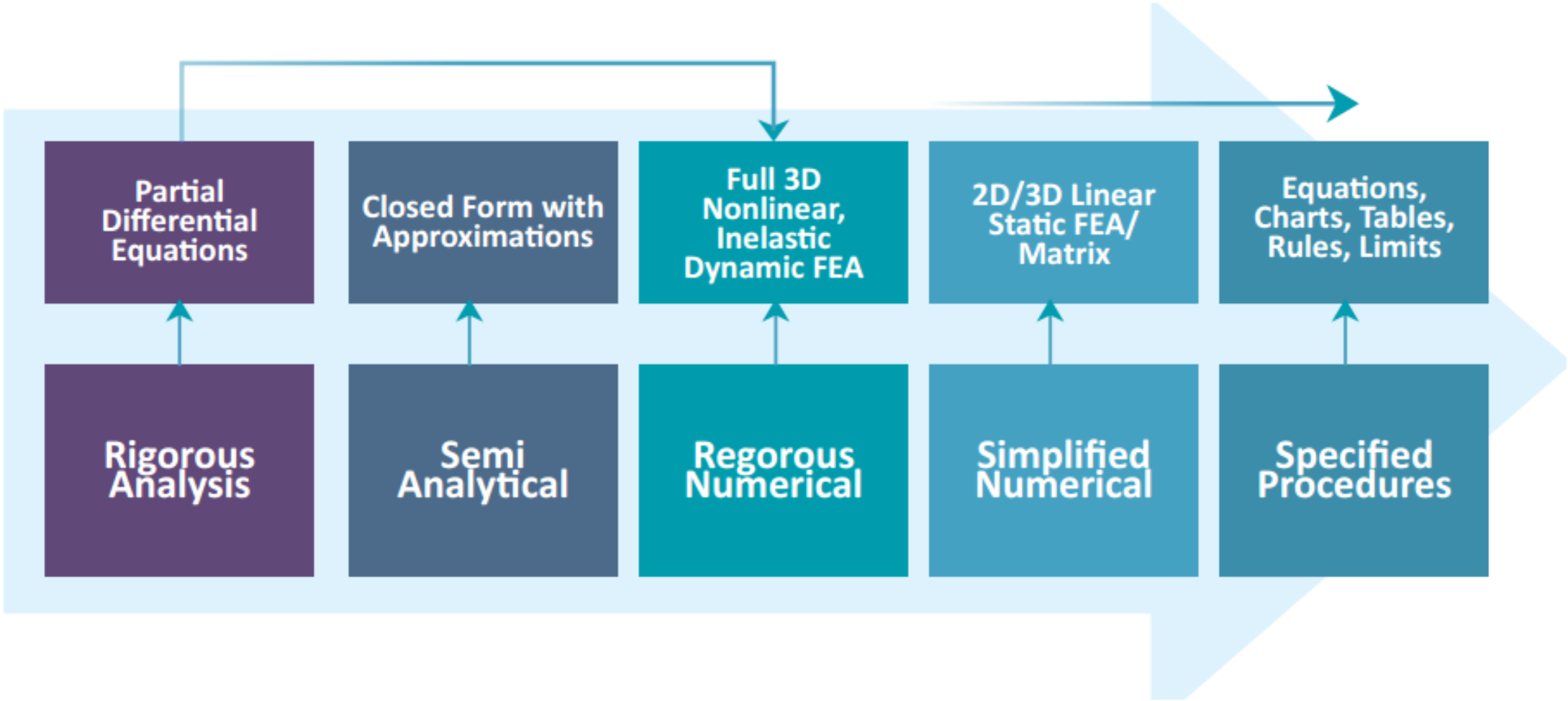
The Response and Design



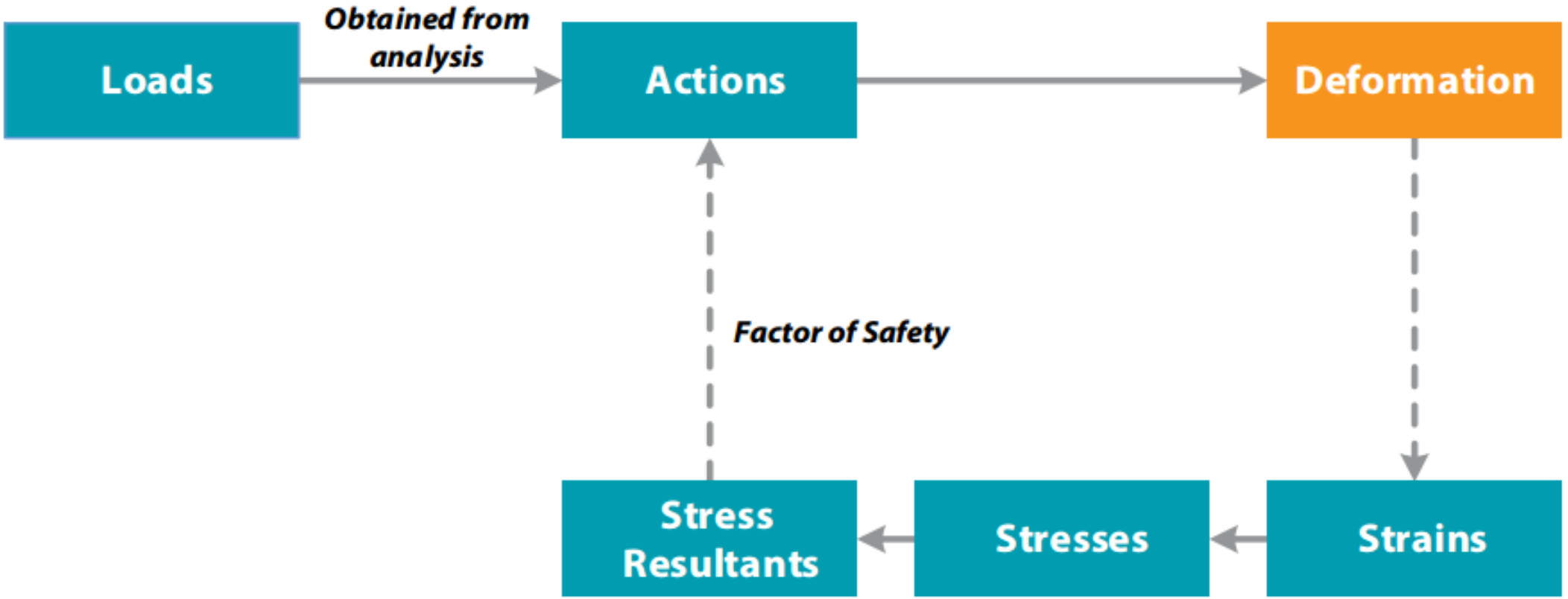
From “Loads” to “Design Actions”



Structural Analysis vs Design



Loads and Stress Resultants



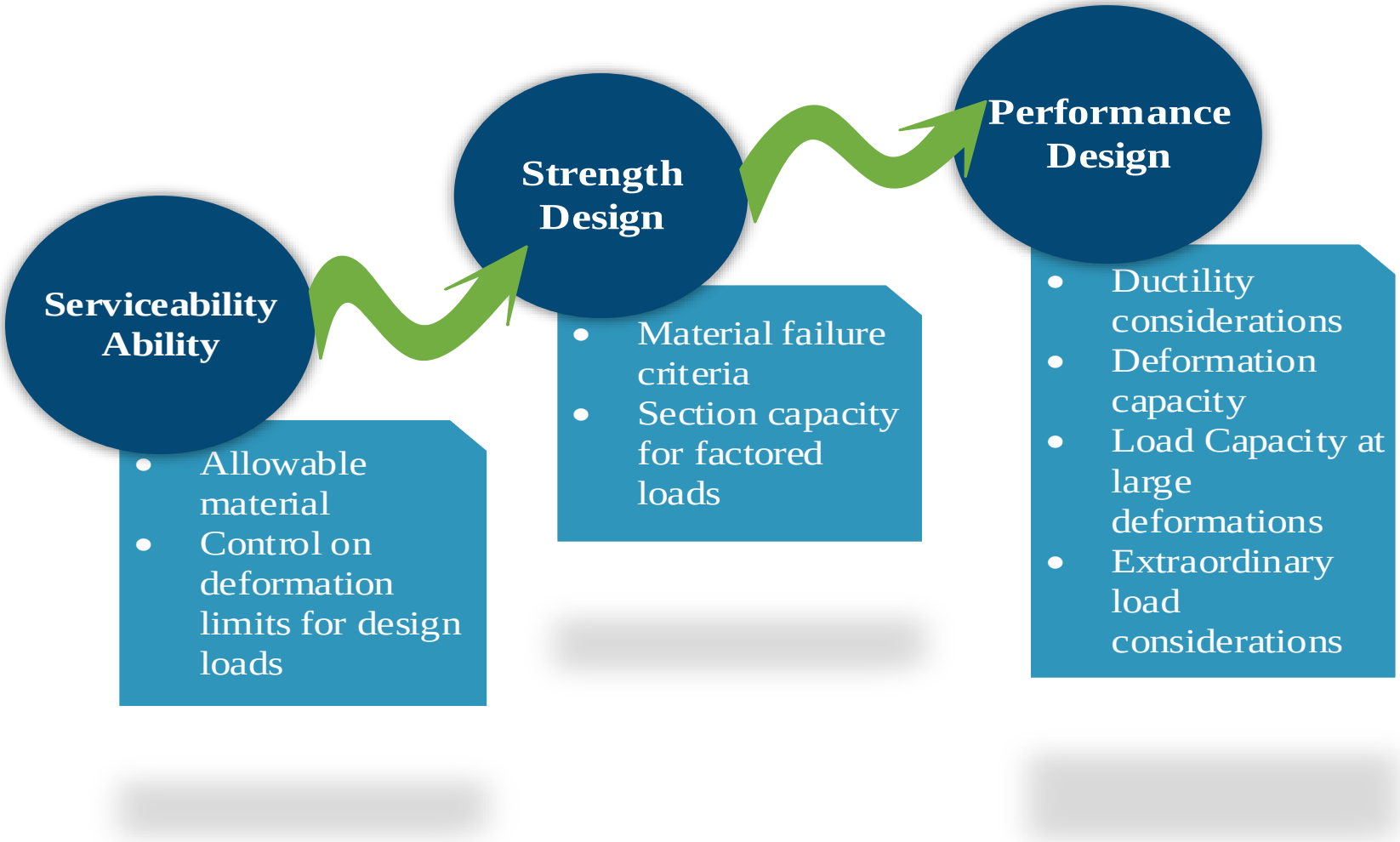
Various Methods of Structural Designs



Various Methods of Structural Design

- Working Stress Design
  - Allowable Stress Design (ASD)
  - Working Stress Design (WSD)
- Load Factor Design
- Ultimate Strength Design
  - Ultimate Strength Design (USD)
  - Strength Design (SD)
  - Load and Resistance Factor Design (LRFD)
- Performance Based Design
  - Pushover Analysis
  - Capacity Based Design

From Serviceability to Performance



From Serviceability to Performance

- Satisfying one design level does not ensure that other design levels will be satisfied
  - **Serviceability design** only ensures that deflections and vibrations etc. for service loads are within limits but says nothing about strength
  - **Strength design** ensures that a certain factor of safety against overload is available within a member or a cross-section but says nothing about what happens if load exceeds design level
  - **Performance design** ensures that structure as a whole reaches a specified demand level. Performance design can include, both service and strength design levels

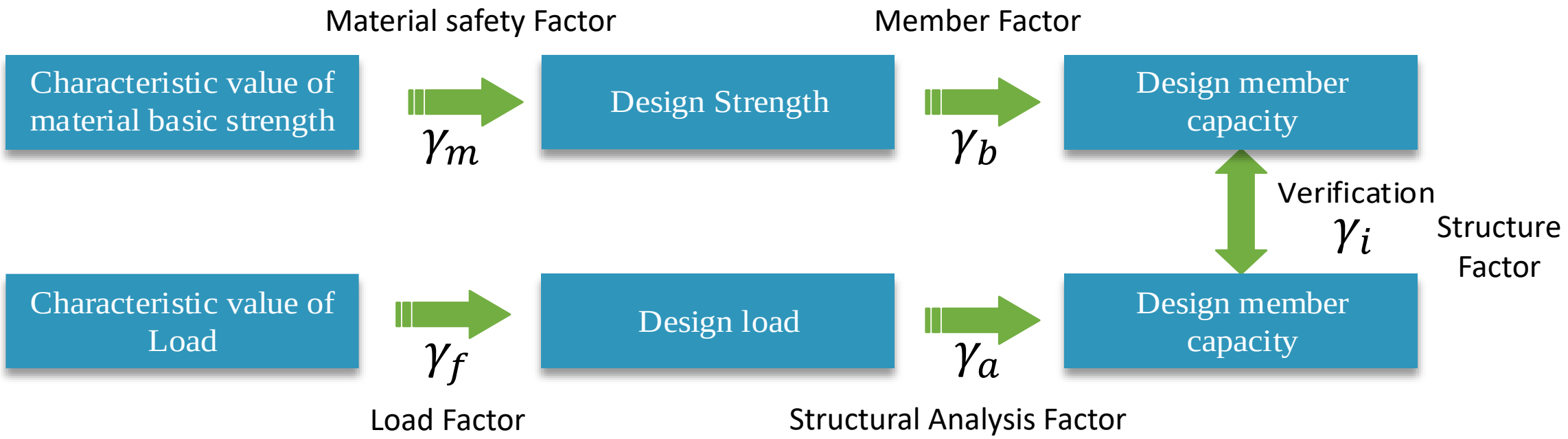
Limit State Design Concept

Types of Limit State	Description
Ultimate Limit states	• Loss of equilibrium • Rupture • Progressive Collapse • Formation of plastic mechanism • Instability • Fatigue
Serviceability limit states	• Excessive deflections • Excessive crack width • Undesirable Vibration
Special limit states	Due to abnormal conditions and abnormal loading such as • Damage or collapse in extreme earthquakes • Structural effects of fire, explosion • Corrosion or deterioration

Limit State Design Concept

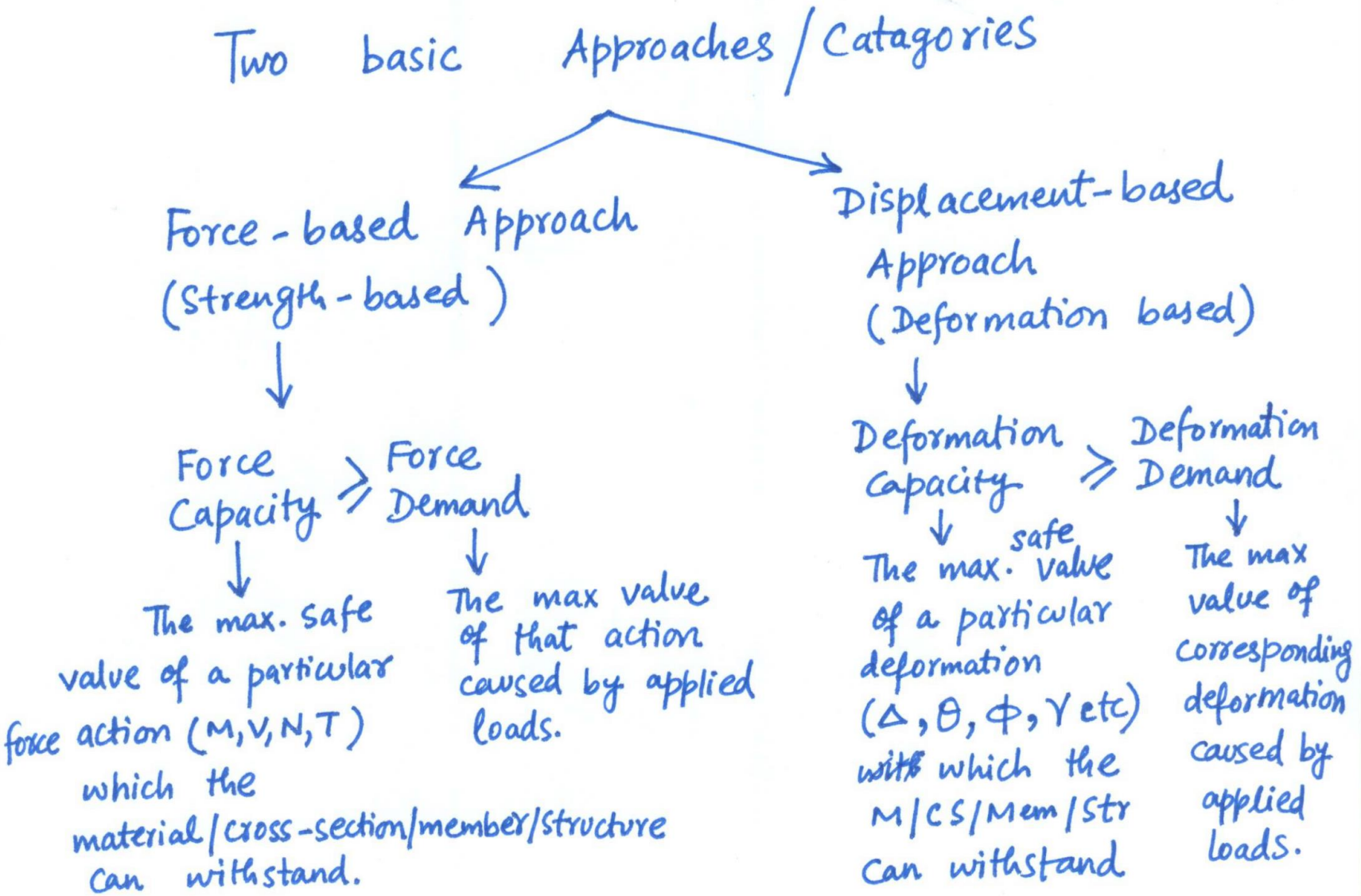
- Limit state design involves
  - Identification of all potential modes of failure (i.e. identify significant limit states
  - Determination of acceptable levels of safety against occurrence of each limit state
  - Consideration by the designer of significant limit states

Section Design Process



Limit State Design Concept

Force-based Design vs. Displacement-based Design

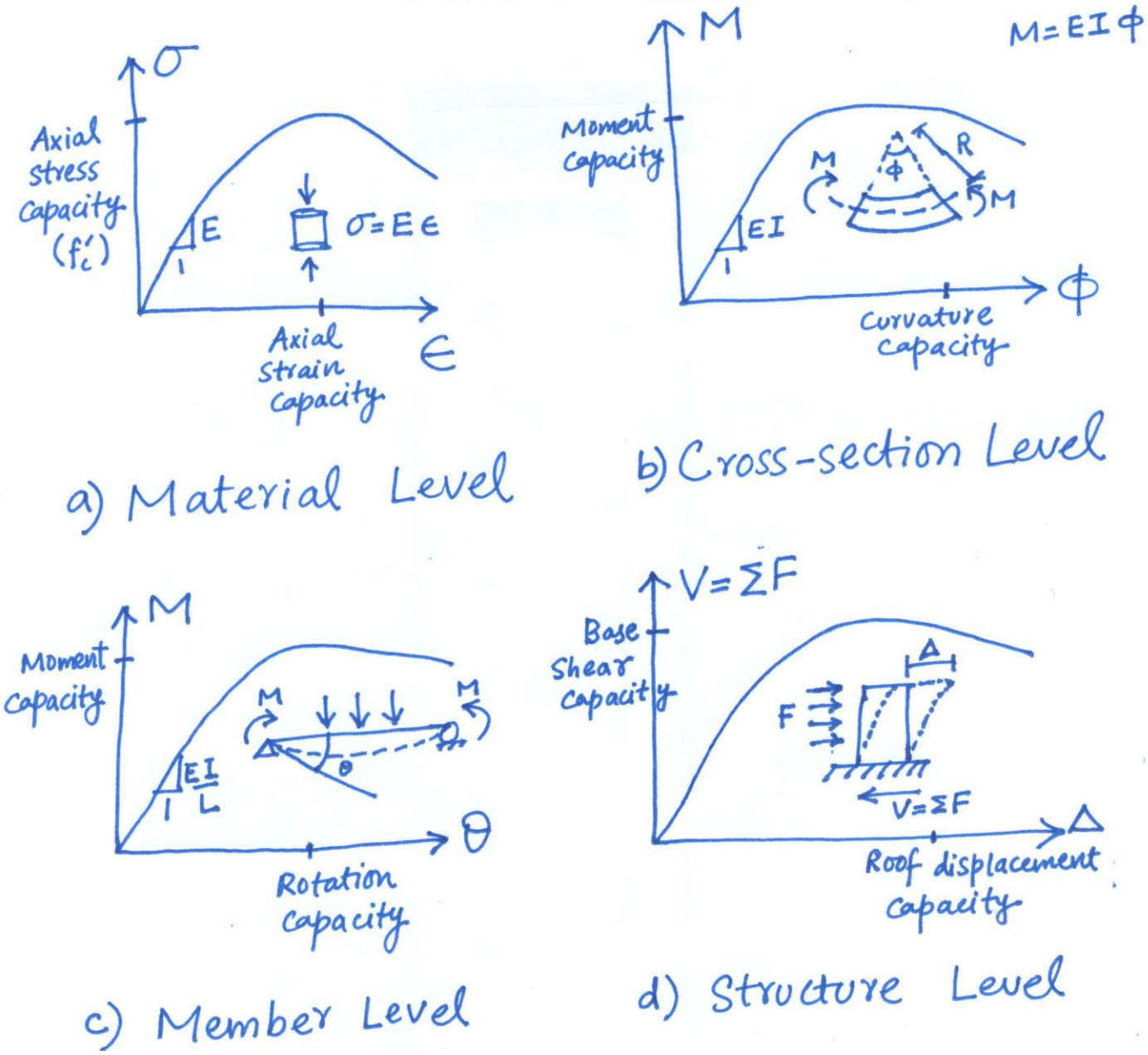


Force-based Design vs. Displacement-based Design

Force-based Approach:
a) Working stress design (Allowable stress design)
b) Ultimate strength design

Displacement-based Approach:
a) Capacity Design (Plastic Design)
b) Direct Disp. based Design (DDBD by Priestley)
c) PBD Approach

Force-based Design vs. Displacement-based Design



ASD/WSD vs. USD

WSD → $\sigma_{demand} \leq \sigma_{allowable}$

($\sigma_{allowable} \cong (0.4\text{-}0.5) f'_c$)

σ_{demand} caused by unfactored loads

USD → $\phi \sigma_n \geq \alpha \sigma_{demand}$

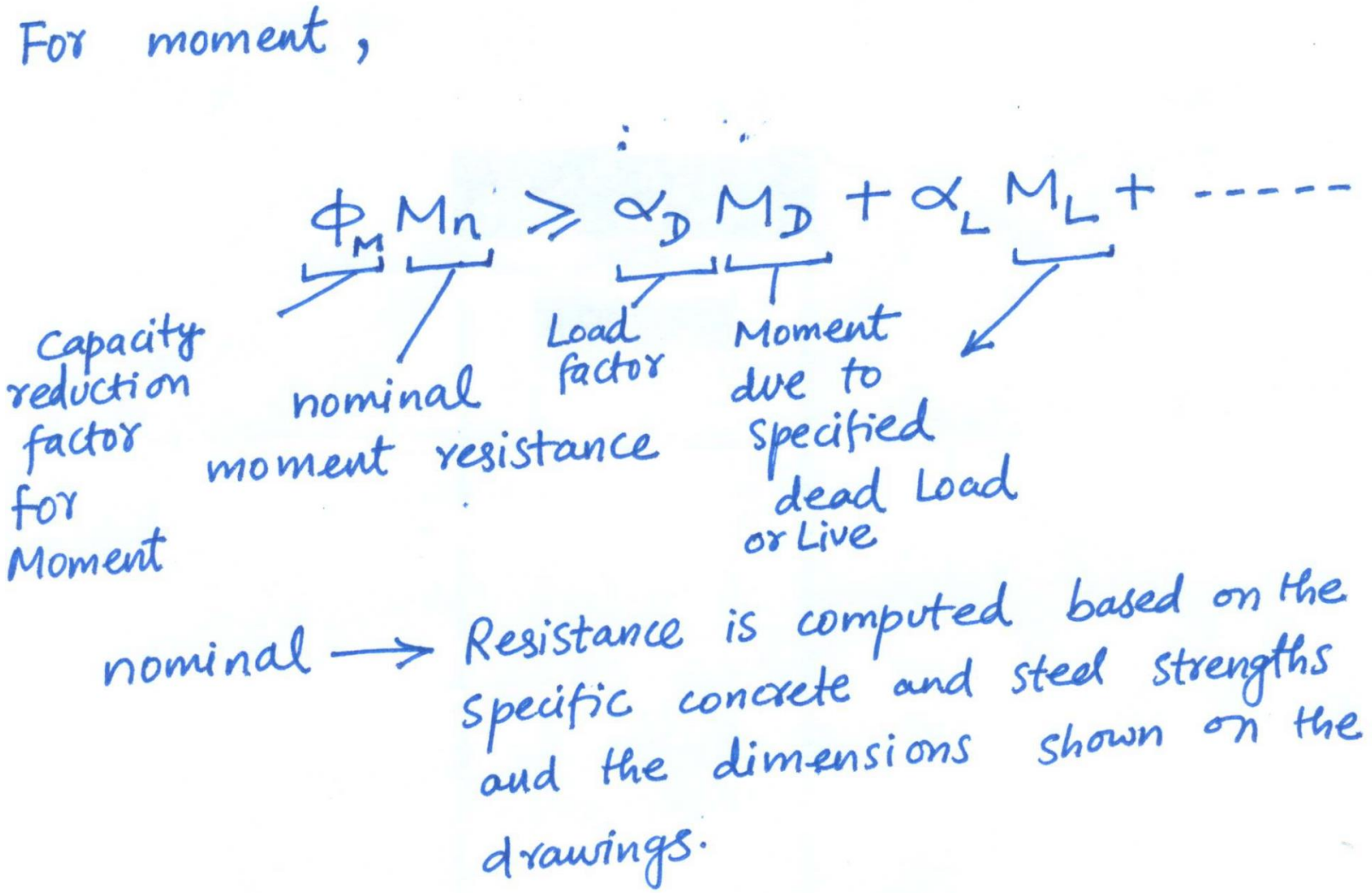
$\phi \sigma_n \geq \sigma_u$

σ_u caused by factored loads

$\phi < 1$ (To allow for the possibility that the capacity can be less than computed)

$\alpha > 1$ (To allow ------- ----- load effects (demand) can be greater than computed)

Ultimate Strength Design (USD)



Ultimate Strength Design (USD)

For shear,

$$\phi_v V_n \geq \alpha_D V_D + \alpha_L V_L + \cdots$$

For axial load,

$$\phi_p P_n \geq \alpha_D P_D + \alpha_L P_L + \cdots$$

$$\alpha_D M_D + \alpha_L M_L + \cdots = M_U$$

$$\alpha_D V_D + \alpha_L V_L + \cdots = V_U$$

$$\alpha_p P_p + \alpha_L P_L + \cdots = P_U$$

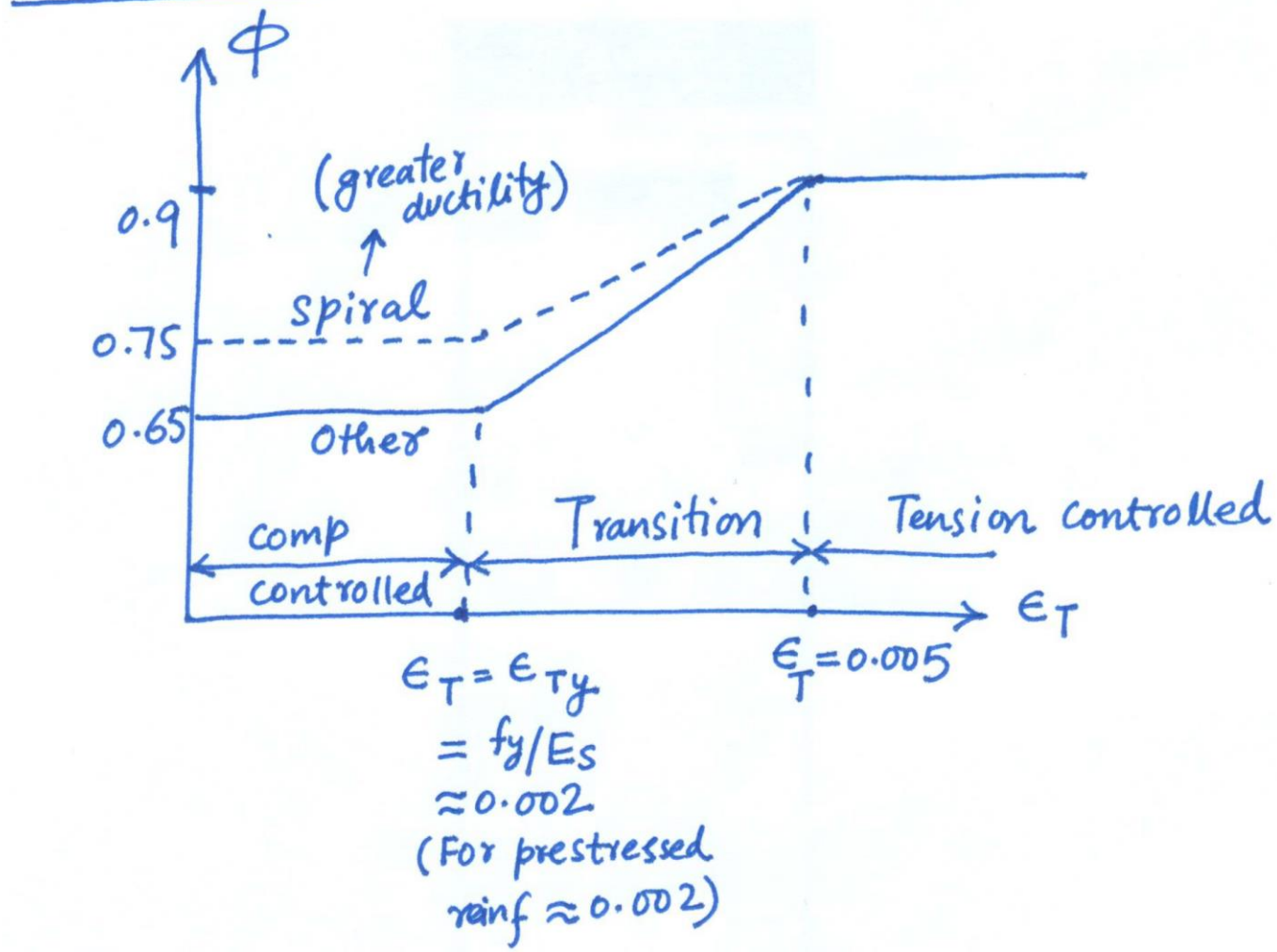
Ultimate Strength Design (USD)

Table 21.2.1 – Strength Reduction Factors ϕ
(ACI 318-14)

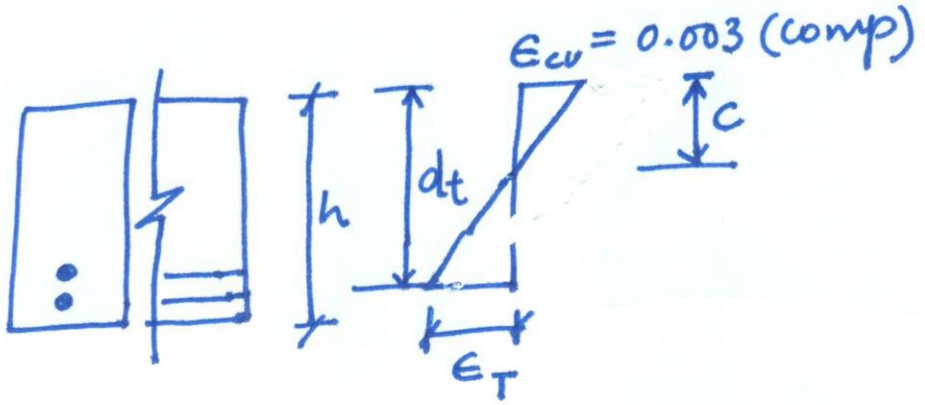
Moment, axial force, or combined moment and axial force	0.65 to 0.9
Shear	0.75
Torsion	0.75
Bearing	0.65
Post-tensioned Anchorage zones	0.85
Brackets and Corbels	0.75
Struts, Ties, nodal zones and bearing area (STM method)	0.75
Components of connections of precast members controlled by tension yielding of steel	0.9
Plain Concrete Elements	0.6
Anchors in concrete elements	0.45 to 0.75

Ultimate Strength Design (USD)

Strength Reduction Factors (ϕ) For Moment, Axial Force and Combined Moment and Axial force



Tension and Compression Controlled Sections

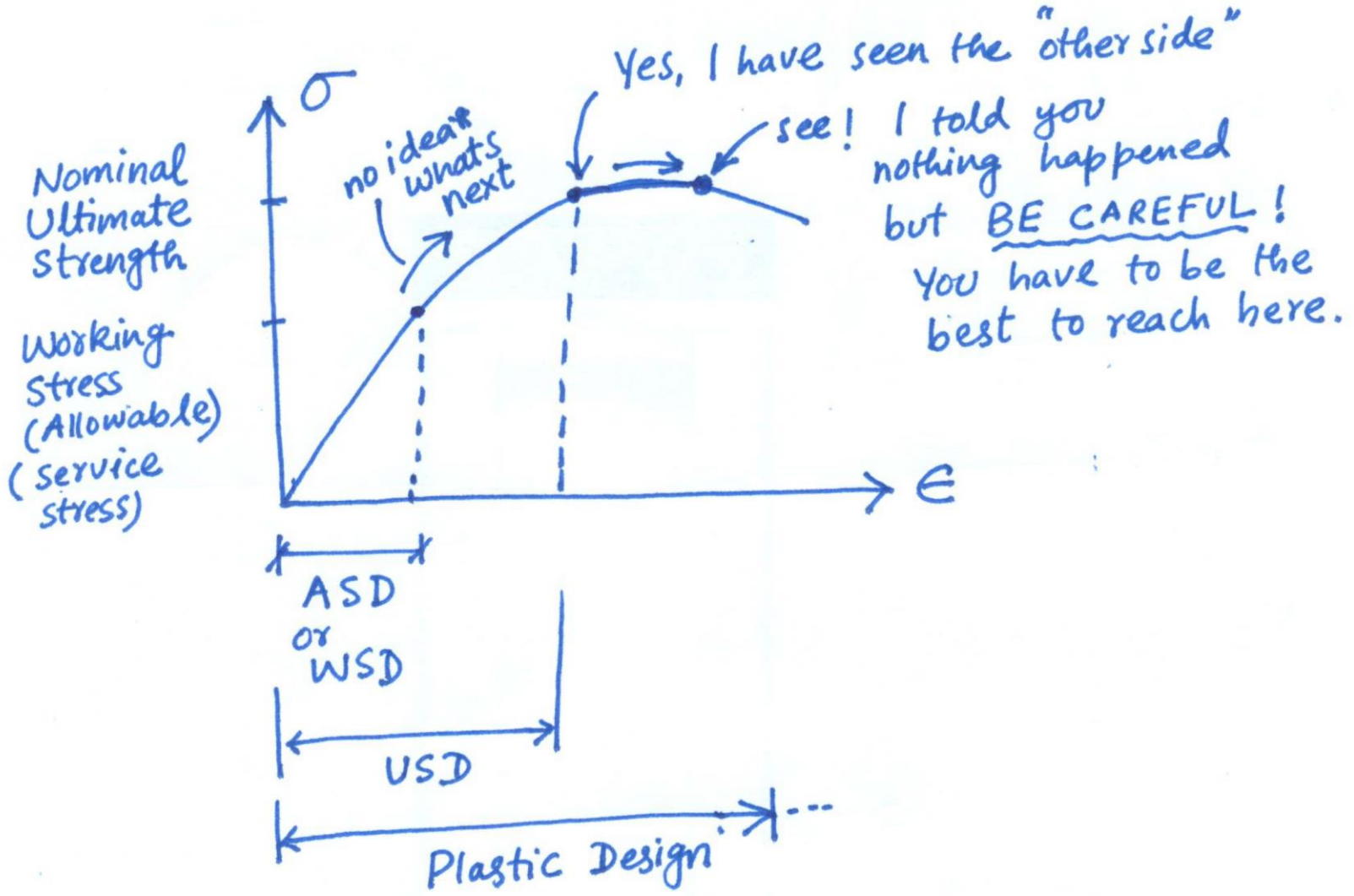


If $\epsilon_T \geq 0.005$ — Failure by excessive deflection and cracking → Tension controlled (Beams, slabs)

If $\epsilon_T \leq \epsilon_{Ty}$ — Brittle compression failure (less ρ) → compression controlled (columns)

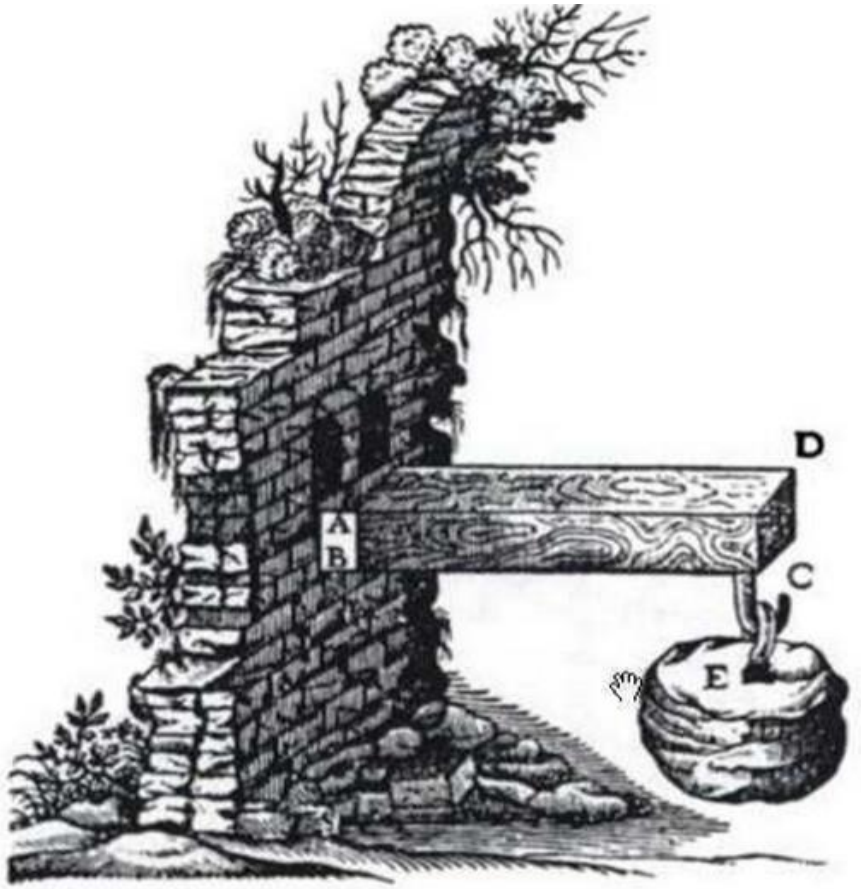
If $\epsilon_{Ty} \leq \epsilon_T \leq 0.005$ — Transition region Linear interpolation.

ASD/WSD vs. USD vs. Plastic Design



Building Codes

The beginning of Structural Engineering



recognition of the loads,

and fact that loads have an effect on members and materials

and that there is a resistance within materials to resist the loads effects

and that there is some relationship between them

Evolution of our Understanding of Structures

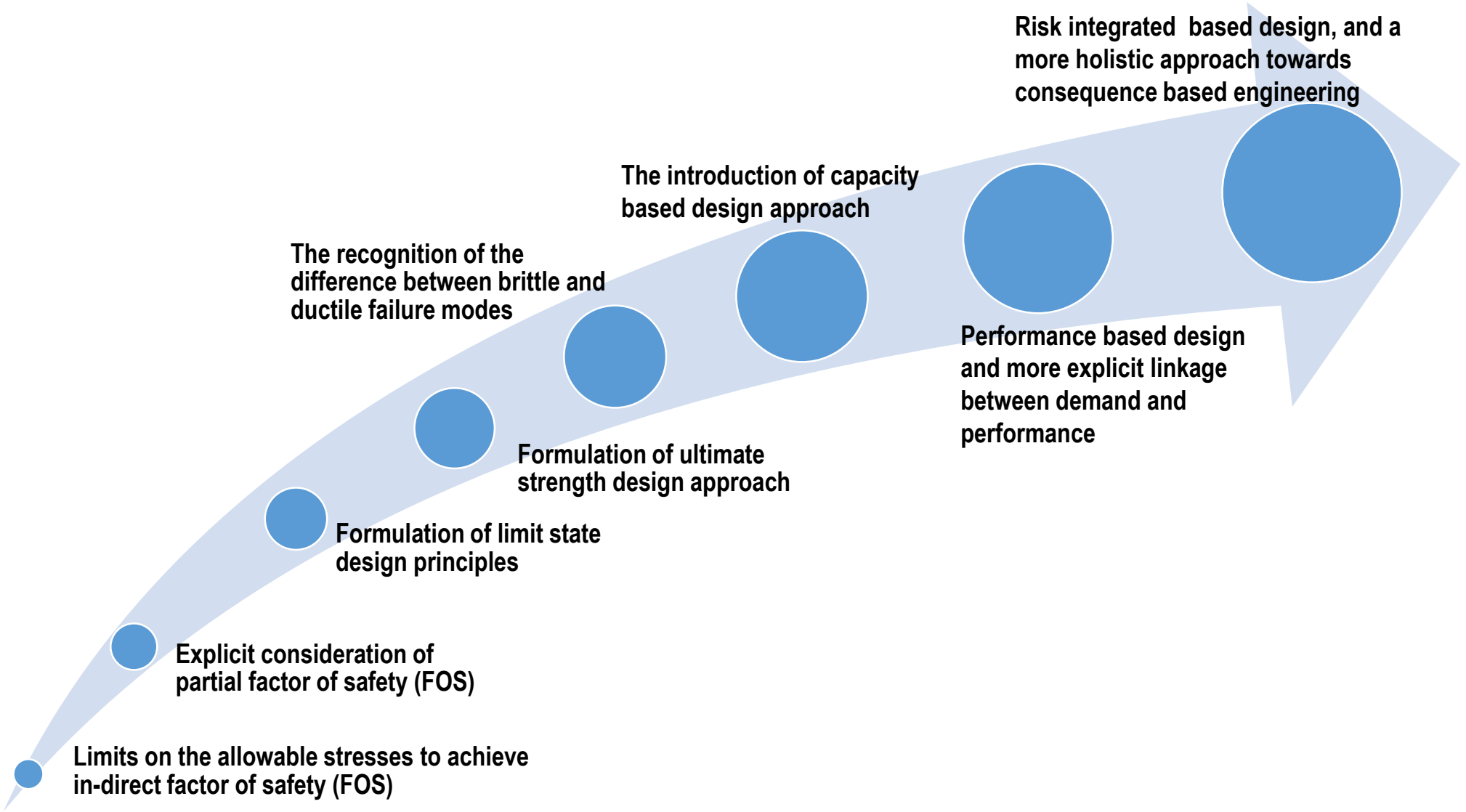
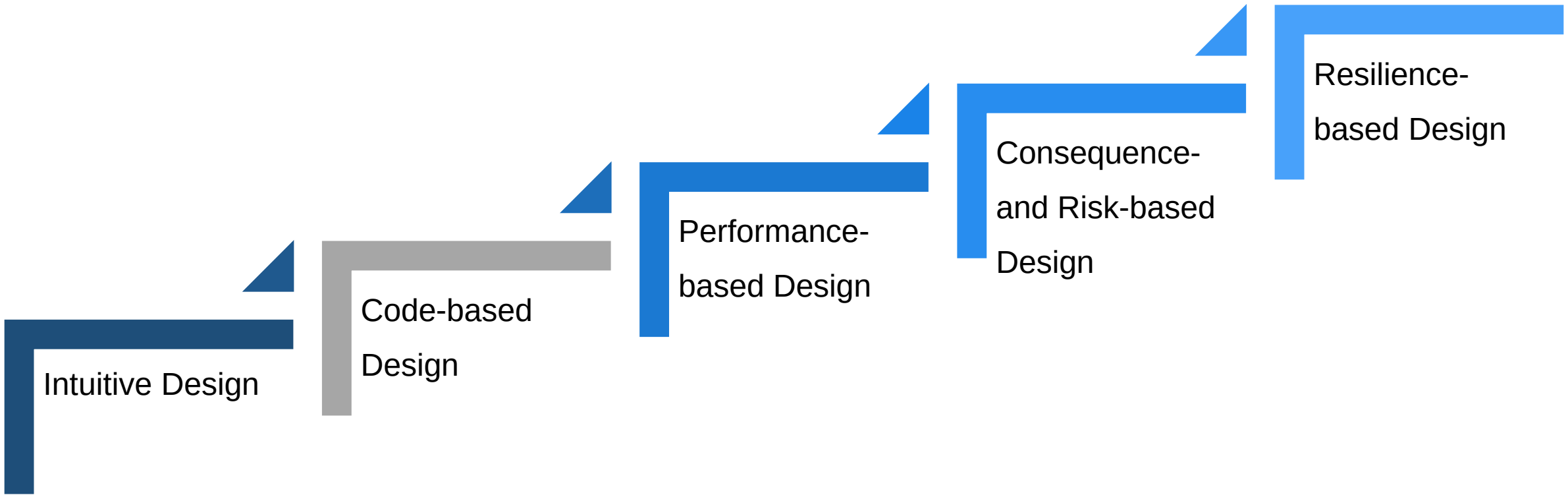


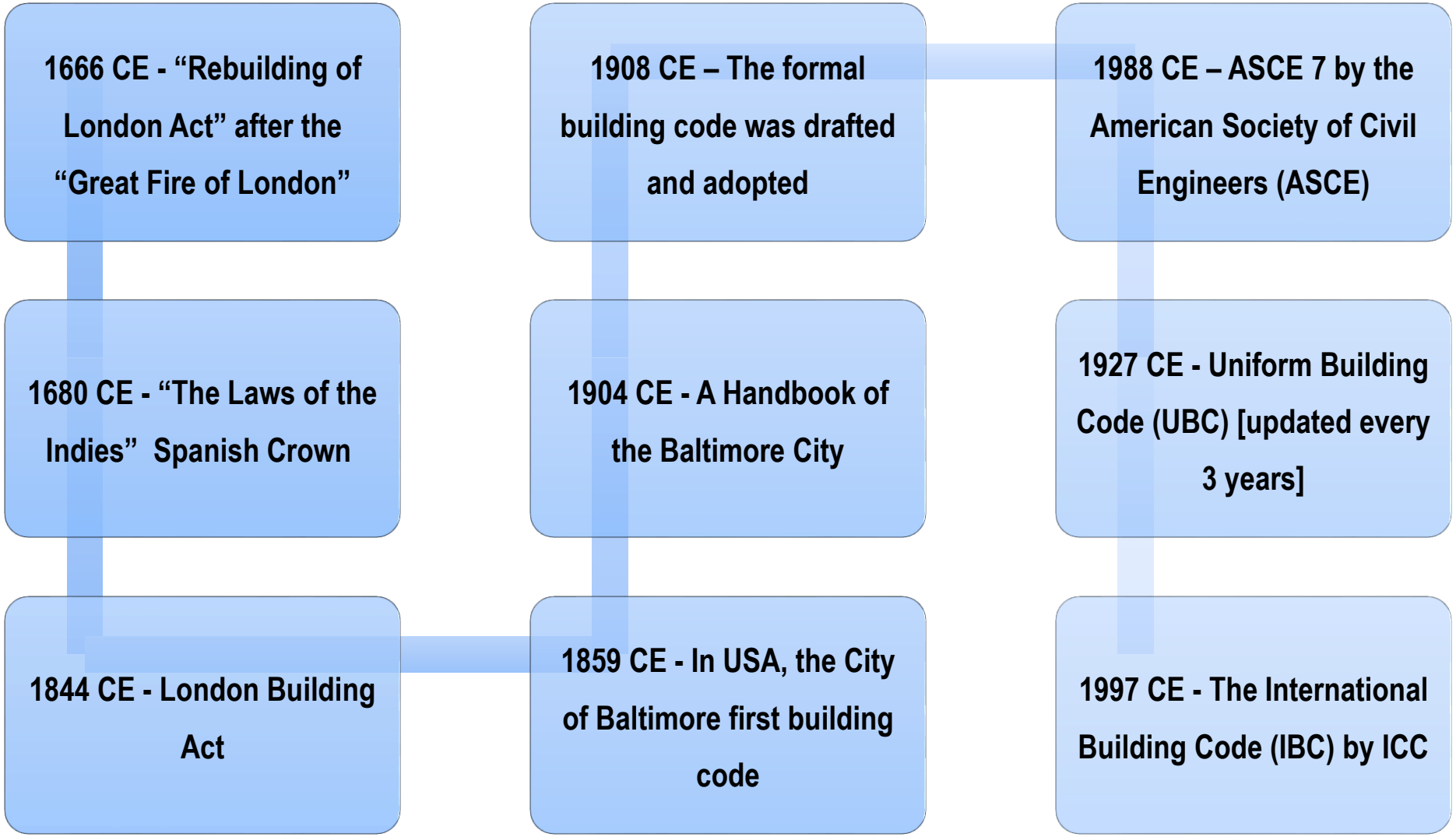
Table 1-1: Important historical developments related to structural analysis and design

Year (CE)	Development
1452–1519	Earliest contributions from Leonardo da Vinci.
1638	Galileo Galilei examined the failure of simple structures and published his book "Two New Sciences".
1660	Robert Hooke presented the Hooke's law which is the basis for elastic structural analysis.
1687	Isaac Newton published his document “Principia Mathematica” containing the famous Newton's laws of motion.
1750	Leonhard Euler and Daniel Bernoulli developed Euler–Bernoulli beam theory.
1700–1782	Daniel Bernoulli introduced the principle of virtual work.
1707–1783	Leonhard Euler developed the theory of buckling of columns.
1826	Claude-Louis Navier published a document analyzing the elastic behavior of structures.
1873	Carlo Alberto Castigliano presented his theorem for computing displacement as partial derivative of the strain energy.
1874	Otto Mohr formalized the idea of a statically indeterminate structures.
1922	Timoshenko corrects the Euler-Bernoulli beam equation and presented “Timoshenko’s Beam Theory”
1936	Hardy Cross developed the moment distribution method, an important innovation in the analysis and design of continuous frames.
1941	Alexander Hrennikoff solved the discretization of plane elasticity problems using the lattice framework.
1942	R. Courant presented solution of problems by dividing a domain into finite sub-regions.
1956	J. Turner, R. W. Clough, H. C. Martin, and L. J. Topp introduces the term “finite element method” and published work which is widely recognized as the first comprehensive treatment of the method.

Evolution of Structural Design Approaches



Development of Formal Buildings Codes



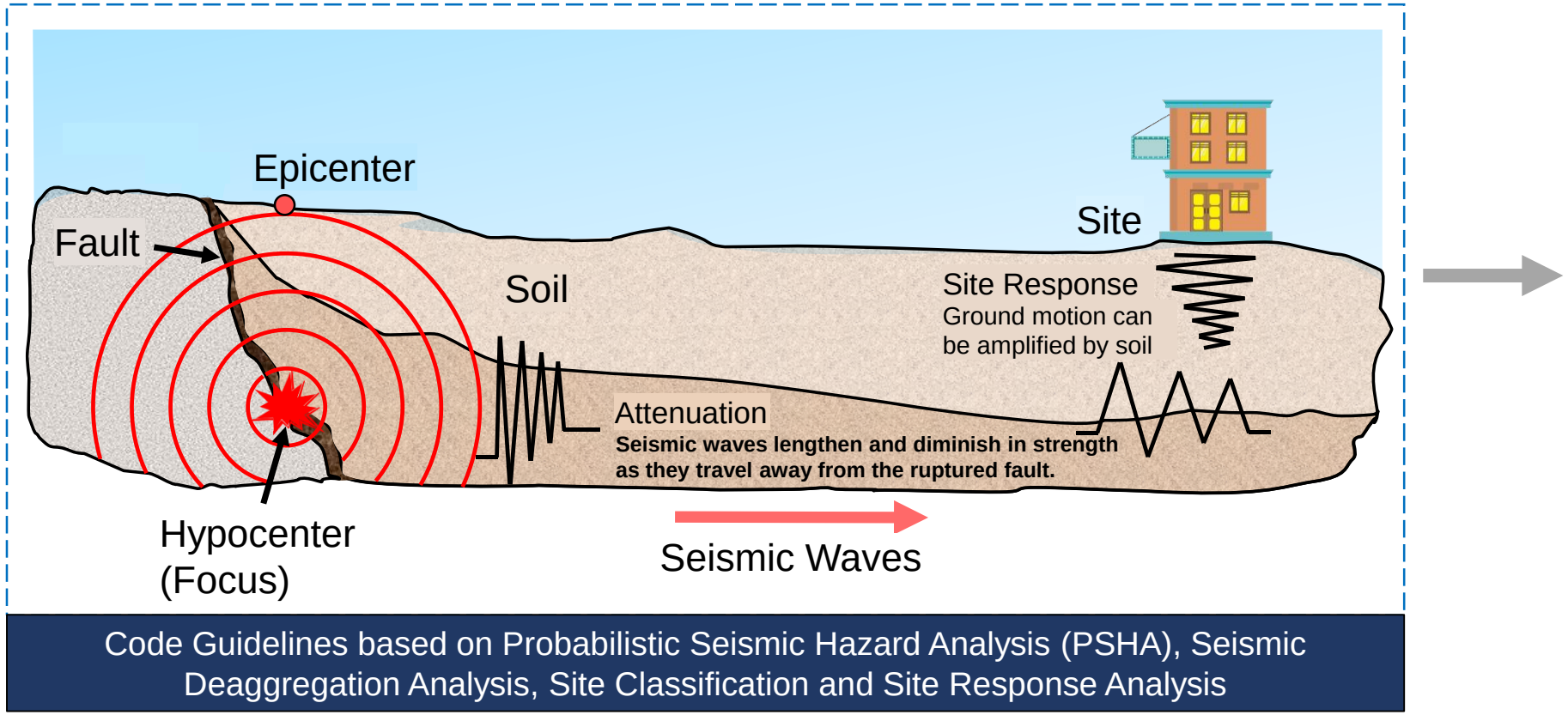
Building Codes

- A building code is a properly documented set of rules and guidelines specifying the minimum standards for constructed facilities.
- The main purpose of building codes is to protect public health, ensure safety and general welfare as they directly govern the construction and occupancy of buildings and other structures.
- The building code becomes law of a particular jurisdiction when formally enacted by the appropriate governmental or private authority.
- The complete process of planning, design, construction and operation of buildings are guided by various building standards, guidelines, codes and design aids.
- Improving the building code quality in terms of addressing real-life problems and enforcement would directly help cities to improve their environmental sustainability and disaster resilience.

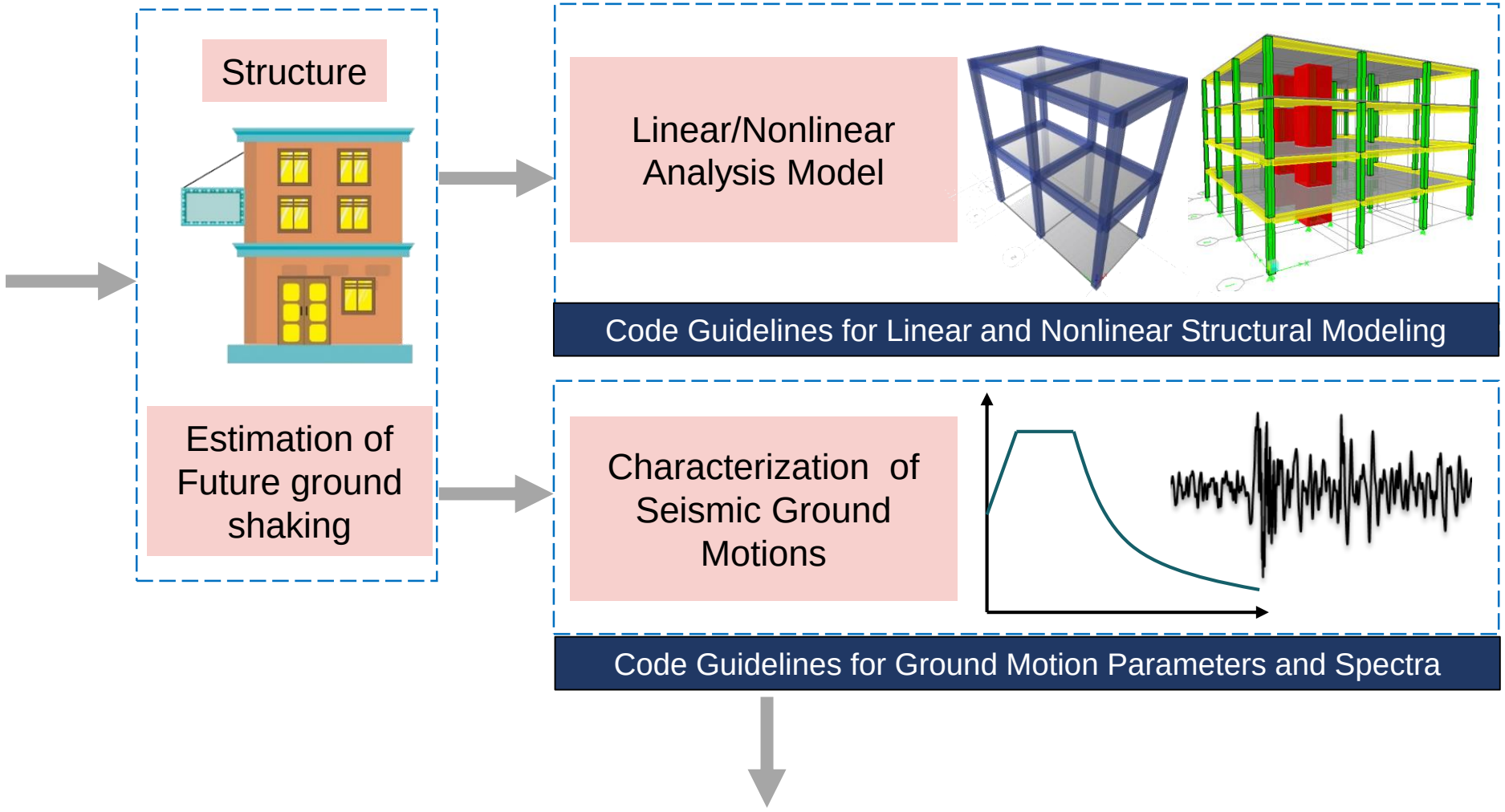
Common Families of Traditional Building Codes

- Uniform Building Codes (UBC), International Building Codes (IBC)
- Guidelines from American Concrete Institute (ACI), American Society of Civil Engineers (ASCE), American Institute of Steel Construction (AISC)
- Euro Codes (EC), British Standards (BS)
- Building Code of Australia (BCA) & Australian Standards, NZ Codes, etc.
- Guidelines from Japanese Society of Civil Engineers (JSCE), Korean Society of Civil Engineers (KSCE), Chinese and Russian Codes, etc.
- Indian Standards (IS), Thailand Codes, Myanmar National Building Code (MNBC), National Building Code of Philippines (NBCP), Building Code of Pakistan (BCP), etc.

Earthquakes and Building Codes (1)



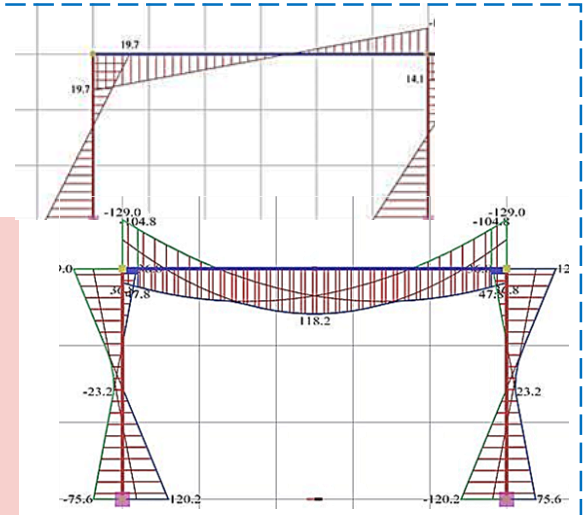
Earthquakes and Building Codes (2)



Earthquakes and Building Codes (3)

Estimation of Linear/Nonlinear Seismic Demands

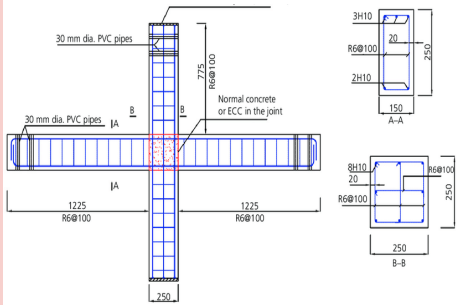
- Global-level Responses
- Inter-story Responses
- Component-level Responses



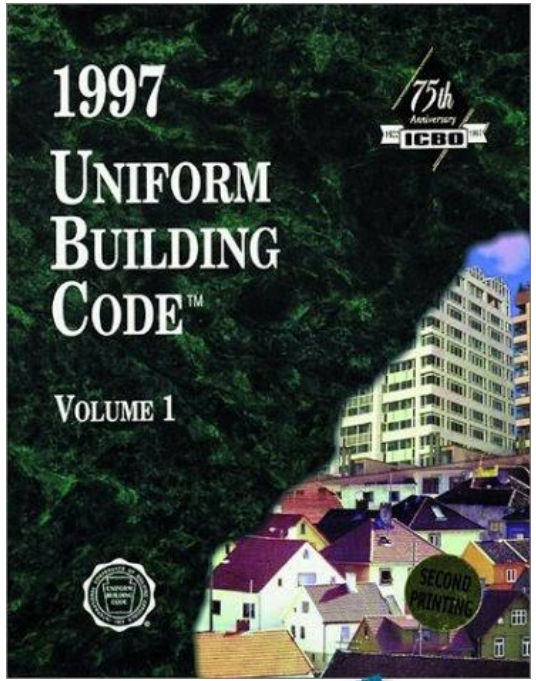
Code Guidelines for Seismic Analysis Procedures

Estimation and Provision of Required Member Capacities

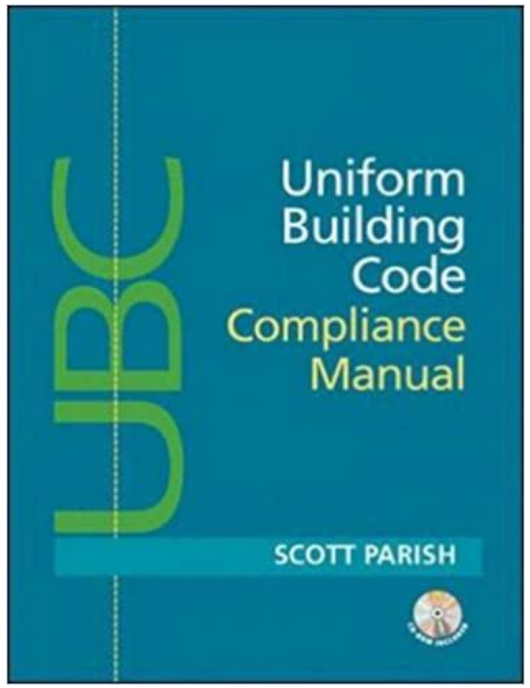
- Reinforcement
- Cross-section sizes
- Detailing rules

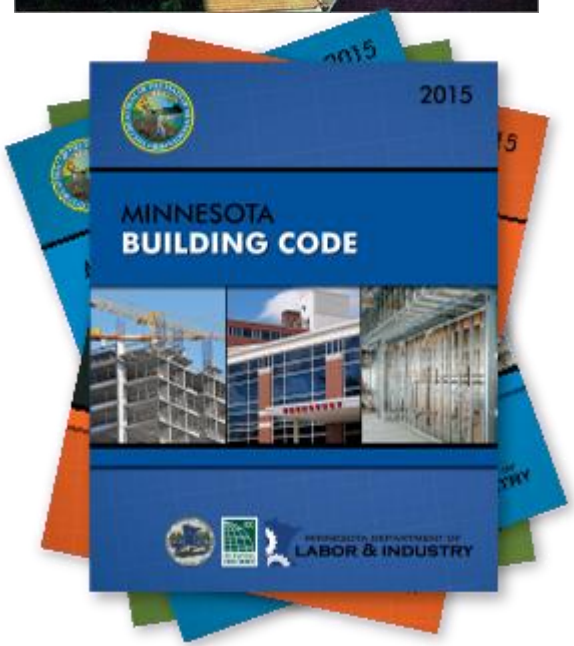


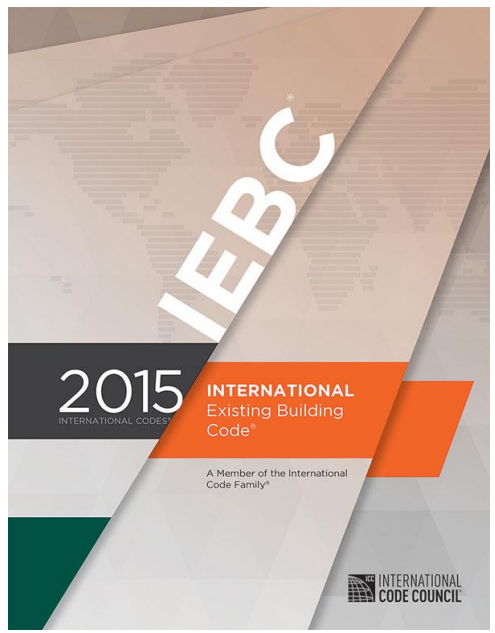
Code Guidelines for Seismic Design and Detailing


1997
UNIFORM
BUILDING
CODE™
VOLUME 1
75th Anniversary
ICBO
SECOND PRINTING

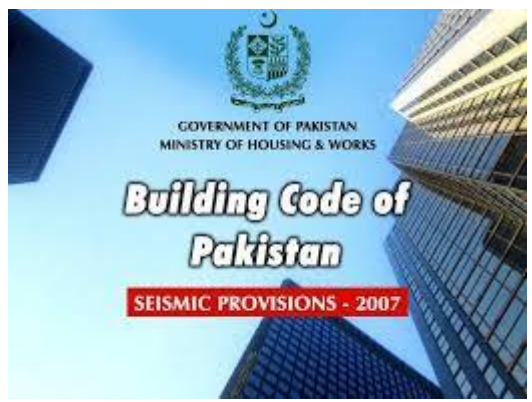

National Construction Code Series
VOLUME ONE
Building Code of Australia 2011
Class 2 to Class 9 Buildings
ABCB


UBC
Uniform Building Code Compliance Manual
SCOTT PARISH


2015
MINNESOTA
BUILDING CODE
LABOR & INDUSTRY


IEBC
2015
INTERNATIONAL
Existing Building Code®
A Member of the International Code Family®
INTERNATIONAL CODE COUNCIL

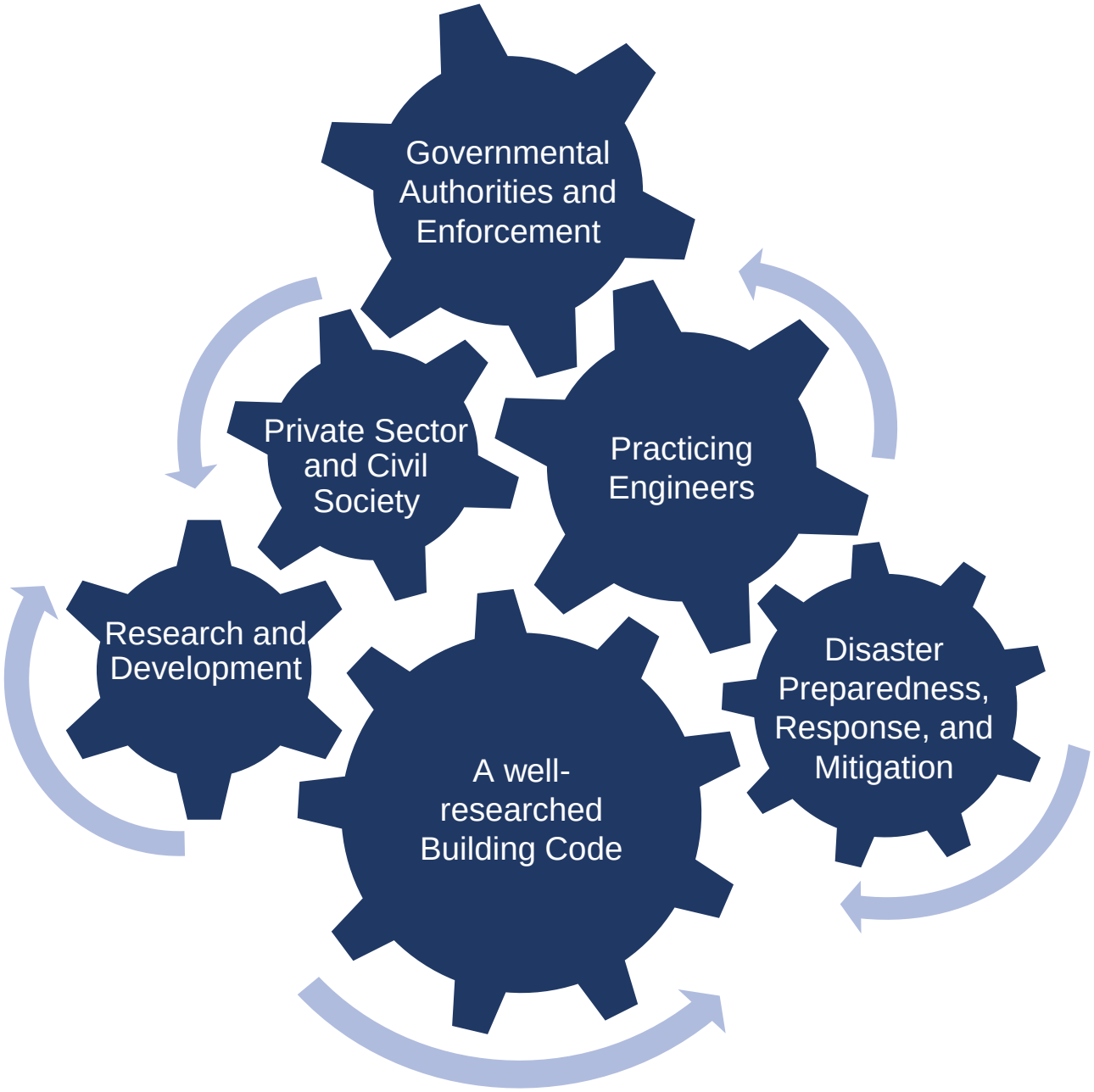

INTERNATIONAL BUILDING CODE
IBC
INTERNATIONAL BUILDING CODE
2018
BBI FAST - TABS
IBC
2018
Builder's Book Inc.


GOVERNMENT OF PAKISTAN
MINISTRY OF HOUSING & WORKS
Building Code of Pakistan
SEISMIC PROVISIONS - 2007

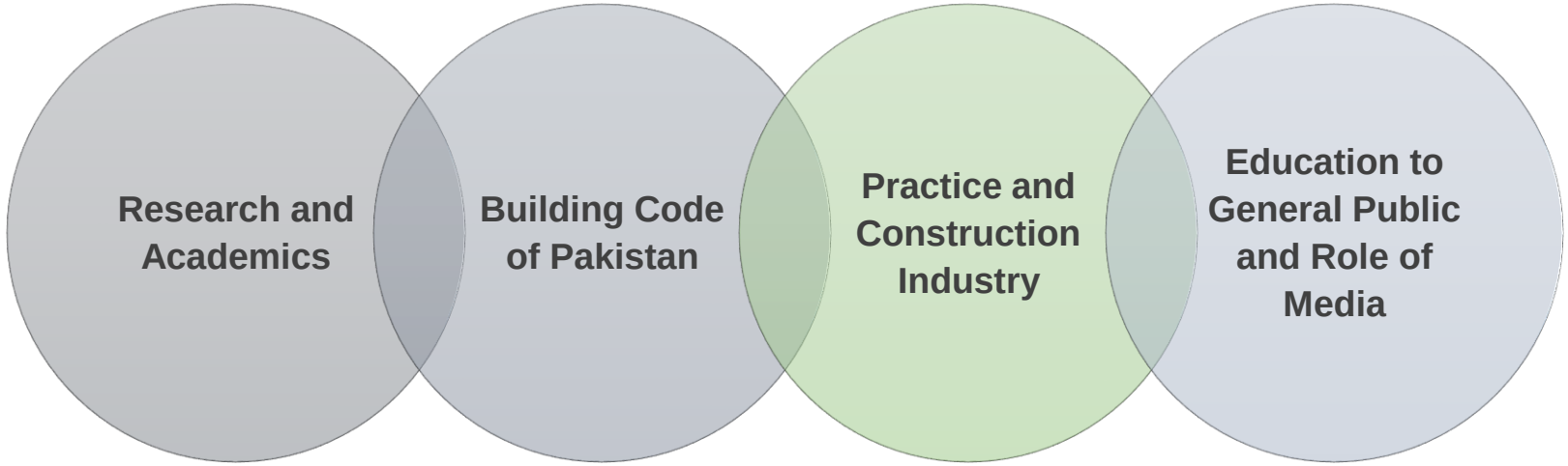
History of Building Codes in Pakistan

Pre-partition	1947 through the 1975	1970s to 1986	1986 to 2007	2007 to 2020	???
• Seismic design provisions were added to the building code of the municipality of Quetta after a damaging earthquake there in 1935	• British building codes remained in use after partition. • Some government agencies continue to use British codes even today, though not necessarily for earthquake design of building structures	• Trained engineers began applying American codes, especially the Uniform Building Code (UBC). • Until the international Building Code (IBC) replaced it in 2000, the UBC was the leading model code throughout California and the U.S. west coast. • It was updated on a three-year cycle through its final edition in 1997.	• Pakistan's first national building code was published in 1986 as "an advisory document" with seismic provisions modeled on contemporary editions of the UBC. • Though perhaps intended for adoption by local bylaw, the 1986 code was not officially adopted or enforced. • While sometimes used as a reference, trained engineers continued to use the UBC.	• After the 2005 Kashmir earthquake, Ministry of Housing & Works hired National Engineering Services Pakistan (NESPAK) to develop seismic hazard maps and earthquake design provisions for a thorough update of the 1986 code. • NESPAK collaborated with a committee of experts convened by the Pakistan Engineering Council (PEC), mostly academics, to produce a consensus document that became the 2007 Building Code of Pakistan. • The provisions of BCP-2007 in conjunction with its model codes are used for the structural design of buildings in Pakistan.	

The Roadmap to Achieve Disaster Resilience in Pakistan

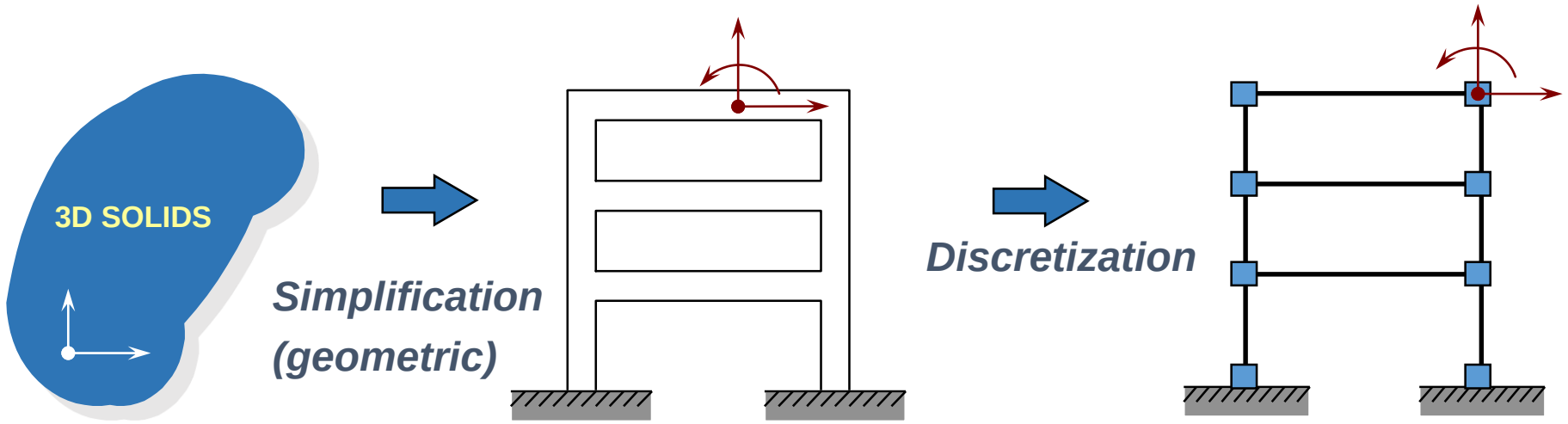


The Roadmap to Achieve Disaster Resilience in Pakistan



Fundamentals of Structural Modelling and Analysis

Solid – Structure - Model



3D-CONTINUUM MODEL
(Governed by partial differential equations)

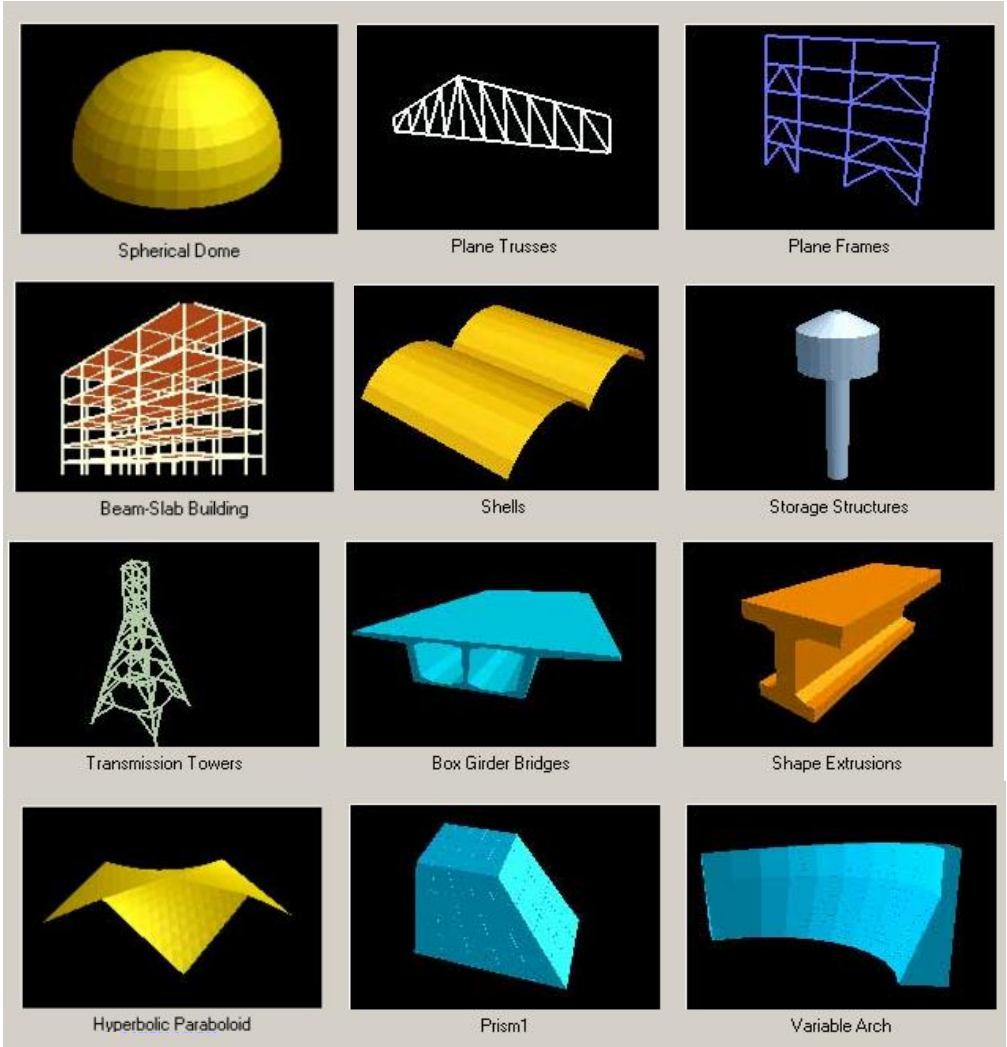
CONTINUOUS MODEL OF STRUCTURE
(Governed by either partial or ordinary differential equations)

DISCRETE MODEL OF STRUCTURE
(Governed by algebraic equations)

Continuum vs. Structure

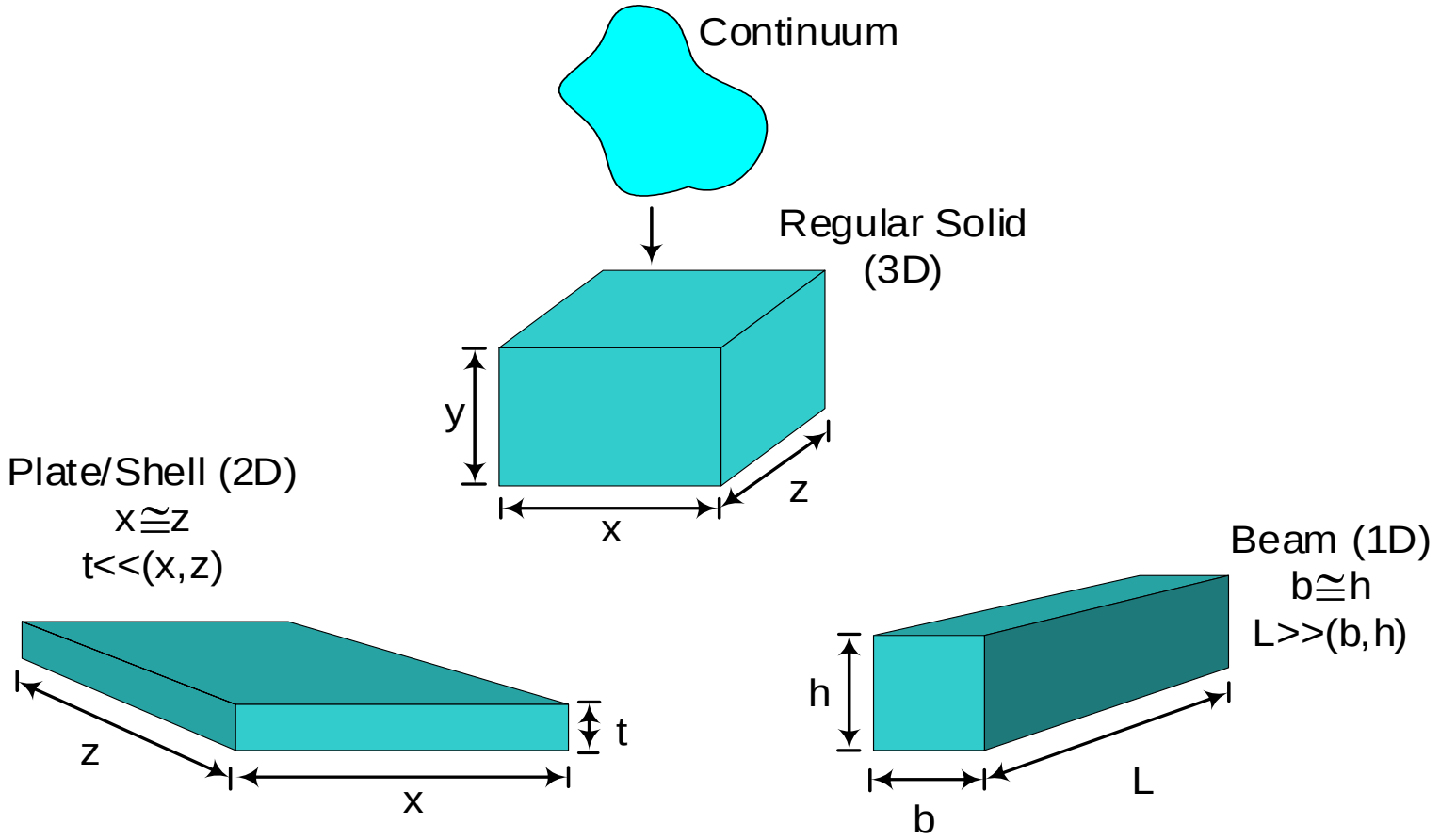
- A **continuum** extends in all direction, has **infinite** particles, with continuous variation of material properties, deformation characteristics and stress states.
- A **structure** is of **finite** size and is made up of an assemblage of substructures, components and members.
- Structures can be categorized in many ways. For modeling and analysis purposes, the overall physical behavior can be used as basis of categorization.
  - Cable or Tension Structures
  - Skeletal or Framed Structures
  - Surface or Spatial Structures
  - Solid Structures
  - Mixed Structures

Structure Types



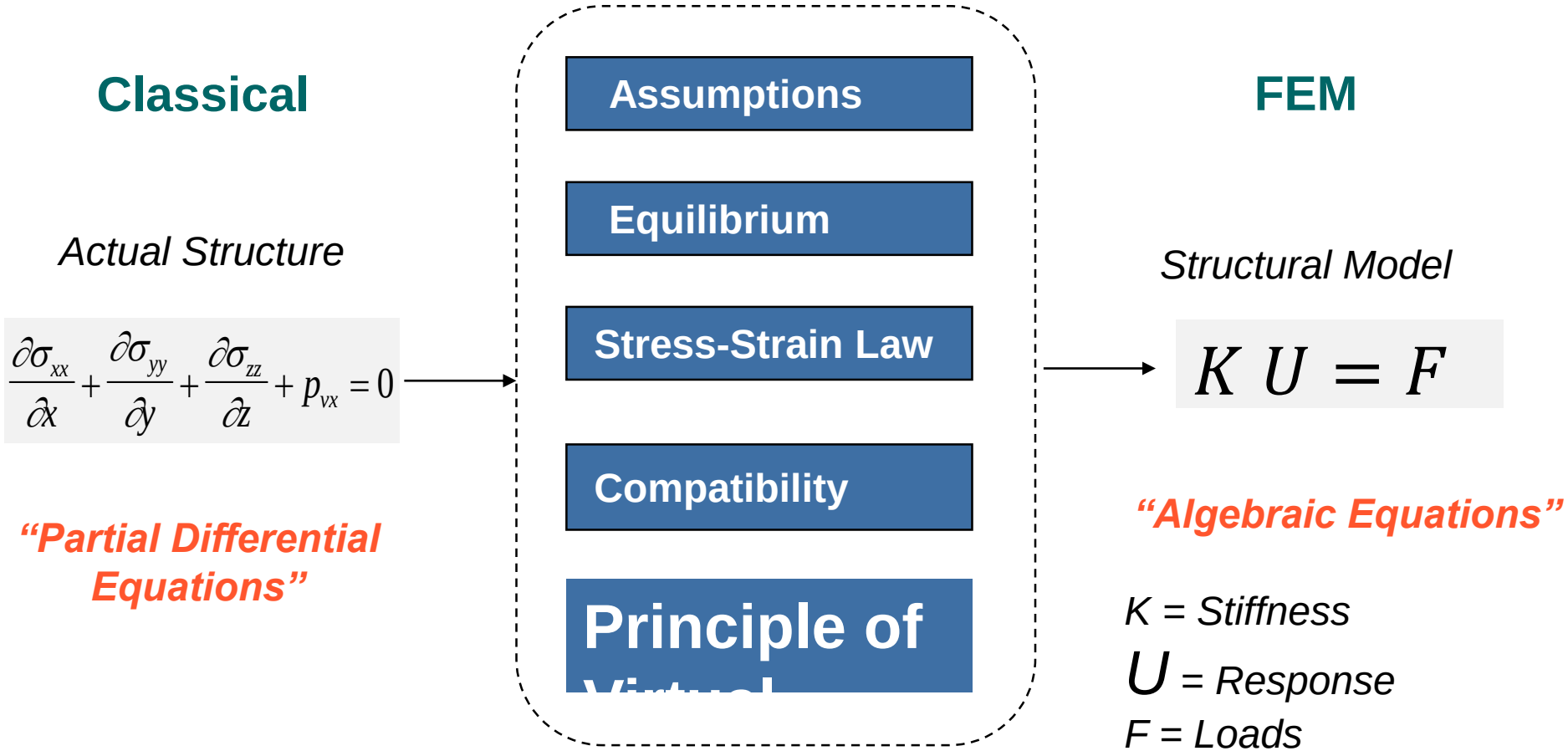
- Cable Structures
  - Cable Nets
  - Cable Stayed
- Bar Structures
  - 2D/3D Trusses
  - 2D/3D Frames, Grids
- Surface Structures
  - Plate, Shell
  - In-Plane, Plane Stress
- Solid Structures

Structural Idealization

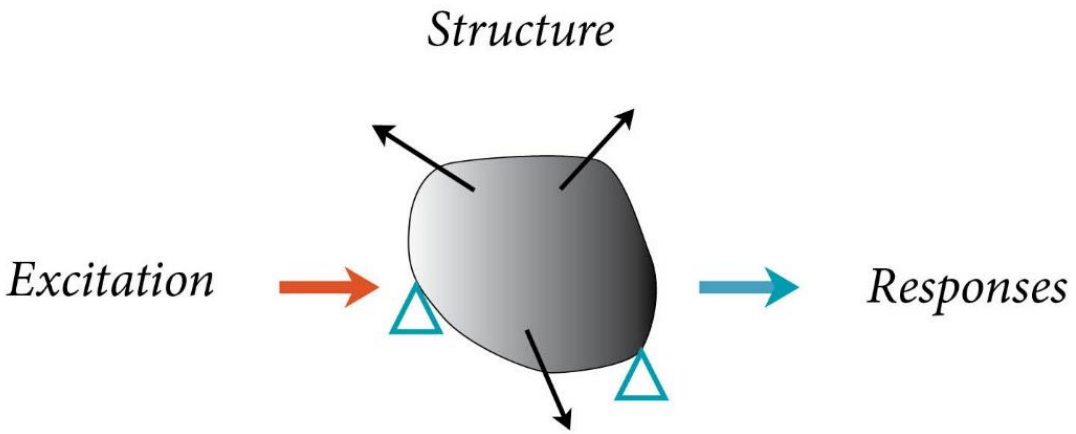


Dimensional Hierarchy of Structural Members

From Classical to FEM Solution

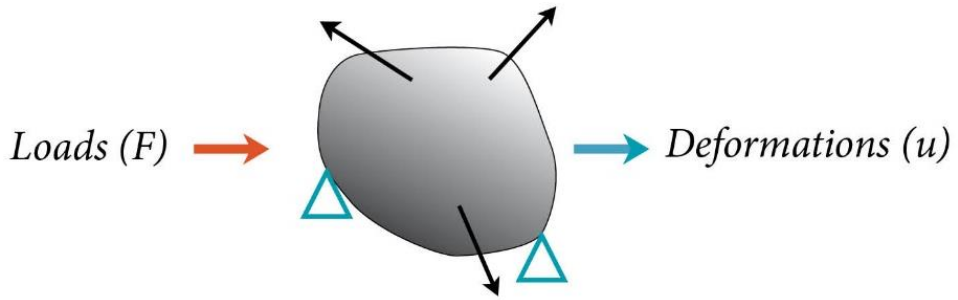


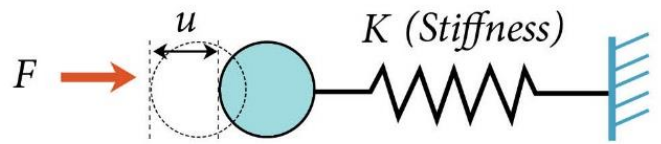
The Concept of Stiffness



- *Static*
- *Dynamic*
- *Elastic*
- *Inelastic*
- *Linear*
- *Nonlinear*

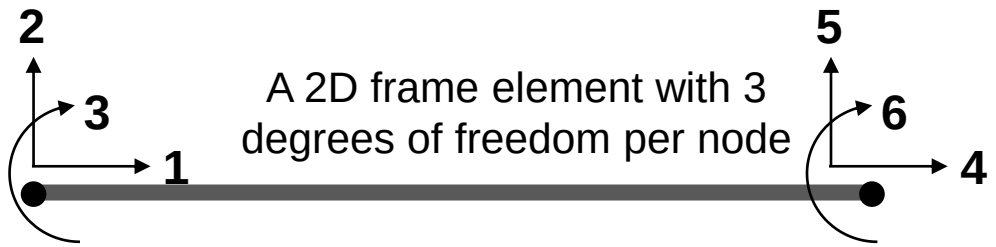
Eigth types of equilibrium equations are possible





Equilibrium Equation → $F=Ku$

What is Stiffness (K) “made off”?

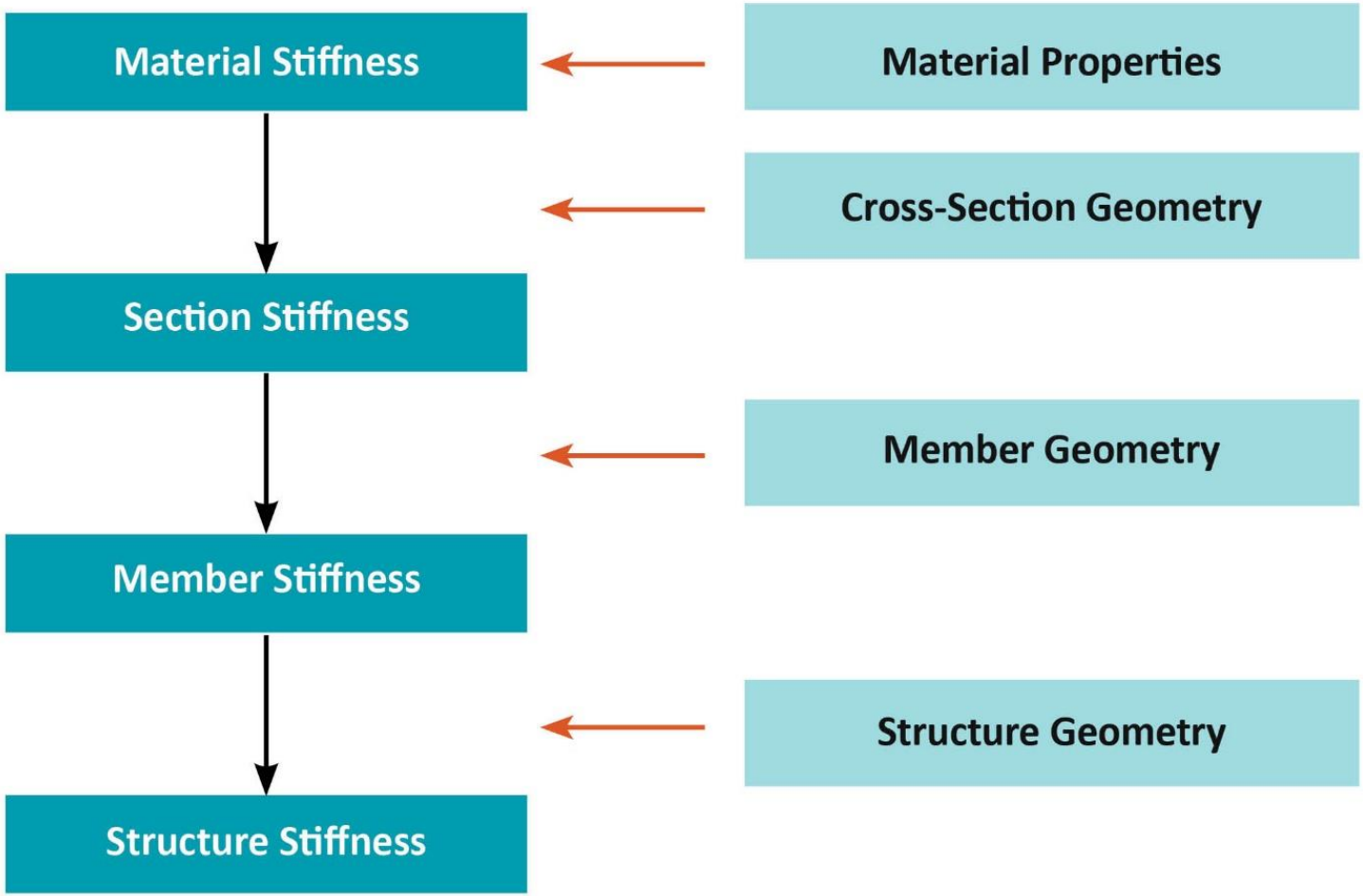


$$K = f(E, A, I, L)$$

Material Property | Cross-section Properties | Member Property

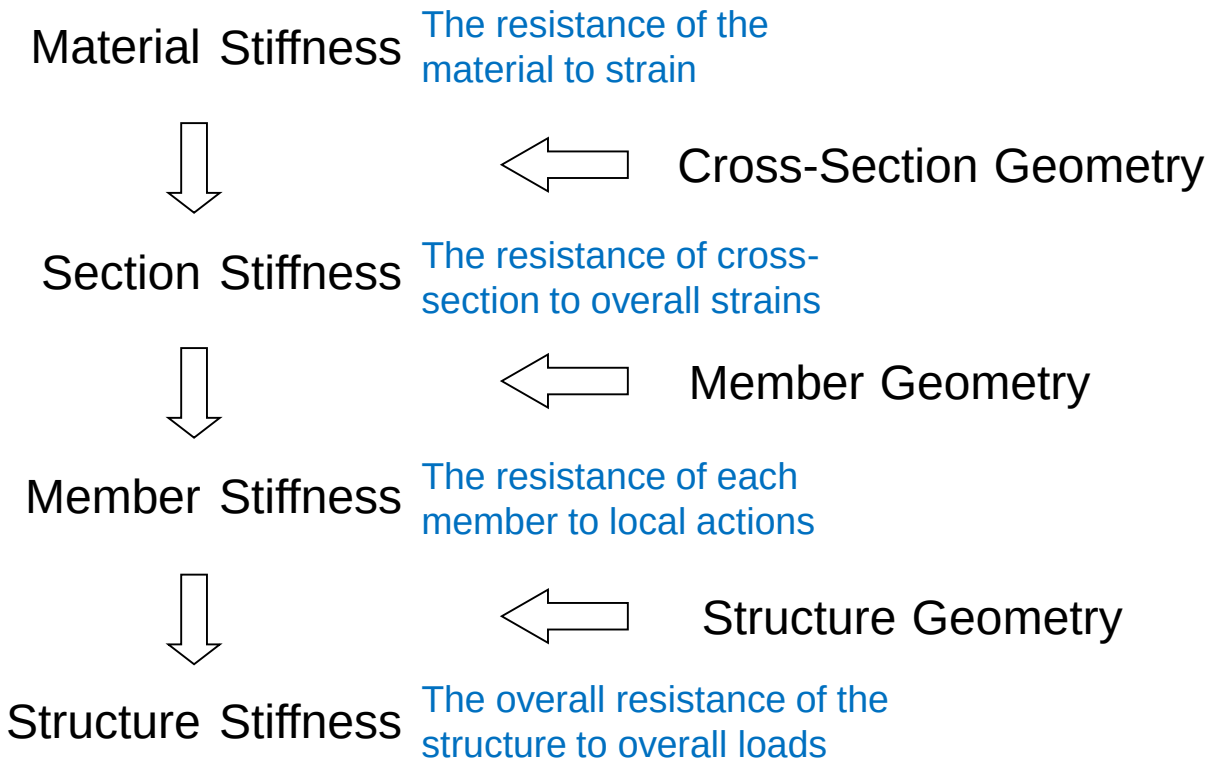
$$K = \begin{pmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{pmatrix}$$

What Stiffness is made of

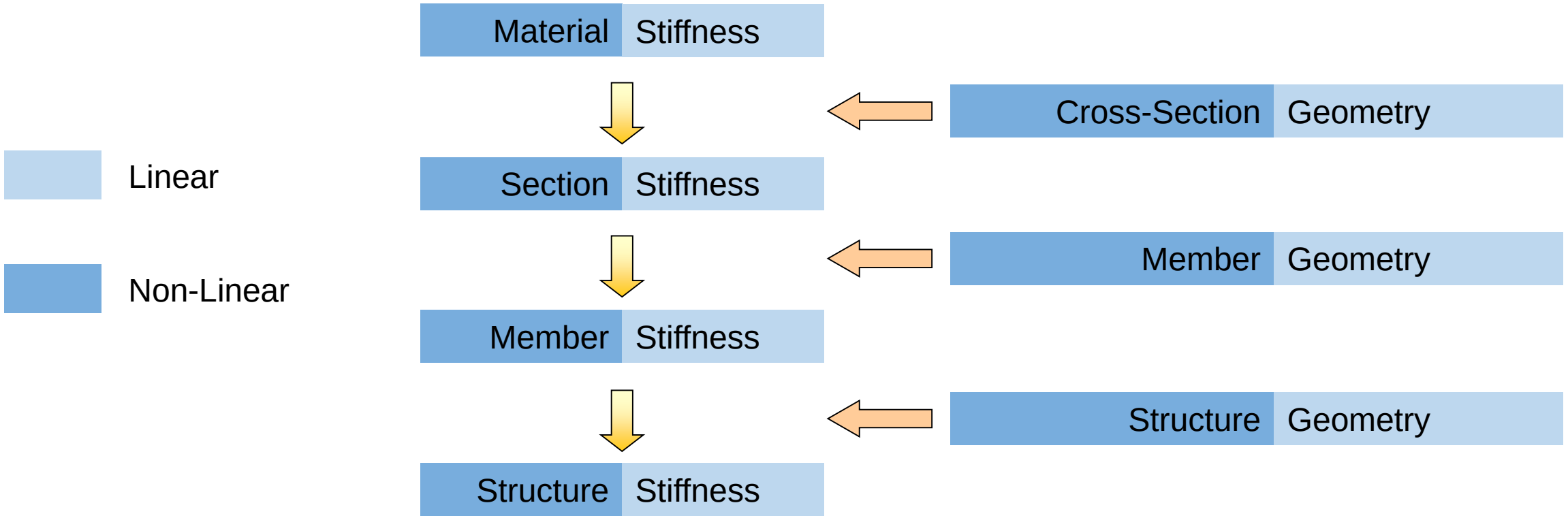


What is Stiffness (K) “made off”?

- The **overall stiffness** of the structure is derived from the overall geometry and connectivity of the members, their stiffnesses, and the boundary conditions.
- The **member stiffness** is derived from the **cross-section stiffness**, and member geometry.
- The **cross-section stiffness** is derived from the **material stiffness** and the cross-section geometry.
- All of these stiffness relationships may be linear or nonlinear.



The Global Structure Stiffness - K



Nonlinear Modeling is an advanced topic

= We provide to the FE software.

Material Modeling or Definition

- Elastic Materials: E, ν, G, α is required.
- Inelastic Materials: Complete stress-strain curve and hysteretic need to be defined.

Cross-section Definition

- Elastic Cross-sections: A, I, shear areas etc. are required.
- Inelastic Cross-sections: Complete moment-curvature curve and hysteretic behavior are required.

1 Material Stiffness

2 Cross-Section Geometry

Section Stiffness

3 Member Geometry

Member Stiffness

4 Structure Geometry

Structure Stiffness

Member Definition

- Elastic Members: Lengths and orientations are required. In several FE software, user can just “draw” members.
- Inelastic Members: Complete moment-rotation curve and hysteretic behavior are required.

Structural Configuration

- Elastic Structure: Overall structural layout, global geometry and orientations are required.
- In several FE software, user can “draw” the whole structure.

Finite Element Modeling, Analysis and Design Process

-1 Idealization

0 Discretization

1 Materials

2 Cross-sections

3 Members

4 Structure

$= K$

5 Loads (F or $F(t)$ vector)

6 Run $\boldsymbol{K\,U = F}$ or $\boldsymbol{M\ddot{U}(t) + C\dot{U}(t) + KU(t) = F(t)}$

7 Post-processing, Local Forces

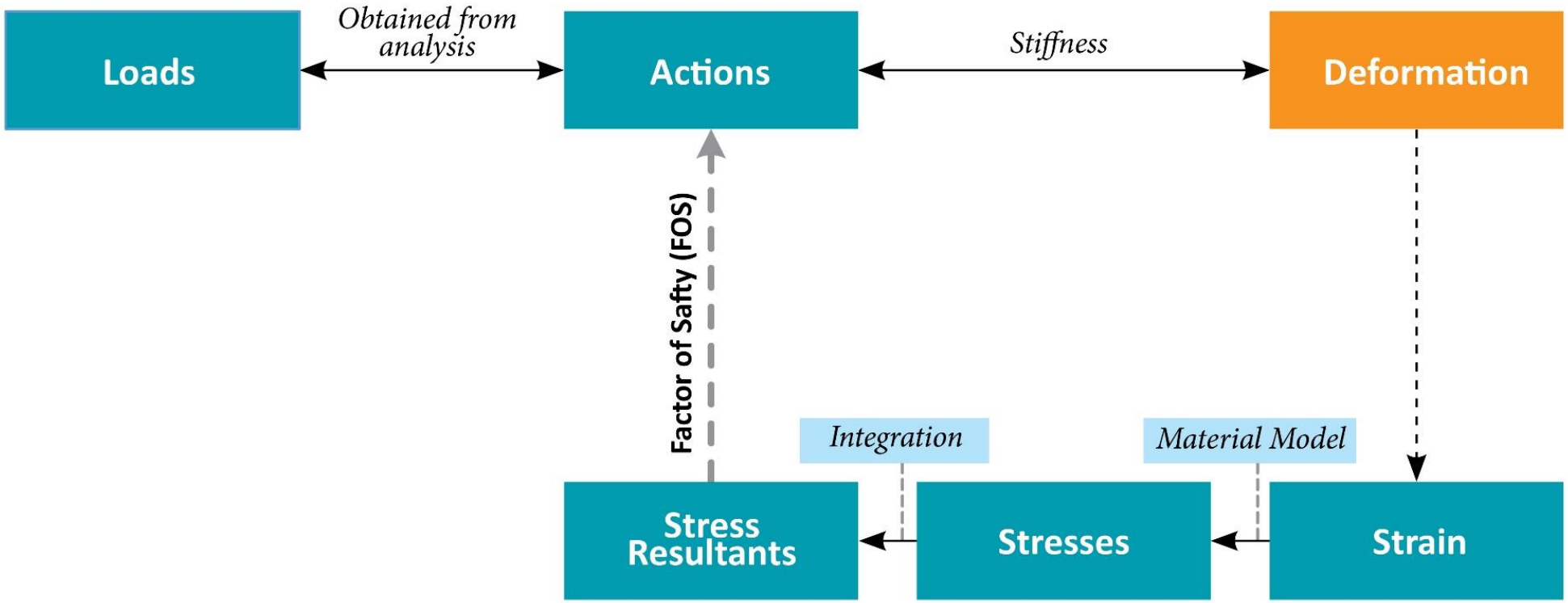
8 Structural Design

= Our task; FE software doesn't help.

= We provide to the FE software.

= FE software does for us.

Fundamental Concepts in Structural Design Process



Fundamental Concepts in Structural Design Process

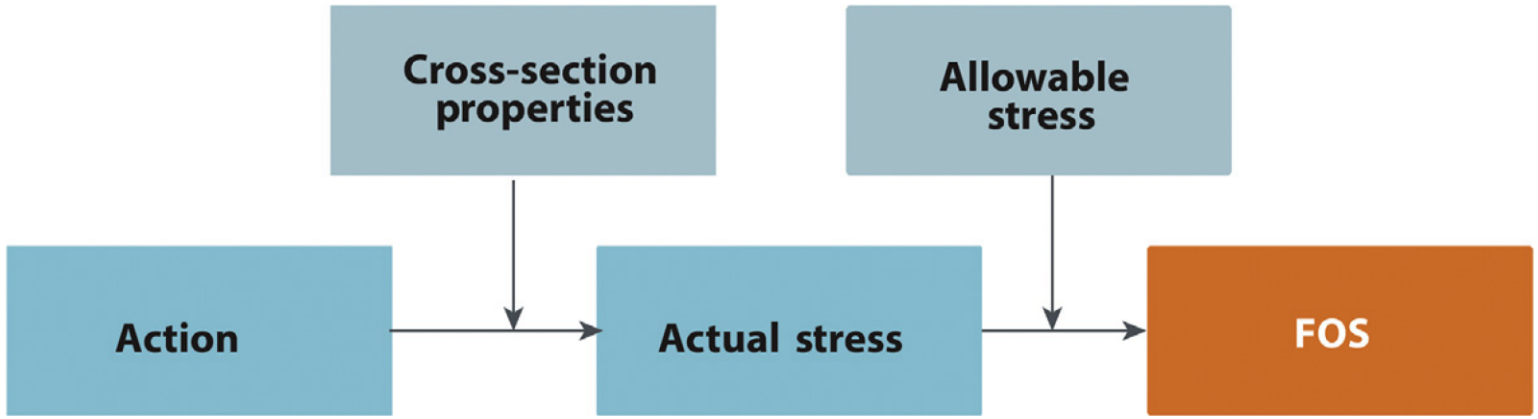


Figure 3.2 Determination of actual stresses due to applied actions as the basis for cross-section design.

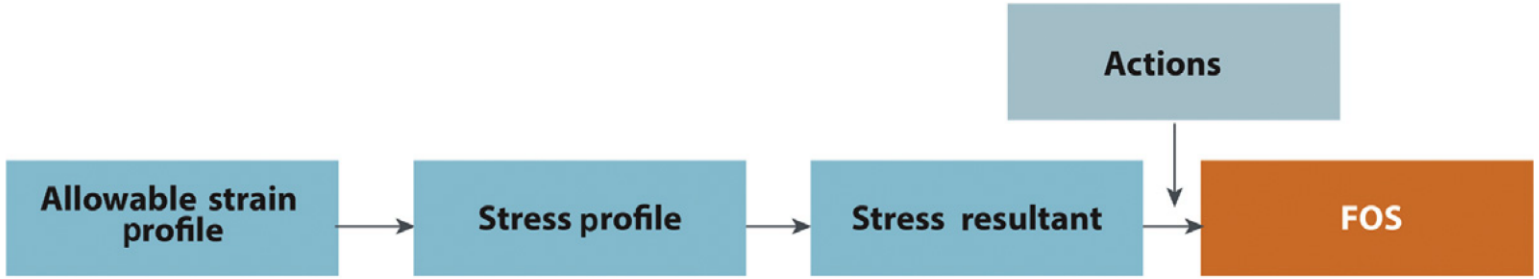


Figure 3.3 Determination of internal stress resultants as the starting point for cross-section design.

Fundamental Concepts in Structural Design Process

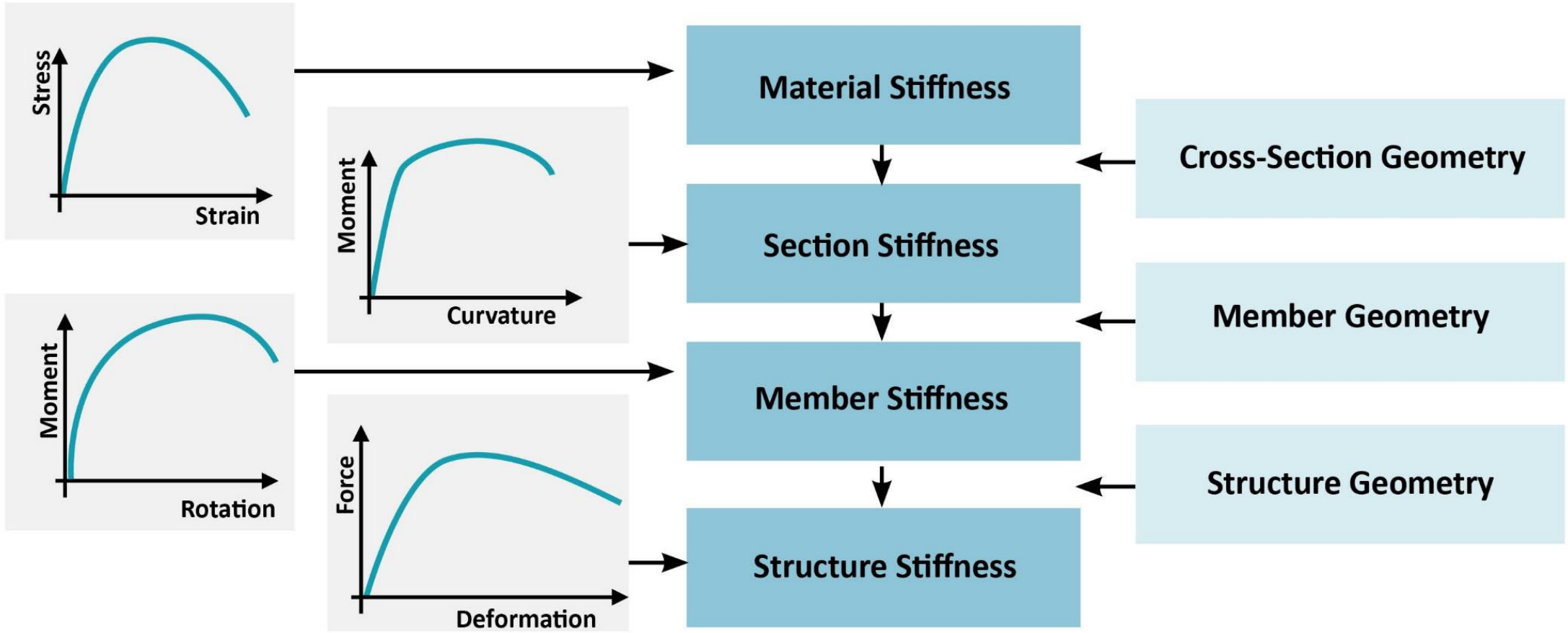
a) **Action-Deformation Relationships:** Defining an action-deformation relationship means linking the deformations produced in a member due to applied actions or linking the restraining actions with applied deformations. These relationships involve the entire stiffness of the member and may be either linear or nonlinear. One action can produce more than one deformation and one deformation may be caused by more than one action.

b) **Deformation-Strain Relationships:** Deformation-Strain relationship means linking deformations with corresponding strains. Each deformation produces a particular strain pattern or profile on the cross-section. A particular strain may be result of several deformations. For example, axial strain may be produced due to axial deformation as well as flexural curvature. This relationship is defined primarily by the cross-section's stiffness and may be linear or nonlinear.

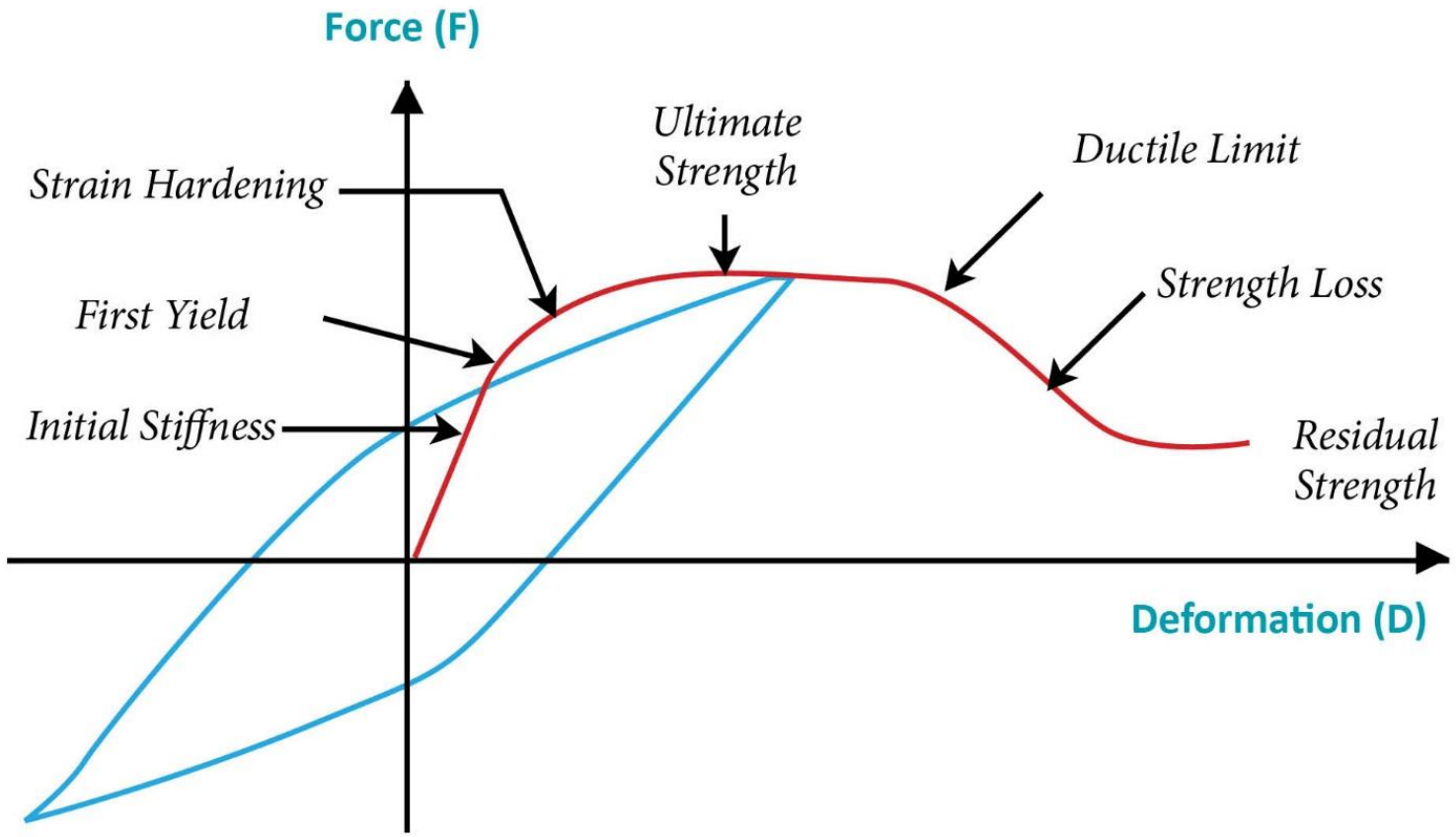
c) **Stress-Strain Relationships:** Stress-Strain relationship means linking strain to corresponding stress. Generally, this relationship is used at material level, indicating material stiffness and its behavior. For example, Hooke's Law describes the stress-strain relationship for a linear elastic material but in general, this relationship is nonlinear for most materials.

d) **Stress Resultant-Action Relationships:** This last relationship is the expression of equilibrium and completes the cycle of all relationships. In fact, this relationship is the basis for strength design of structural members, which states that "the internal stress resultants should be in equilibrium with external actions with adequate margin for safety".

Action-Deformation Relationships

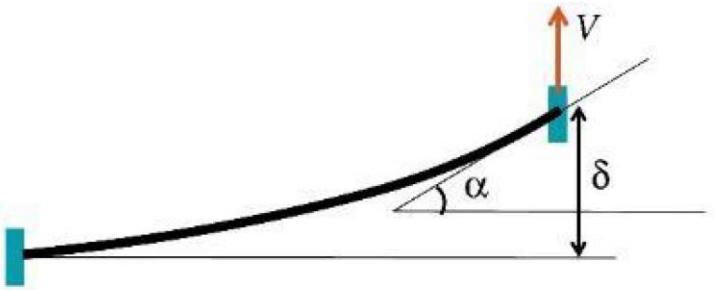
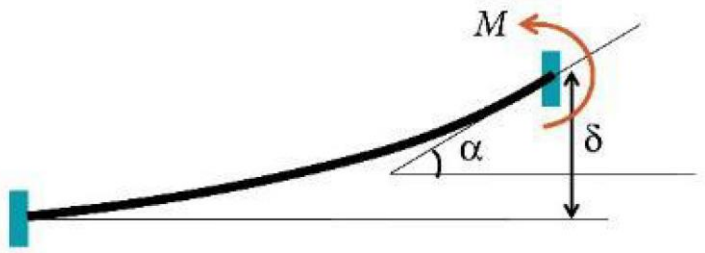


The Nonlinearity of Response and Stiffness (Action-Deformation Relationships)



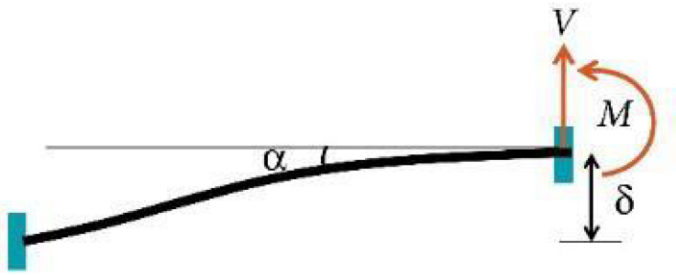
Deformations for Applied Actions: Flexibility Relationships

Table 1-3: Some linear elastic flexibility relationships for a simple beam element

Case	Illustration	Flexibility Relationships
Axial deformation Δ under lateral force P		$\Delta = \frac{PL}{AE}$
Vertical deformation "δ" and rotation "α" under vertical force "V" only		$\delta = \frac{VL^3}{3EI}$ $\alpha = \frac{VL^2}{2EI}$
Vertical displacement "δ" and rotation "α" under Moment (M) only		$\delta = \frac{ML^2}{2EI}$ $\alpha = \frac{ML}{EI}$

Deformations for Applied Actions: Flexibility Relationships

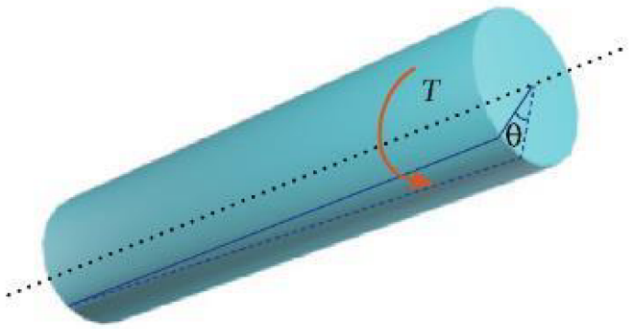
Vertical displacement “δ” and rotation “α” under combined shear force (V) and Moment (M)



$$\delta = \frac{L^3}{6EI}\left(2V - \frac{3M}{L}\right)$$

$$\alpha = \frac{L^2}{2EI}\left(\frac{2M}{L} - V\right)$$

The rotation along the member (θ) due to torque (T) applied at the ends, excluding the effect of warping



$$\theta = \frac{T\,L}{GJ}$$

It is important to note that in all flexibility relationships, actions are related to deformations by various cross-sectional properties (e.g. A, I and J), material properties (e.g. G and E) and length of member (L).

Restraining Actions for Assumed Deformations

Table 1-4: Some restraining actions related by corresponding deformations through linear elastic stiffness

Case	Actions for assumed deformations
Axial force P due to axial deformation Δ	$P = \frac{E A}{L} \Delta$
Shear force V at the restraining end for deflection "δ" and rotation "α" at the other end	$V = \frac{12 E I}{L^3} v + \frac{6 E I}{L^2} \alpha$
Moment M at the restraining end for deflection "δ" and rotation "α" at the other end	$M = \frac{6 E I}{L^2} v + \frac{4 E I}{L} \alpha$
Restraining torque T due to axial rotation θ	$T = \frac{G J}{L} \theta$


Thank You!